

The Great Silent Crash of the 21st Century*

Dr. John Taskinsoy ^a

ABSTRACT

A mysterious creator under the alias Satoshi Nakamoto (a pseudonym) launched the world's first successful cryptocurrency in early January 2009 which, not only was a historic moment, but was one that cultivated a technology revolution and money's evolution into a digital form. However, technical issues (inherent flaws) in the design of Bitcoin blockchain, non-technical issues (political backlash, regulatory hurdle, and environmental hazard), plus the opaqueness surrounding the launch of Bitcoin have opened the door for an endless debate, incessant criticism, spurious claims, heated arguments, plethora of articles, and media frenzy contemplating what Bitcoin really is (crypto-asset, commodity, or investment vehicle) or it is not (currency). On the technical side, blockchain that made Bitcoin a household name fails miserably; high latency (8 minutes or more), low transaction throughput (7 per second), low-scalability (mining, proof-of-work based validation using consensus and cryptography), and high energy cost make Bitcoin unfit to compete with new and fast-scalable cryptocurrencies such as Solana's transaction speed (50,000 per second) or XLM's \$0.00001 fee per transaction. Although technical problems are not without a solution, non-technical issues are not easy to resolve because their resolution depends on politicians, law makers, regulators, and various government agencies (central banks, the Fed and the ECB in particular) who choose to run headlong into backlash to Bitcoin and other cryptocurrencies. On Tuesday (October 8, 2022), prices of cryptocurrencies tanked, citing the industry-shaking collapse of FTX (second largest after Binance), some even dubbed the event as "*Crypto's Lehman moment*". But erratic price movements is not something new in the crypto industry which has been on a roller-coaster since December of 2021, i.e. after Bitcoin price hit almost \$68,000 and its market cap \$1.24 trillion, jittery investors in a hurry began to cash out their hefty gains. The inability of FTX's CEO Sam Bankman-Fried to handle his plan to sell his company (which was regarded as one of crypto's "*blue chip*" companies) to the rival crypto exchange Binance set off a widespread selling panic, as a result, cryptocurrency market shed a mindboggling \$236.7 billion (\$81 billion by Bitcoin) in just two days (Tuesday and Wednesday), which by any standard was insanely bonkers.

Keywords: Bitcoin; Market Crash; Cryptocurrency; Digital Revolution; Environmental Hazard

JEL classification: G12, G21, G32, E42, E44, E51

* This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

^a Corresponding author email address: johntaskinsoy@gmail.com

Faculty of Economics & Business – Universiti Malaysia Sarawak (Unimas), 94300 Kota Samarahan, Sarawak, Malaysia.

1.0 Introduction

Ever since money was invented¹, economies where coinage or paper money was used as a medium of exchange have become increasingly susceptible to shocks in form of economic, financial and currency crises (for the history of money, see Davies, 2002). Before the invention of money, there was bartering which dates back to the prehistoric people; it is a form of trade without the use of a monetary medium such as metallic (silver, gold, copper) money, paper (fiat) money, plastic money (credit, debit, store, gas), digital coin or electronic cash (Bitcoin) to facilitate exchange. A bartering system both in theory and practice is distinctly different than a monetary-based economy; for instance, a barter transaction is extremely simple, executed unilaterally or multilaterally without the need of a trusted third party intermediation (i.e. banks, credit cards, and no transaction costs). Also in barter economy, only goods or services are used for exchange of other goods and services; moreover, immediate delivery takes place at the conclusion of a mutual agreement. Unlike the merciless capitalism where exploitation of labor supersedes social welfare, barter is a purely peer-to-peer system of exchange, and this aspect of bartering makes it free of the capitalist system's complex problems and its inflationary monetary mechanisms. Under barter, agents of the economy are not concerned with frictional costs, asymmetry of information, economies of scale, exploitation of labor, and forms of employment. More importantly, the barter system does not revolve around expanding or contracting of the aggregate money supply through credit (printing and unprinting money), therefore there is no debt hangover.

Although the probability of a high-magnitude economic or financial crisis to occur is circa 5% (once every 20 years or so, about every 10 years in the U.S.), the recurrence of contemporaneous modern crises is nothing of new, they had occurred in the past but more disruptive and longer-lasting crises promise to strike in the future. Despite the world undergoing several hundred cases of financial folly (Reinhart & Rogoff, 2009, see Table 1²), the human race is still very much incapable of recognizing warning signs (writings on the wall) of the next crisis, eradicating causes, or predicting in advance the time of a future crisis. Douglas and Wildavsky (1983) asked; "Can we know the risks we face, now or in the future? No, we cannot: but yes, we must act as if we do". Banks, an inextricable part of any economy, facilitate and allocate real and scarce resources from savers to investors who enter into intertemporal contracts through space and time amid uncertainties. Therefore, banks not only have an endless propensity to adopt financial innovation to cope with asymmetry of information, imperfect frictions and transaction costs (Arrow, 1964), but a normal-functioning banking system can provide signs of stability or instability and be a good predictor of future prospects (Levine, 1997).³

¹ First paper money was used by the Chinese Tang Dynasty during A.D. 618-907 (for a historical perspective, see Cipolla (1976); Cook (1958); Dalton (1967)); however Excessive issues of printed money in China caused hyperinflation, leading to the elimination of paper money in 1455 (also see Taskinsoy, 2019a).

² A small list of important economic, financial and currency crises.

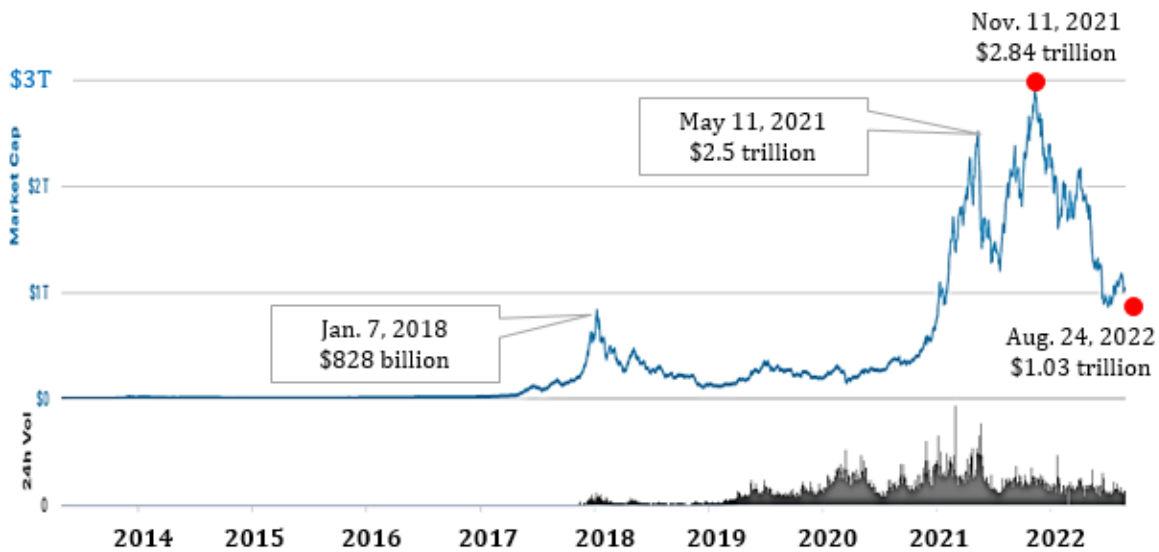
³ For a historical perspective and longer discussions on this topic and other related topics, interested readers are welcome to check out a list of articles by Taskinsoy (2012; 2013; 2018; 2019; 2020; 2021; 2022).

Table 1: List of economic crises and depressions

1st Century	The Financial Panic of AD 33 (unsecured loans by main Roman banking houses).
3rd Century	Crisis of the Third Century (Imperial crisis, Roman Empire nearly collapsed).
7th Century	Coin exchange crisis of 692 (Byzantine emperor Justinian II refuses to accept tribute from the Umayyad Caliphate with new Arab gold coins).
14th Century	Banking crisis (the crash of the Peruzzi and the Bardi family in 1345).
17th Century	Kipper und Wipper (1618–22) financial crisis at start of Thirty Years' War Tulip mania (1637) an economic bubble that burst in 17th century Holland The General Crisis (1640) Arguably the largest worldwide crisis in history
18th Century	Great Tobacco Depression (1703) (British America)[2] South Sea Bubble (1720) (UK) Mississippi Company (1720) (France) Crisis of 1763 – started in Amsterdam Great East Indian Bengal Bubble Crash (1769) (India) Crisis of 1772 – started in London and Amsterdam (collapse of the bankers) War of American Independence Financing Crisis (1776) (United States) Panic of 1785 – United States Panic of 1792 – United States Panic of 1796–1797 – Britain and United States
19th Century	Danish state bankruptcy of 1813 Post-Napoleonic depression (post 1815) (England) Panic of 1819, a U.S. recession with bank failures Panic of 1825, a pervasive British recession in which many banks failed Panic of 1837, a U.S. recession with bank failures, followed by a 5-year depression Panic of 1847, started as a collapse of British financial markets Panic of 1857, a U.S. recession with bank failures Panic of 1866, was an international financial downturn Great Depression of British Agriculture (1873–1896) Long Depression (1873–1896) Panic of 1873, a US recession with bank failures, followed by a four-year depression Panic of 1884; Panic of 1890; Panic of 1893, a US recession with bank failures Australian banking crisis of 1893; Panic of 1896
20th Century	
1900s	Panic of 1901, a U.S. economic recession (control of the Northern Pacific Railway) Panic of 1907, a U.S. economic recession with bank failures
1920s	Depression of 1920–21, a U.S. economic recession following the end of WW1 Wall Street Crash of 1929 and Great Depression (1929–1939) the worst depression
1970s	1970s energy crisis; OPEC oil price shock (1973); 1979 energy crisis (1979); Secondary banking crisis of 1973–1975 in the UK; Latin American debt crisis (late 1970s, early 1980s) known as "lost decade"
1980s	Early 1980s Recession; Chilean crisis of 1982; Bank stock crisis (Israel 1983); Japanese asset price bubble (1986–1992); Black Monday (1987) (1987) (US); Savings and loan crisis, 1,043 out of the 3,234 S&Ls failed from 1986 to 1995 in the U.S
1990s	Special Period in Cuba (1990–1994); Early 1990s Recession; 1991 Indian economic crisis; Finnish banking crisis (1990s) (1991–1993); Swedish banking crisis (1990s) Black Wednesday (1992); 1994 economic crisis in Mexico; 1997 Asian financial crisis 1998 Russian financial crisis; 1998–1999 Ecuador economic crisis; 1998–2002 Argentine great depression; Samba effect (1999) (Brazil)
2000s	2001 Turkish economic crisis; dot.com crisis (2001-02); burst of US housing bubble and mortgage debacle of 2006; global financial crisis (2007-08); Euro sovereign debt crisis (2010-12); Turkish currency crisis (2018); COVID-19 pandemic and health crisis.

Source: Wikipedia, https://en.wikipedia.org/wiki/List_of_economic_crises

With no scarcity of jargon in the economic literature (i.e. great depression, financial meltdown, great recession, sovereign debt crisis, etc.), it seems very odd that the cryptocurrency market was treated as an orphan and no special term was coined for its recent massive crash, the market cap⁴ of which has taken a nosedive from almost \$3 trillion in November 2021 to barely \$900 billion in July 2022, that is a wealth evaporation of over \$2 trillion (see Figure 1). The world may choose to ignore it, but this paper will not do that by coining the term the “Great Silent Crash” (GSC) of the 21st century.



Source: CoinMarketCap, <https://coinmarketcap.com/charts/>

Figure 1: Total cryptocurrency market capitalization (\$trillion)

The Dutch tulipmania (1634-38), the Mississippi bubble (1719-20), the South Sea bubble (1720), the savings and loan (S&L) crisis (1989), the Asian crisis (1997-98), the dot.com crisis (2001-02), and the mortgage debacle (2006) have not only generated more hype and frenzy but greater media coverage than the sudden (but anticipated) collapse of the cryptocurrency market in 2022 which is now used to undergoing erratic ups and downs. Despite centuries of financial folly (Reinhart & Rogoff, 2009), instability-inflicting triggers, actions and measures taken as a remedy to restore investor confidence have remained somewhat unchanged. Akin to the Federal Reserve’s (the Fed) tight-money policy (contractionary) in the late 1920s and quantitative easing (expansive monetary policy/dollar glut) in the late 1990s and early 2000s, Roman Emperor Julius Caesar expanded the money supply via cash handouts and credit expansion; moreover, the government-funded public projects also increased its sovereign debt significantly, which forced Julius Caesar to pass a law in 49 BC to regulate usury with the hidden objective of preventing capital flight from Italy; nevertheless, fast falling interest rates

⁴ <https://coinmarketcap.com/charts/>

from 12% to 4% caused a surge in capital flight. Almost a century later, the invocation of the old law by the Emperor Tiberius in 33 AD resulted in a substantial drop in the aggregate money supply (i.e. credit crunch), which prompted early recalls of loans and ultimately a crash of real estate prices (i.e. similar to a cascade of defaults on loans during the mortgage debacle of 2006) that led to a full-blown financial crisis⁵, modern comparisons of which were made with the 2008 global financial crisis “GFC”. Another similar remedy, as the Fed used during the GFC and the COVID-10 pandemic (i.e. stimulus checks sent by President Trump), Tiberius injected liquidity in form of interest-free loans.

Nearly 2,000 years after the Roman event, still agents of the economy (economists, scholars, financial authorities, regulators, practitioners, and multilateral organizations), economic theories and models, analytical tools, banking regulation/supervision, Basel I and II standards, enhanced techniques, and risk mitigation frameworks failed to prevent disruptive systemic shocks that began in the late 1990s. A chorus of economists and financial experts believe that the Fed’s misleading (misguided) policy errors turned an ordinary-looking recession both in the late 1920s (Friedman & Schwartz, 1963) and the early 2000s into a global scale crisis. Great Recession (Friedman & Schwartz, 2008), caused by many of the same triggers that instigated the Great Depression, is a naked truth that Bernanke (and the Fed) failed to keep a promise that he made in a speech in memory of the great monetarists Milton Friedman and Anna Schwartz; as a reflection on the most unprecedented credit (contraction) crisis in the history of humankind, this paper is compelled to rephrase the last part of Bernanke’s speech;

Milton and Anna: Oops, we did it again. We're sorry, but this won't change anything. Thanks to you, but we will most likely do it again, again and again because this is what we do best.

Monetarists postulate that the Fed’s misguided and unjustified tight-money policy error (intended to curb the stock market bubble) at the end of roaring 1920s caused a substantial drop in the US money stock, and the ensuing severe contraction led to the 1929 stock market crash, followed by recession and the inevitable Great Depression (Bernanke & Gertler, 1995; Eichengreen, 2002; Bernanke, 2004; Friedman & Schwartz, 1963). Epstein and Ferguson (1984) defended the Fed’s tight-money policy action, claiming the Fed’s main objective was to curtail speculation on Wall Street in order to avoid stock market crash, but the Fed gravely failed and left the economy debilitated. The Great Depression epoch (1930s), an economic and financial tsunami that the world had never experienced before⁶, was triggered by numerous causal effects (see Bernanke, 1983; Bernanke & James, 1991; Hamilton, 1987;

⁵ See Wikipedia, https://en.wikipedia.org/wiki/Financial_crisis_of_33

⁶ Between 1929 and 1933, about 2,000 banks failed annually (about 9,000 banks); nominal gross national product (GNP) fell nearly 50% whereas the real output fell over 25%; wholesale and consumer prices plummeted, over 30% and 25% respectively; unemployment skyrocketed over 20% (fell below 10% only after WWII); and real interest rate nearly tripled (jumped from little over 4% to 11.50%). For further discussion, see Friedman and Schwartz (1963); Wheelock (1998).

Hall & Ferguson, 1998; Meltzer, 1976; Temin, 1976; Wheelock, 1998). Beside Fed policy errors, the following events contributed; the death of Benjamin Strong in 1928 (head of the New York Federal Reserve Bank)⁷, the 1929 stock market crash (Black Tuesday), *beggar-thy-neighbor*⁸ policies (Eichengreen & Irwin, 2010), the U.S. Smoot-Hawley tariff of 1930 (e.g. Irwin, 1998), the collapse of international trade (Eichengreen & Irwin, 1995), banking panic and crashes (Kindleberger, 1978, 1986), a lack of gold reserves at the Fed (Epstein & Ferguson, 1984), Great Britain's decision in 1931 to leave the gold standard due to speculative attacks on the British pound (Cooper, 1982), and finally the Glass-Steagall Act⁹ known as the Banking Act of 1933 which barred US commercial banks from engaging in investment banking activities (Crawford, 2011; Lardner, 2009; Mester, 1996). To turn the economic tide, President Roosevelt reduced the value of dollar to gold from \$20.67/oz to \$35/0z.

Table 2: High-magnitude financial crises and their common causes

Shocks (events)	Causes (triggers)
The Great Depression 1929-1939	Stock market bubble (the roaring 1920s); newly established young Fed's unwarranted tight-money (contractionary) policy to curtail speculation on Wall Street and to avoid stock market crash; a decade-long Dust Bowl that contributed to famine and homelessness; Britain's decision to leave the gold standard; and Fed's policy errors created deflation and high unemployment.
The Savings & Loan (S&L) Crisis 1986-1989	Stagflation in the 1970s (OPEC oil embargo); failure of Fed Chairman Paul Volcker's contractionary monetary policy to tame (control) high inflation; worst recession in early 1980s; risky real estate investments by banks; Reagan cut taxes and increased spending which doubled national debt.
The Asian Crisis 1997-1998	Credit and economic bubbles; hot money as the double-edged sword; short-term capital flow and reversals; fixed exchange rates; dollar-denominated loans (maturity mismatch); a series of external shocks (the devaluation of the Chinese renminbi and the Japanese yen); the Fed began to raise interest rates (capital flight to safety); and speculative real estate bubbles.
The dot.com Crisis 2001-2002	Low interest rates; a surge in tech start-ups; stock market bubble and the Internet bubble; a majority of dot.com companies did not have a realistic business plan or future revenue prospects; overcapacity issue; fast cash-burn rate; a handful of companies found success (i.e. Google, eBay, Amazon).
The Subprime Mortgage Crisis 2006-2007	Lending bubble (low-income people were encouraged to buy homes); Lax credit conditions; predatory lending; no down payment (100% of the loan was financed); financial innovation (adjustable rate mortgages – ARM); loan incentives and risky mortgages; and mortgage-backed securities (MBS).
The Global Financial Crisis 2007-2008	Fed's expansive/accommodative monetary policy; historically low interest rates (dollar glut); bursting of the housing bubble; defaults and foreclosures; subprime mortgage crisis (too many sub-prime borrowers); recalls of loans by lenders resulted in bankruptcies of insurance and mortgage companies.

Source: Author

⁷ Although the New York Federal Reserve Bank had no prior authority granted by the federal government, Benjamin Strong allowed the bank to engage in the open market bond operations to keep the money supply stable despite volatility in gold inflows and outflows. This meant bending the rules of banking rules which was not legal.

⁸ Countries passed protectionist policies to resolve their own economic problems at the expense of other nations.

⁹ As a reactionary response to drafted by Senator Carter Glass and Representative Henry Steagall

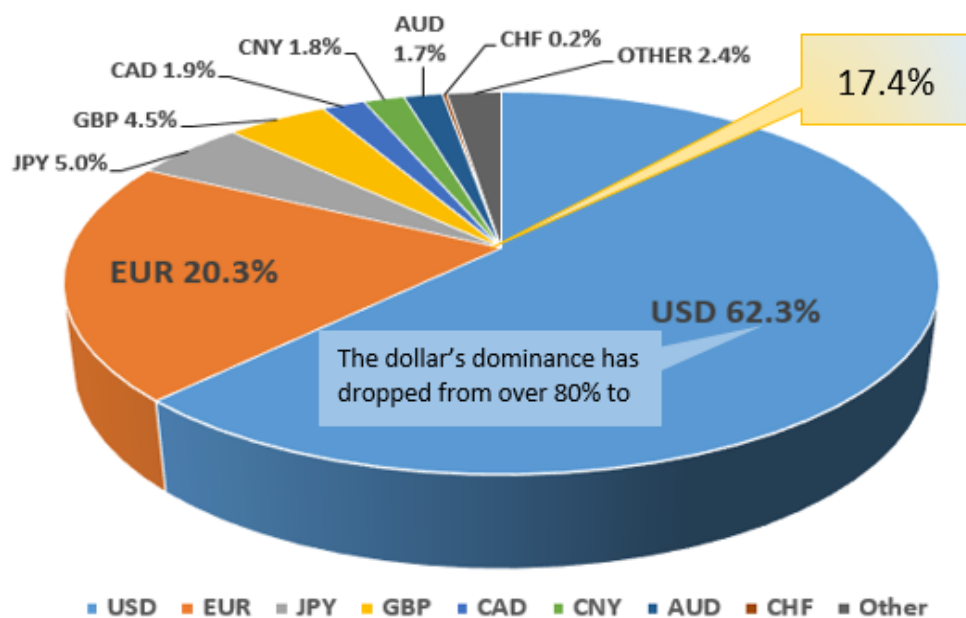
1.1 The Silent Crash

The birth of Bitcoin in January 2009 (first successful cryptocurrency), as its dramatic crash in recent months, was also silent; it was just a phenomenon among some computer engineers and enthusiasts. It is mindboggling that, to this day, the creator of a digital coin (electronic cash) whose market cap¹⁰ has passed \$1 trillion (\$1.28 trillion on 8 November 2021) is still anonymous. In the crypto-currency literature, it is commonly found that Bitcoin was invented by a mysterious creator under the alias Satoshi Nakamoto (a pseudonym), but the opaqueness surrounding the launch of Bitcoin has opened the door for an endless debate, incessant criticism, spurious claims, heated arguments, plethora of articles, and media frenzy contemplating what Bitcoin really is. Satoshi Nakamoto (2008) in his white paper entitled *Bitcoin: A Peer-to-Peer Electronic Cash System* explains it as a secure, permissionless (decentralized) network of electronic cash that runs on blockchain as its infrastructure technology using a distributed ledger and cryptography. Bitcoin is a mineable cryptocurrency, divisible by eight decimals and the smallest denomination is called a Satoshi (i.e. 0.00000001 BTC). Unlike gold (the densest, conductor, malleable and ductile metal), Bitcoin has no intrinsic value, no physical form (only exists digitally), and is stateless (i.e. no de facto Fed-like central authority to set its value).

The dollar's century-long hegemony ('exorbitant privilege') underpinned by American imperialism has prompted a search for alternative currencies (i.e. Bitcoin). Many historians argue that imperialism is in America's DNA; in fact, the evolution of American imperialism and the dollar's hegemony are tightly intertwined. The seeds of an imperial nation were sowed by the European settlers arriving in the new world after 1492, thrived with blood via the use of Monroe Doctrine with "manifest destiny" which made the US territorial expansion throughout the Americas justified and inevitable (Dunning, 2001; McDonough, 2011). The Revolutionary War (1775-1783) was a defining moment in US history, igniting the spark of American imperialism; but the U.S. was poles apart from its humble beginnings, the U.S. expansionism via intervention and annexation became a tradition with winning the Spanish-American War of 1898, the treaty of Paris (December 10, 1898) ended the Spanish colonial rule in Americas and resulted in the U.S. acquisition of new territories in western Pacific and Latin America such as Puerto Rico, Philippines, and Guam (e.g. Corbin & Levitsky, 2003). The emergence of the U.S. as an imperial power made her crave more, the next was winning insular possessions in the Caribbean and Hawaii and playing a leading role in the restructuring of power in Europe (i.e. power was shared by Germany, France, and the UK which were also each other's chief enemy). In the post-WWII, while Europe has moved towards "perpetual peace", the U.S. on the other hand has turned into an anarchic Hobbesian backed by its military might, weaponized dollar, and the abuse of sanction power.

¹⁰ <https://coinmarketcap.com/currencies/bitcoin/historical-data/>

The Imperial Japanese Navy's attack on Pearl Harbor on 7 December 1941 ended the isolationist (i.e. non-intervention) sentiment in the U.S., which immediately reciprocated by declaring war upon Japan on 8 December 1941 (see Schaller, 1976; Steiner, 2011). This surprise military strike by the Imperial Japanese Navy gave the United States a rare opportunity to extend its influence beyond Americas and to re-structure the existing power-sharing in Europe. Leading up to the WWII (1939-45), the US dollar and the British pound were the only two reserve currencies which quickly changed before conclusion of the WWII, dollar's ascendancy to the throne¹¹ as a main reserve currency was inaugurated at the 1944 Bretton Woods Conference (see Bordo, 1993; Garber, 1993). Since the collapse of the Bretton Woods system of fixed exchange rates and President Nixon's decision to cut the dollar's link to gold in 1971, there have been countless efforts to kill off, deflate, or end the dollar hegemony (Figure 2). The Euro (introduced on 1 January 1999), Bitcoin (launched in January 2009), and Chinese renminbi aka yuan (became a reserve currency and was added to the IMF's SDR¹² basket in end-2016) were the latest attempts that fell short of expectations; besides, China and Japan happen to be the two largest foreign investors – addicted to dollars to keep the value of their national currencies low.



Source: IMF; <http://data.imf.org/?sk=E6A5F467-C14B-4AA8-9F6D-5A09EC4E62A4>

Figure 2: Composition of global foreign exchange reserves

¹¹ By the late 19th century, the U.S. succeeded Britain as the dominant economic power, and by the onset of the 20th century, there were only dollar and pound as reserve currencies; the former singly ascended to the throne when the influential British economist Maynard Keynes ridded the UK of the reserve currency status at the Bretton Woods (Eichengreen & Flandreau, 2009) and its imperial grandeur (see Aizenman & Lee, 2005; Eichengreen, 2009; Ikenberry, 2001).

¹² At the IMF, General Arrangements to Borrow (GAB) and the Special Drawing Rights (SDR) were established to provide the needed global liquidity (also central banks were allowed to make currency swaps as necessary). For official gold sales, a mechanism called Gold Pool was established (see Bordo et al., 2017), but it was terminated in 1968 due to a speculative run on gold. The SDR's role of international currency did not materialize as planned. Most financial crises in systemic nature, including the Asian crisis of 1997-98 and the GFC of 2007-08, were due to insufficient liquidity.

Macro events set the stage for the emergence of Bitcoin; as such, 9/11 terrorist attack (2001) carried out by the militant Islamic extremist al-Qaeda, the U.S. war in Afghanistan (2001-21), dot.com crisis (2001-02), mortgage debacle and subprime crisis (2006-07), and global financial crisis (2007-08). Schwartz (2009) argues that the lax credit conditions and the resultant formation of the credit balloon in the late 1990s and early 2000s was a spinoff of the Fed's prolonged expansive (accommodative) monetary policy error with the objective of a short-term gain but ignoring the unthinkable cost of the long-term agony as observed before, during and in the aftermath of the GFC of 2008. Bitcoin's colossal success was not instantaneous; in fact, Bitcoin was still an arcane topic a few years after its launch. For example, a survey by Bloomberg (2013) revealed that less than half of those polled knew what Bitcoin was and 6% of the survey takers thought it was actually a video game.¹³ In little over a decade, the price of Bitcoin has gone from practically \$0.00 at its launch (Jan. 2009) to \$67,566¹⁴ on 8 Nov. 2021, but of course not without extreme volatility. Bitcoin's silent crash between November 2021 and July 2022 was the worst financial mayhem after the GFC of 2008, i.e. more than two-thirds of Bitcoin's value (close to \$1 trillion) vanished (Table 3 shows a list of price corrections).

Table 3: Historical crashes of Bitcoin (2011-2022)

No	Correction start date	Correction end date	# Days in correction	Bitcoin high price \$	Bitcoin low price \$	Decline %	Decline \$
1	8 Jun 2011	15 Dec 2011	185	31.00	2.00	-94%	29.00
2	12 Jan 2012	27 Jan 2012	16	7.38	3.80	-49%	3.58
3	17 Aug 2012	19 Aug 2012	3	16.41	7.10	-57%	9.31
4	6 Mar 2013	7 Mar 2013	2	49.17	33.00	-33%	16.17
5	21 Mar 2013	23 Mar 2013	3	76.91	50.09	-35%	26.82
6	11 Apr 2013	12 Jun 2013	60	266.34	70.00	-74%	196.34
7	19 Nov 2013	19 Dec 2013	30	1,242.00	600.00	-52%	642.00
8	5 Jan 2014	11 Apr 2014	95	1,000.00	440.00	-56%	560.00
9	16 Sep 2014	29 Mar 2015	200	465.86	252.74	-46%	213.12
10	30 Nov 2013	14 Jan 2015	411	1,163.00	152.40	-87%	1,010.60
11	10 Mar 2017	25 Mar 2017	16	1,350.00	891.33	-34%	458.67
12	25 May 2017	27 May 2017	3	2,760.10	1,850.00	-33%	910.10
13	12 Jun 2017	16 Jul 2017	35	2,980.00	1,830.00	-39%	1,150.00
14	2 Sep 2017	15 Sep 2017	14	4,979.90	2,972.01	-40%	2,007.89
15	8 Nov 2017	12 Nov 2017	5	7,888.00	5,555.55	-30%	2,332.45
16	17 Dec 2017	22 Dec 2017	5	19,783.06	13,800.00	-31%	5,983.06
17	22 Dec 2017	5 Feb 2018	14	13,800.00	6,200.50	-55%	7,599.50
18	5 Sep 2018	16 Dec 2018	100	7,361.46	3,230.15	-56%	4,125.19
19	17 Dec 2017	16 Dec 2018	365	19,783.06	3,236.27	-84%	16,547.79
20	27 Jun 2019	15 Dec 2019	195	13,017.12	7,116.28	-45%	5,900.84
21	8 Nov 2021	13 Jul 2022	231	67,566.83	18,999.95	-72%	48,566.88

Source of data: <https://coinmarketcap.com/currencies/bitcoin/historical-data/>

¹³ <https://www.edelmanfinancialengines.com/education-center/articles/will-bitcoin-replace-the-dollar>

¹⁴ <https://coinmarketcap.com/currencies/bitcoin/historical-data/>

Although Bitcoin has no intrinsic value as gold, its value (price) is largely driven by trust, acceptance, and a degree of speculation; therefore, Bitcoin price can be anything in its eco system as long as people are willing to pay for it. Even when the price of each Bitcoin was practically \$0.00 at its launch in 2009 (i.e. the price hit \$0.06 on 26 July 2010, see Figure 3), people still accepted the digital coin for its future prospects. When the price of Bitcoin topped \$1 in February 2011, more people accepted it; as Bitcoin hit \$100 (April 2013) and then \$1,000 (February 2017), people throughout the world became Bitcoin frenzy, but when the price of Bitcoin for the first time passed \$10,000 and \$20,000 levels, the frenzy like the Dutch tulipmania turned into Bitcoin mania. Even ordinary people became avid buyers with the hope that Bitcoin price would continue its upward trajectory. Based on this logic, the eventual price of \$100,000 will give Bitcoin a market cap of \$2.1 trillion, and a price of \$1,000,000 will increase its market cap by tenfold to \$21 trillion, almost the size of US economy). Or this may be, as suggested by some, a heterogeneous dream since ongoing price volatility plus imminent regulation threat by the U.S. government will put strain on Bitcoin's true potential to set new price records.



Source of data <https://www.buybitcoinworldwide.com/price/>

Figure 3: Bitcoin price history (2010-2022)

With the price valuation of Bitcoin (i.e. over 1.12 million-fold), \$1,000.00 invested in Bitcoin in 2010 at the price of \$0.06 would have grown to equal \$1.13 billion. During the silent crash, the total market cap of the cryptocurrency market (i.e. Bitcoin and 20,800 altcoins) has plunged from \$2.8 trillion on 7 November 2021 to \$901 billion on 19 June 2022, which translates to an astounding loss of about \$2 trillion by the top 10 cryptocurrencies, i.e. the market cap of Bitcoin alone shrank nearly \$1 trillion (see Table 4). Despite extreme volatility, wishful folks still believe Bitcoin price can hit \$100,000.

Table 4: Bursting of the second cryptocurrency bubble (ranked by market cap)

Rank	Symbol	Code	Name	November 13, 2021		June 30, 2022		
				Price (\$)	M. Cap (\$B)	Price (\$)	M. Cap (\$B)	Price Fall
1.		BTC	Bitcoin	67,566	1,270.3	19,000	368.9	-71.88%
2.		ETH	Ethereum	4,812	569.1	1,038	126.2	-78.43%
3.		BNB	Binance	654	109.1	197	32.2	-70.88%
4.		ADA	Cardano	2.96	92.3	0.42	14.7	-85.81%
5.		DOGE	Dogecoin	0.68	88.7	0.06	8.0	-91.18%
6.		USDT	Tether	1.00	83.2	1.00	65.8	-20.91%
7.		XRP	XRP	1.79	83.0	0.32	15.2	-82.12%
8.		DOT	Polkadot	53.4	53.2	6.29	6.2	-88.22%
9.		BCH	Bitcoin Cash	1,506	28.5	103	2.1	-93.16%
10.		LTC	Litecoin	346	17.9	52.9	3.1	-70.23%
Market cap of top 10 coins					2,395.3		642.4	-73.18%

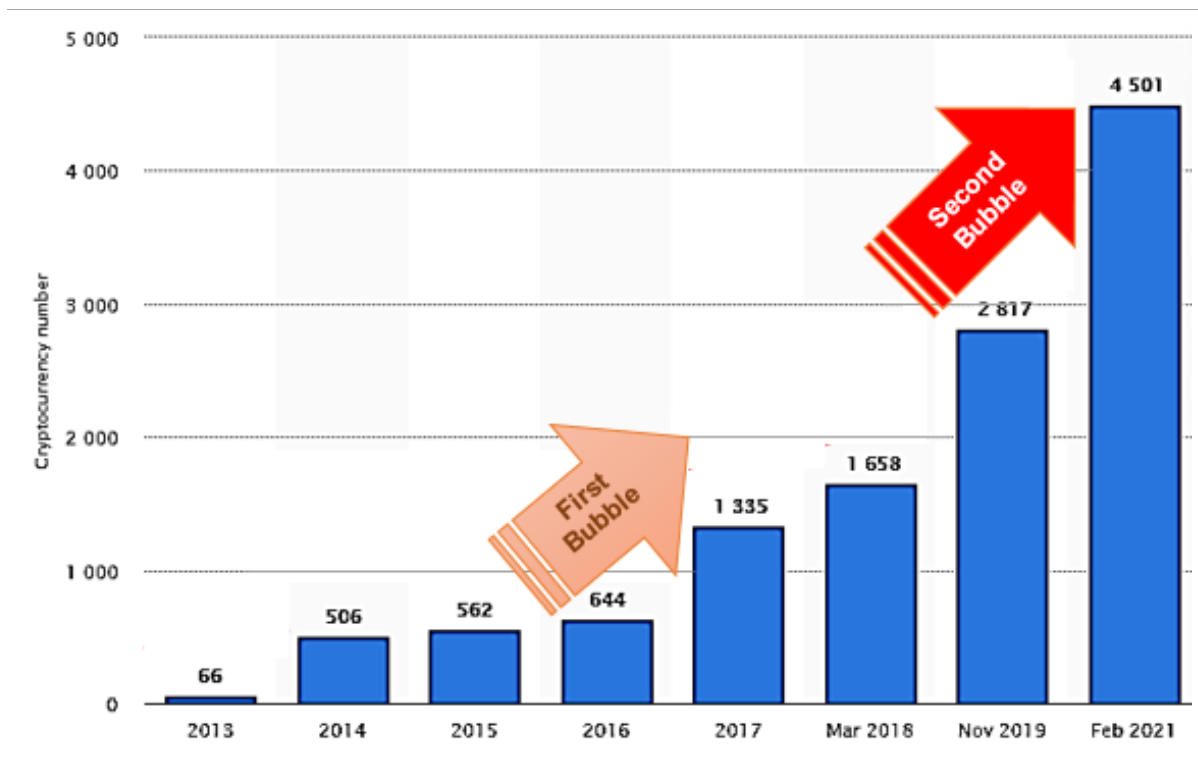
Source: CoinMarketCap; <https://coinmarketcap.com/currencies/binance-coin/historical-data/>

Notes: \$B: Billion; ADA peaked on 31 August 2021, lowest on 12 July 2022; DOGE peaked on 6 May 2021 and lowest on 18 June 2022; USDT peaked on 8 May 2021 and lowest on 26 July 2022, since the price of Tether (USDT) hardly changes, instead the change in market cap is calculated.; XRP peaked on 13 April 2021, and lowest on 30 June 2022; LTC peaked on 6 May 2021, and lowest on 19 June 2022;

The market capitalization of Bitcoin and 20,800 altcoins reached all time high of \$2,834 trillion on 13 November 2021, just months before its dramatic crash on 30 June 2022; the cryptocurrency market silently crashed, shedding nearly \$2 trillion. As illustrated in Table 4, the top digital coins owned more than two-thirds, to be exact, 84.52% or \$2.4 trillion (i.e. the half of which, \$1.27 trillion belonged to Bitcoin alone). The Asian crisis of 1997-98 generated more hype even though American investors lost only \$30 billion (circa 5% of \$700 billion lost by global investors); US President Clinton described the event as a “major glitch” (Nanto, 1998), some even categorized the crisis as an “Asian Flu”. This time is different; conversely, former President Trump has said that he is not “a fan of Bitcoin and other cryptocurrencies”; moreover, he strongly feels agitated and argues that Bitcoin and Libra create an undesired situation of diminishing US (the Fed) power. Due to US regulators and law makers running headlong into backlash to Facebook’s plan to launch a stable coin, first dubbed Libra in 2019 and a year later was rebranded as Diem, the three-year odyssey of launching a digital currency by a private company has been derailed on account of stoked fears by the Fed and the ECB.

2.0 Literature Review

There have been at least 50 price corrections (i.e. decline by 20% or more) in cryptocurrency market, but only two severe crashes; in the first boom-and-bust cycle (2016-17), initial coin offerings (ICOs) had soared (Figure 4 & 5) to take advantage of a bull digital coin market. In the first speculative bubble period (last three months before the inevitable crash in early January 2018), the price of Bitcoin had surged from \$3,733 on 15 September 2017 to an intraday high of \$20,089 on 17 December 2017 (i.e. \$3,851 on January 5, 2018). In the second boom-and-bust cycle (2019-21), a spike in ICOs looked and felt like a déjà vu, pointing to the writings on the wall. The speculative rampant, madness of crowds, irrational exuberance, and herd behavior once plagued the first bubble and its subsequent burst found a new life in the second bubble led by Bitcoin (alpha coin) and its progenies (altcoins).



Source: Statista; <https://www.statista.com/statistics/863917/number-crypto-coins-tokens/>

Figure 4: Number of cryptocurrencies worldwide from 2013 to 2021

As illustrated in Figure 4, the number of ICOs more than doubled during both bubble periods, 2016-17 and 2020-21 respectively), which was in many aspects similar to the dot.com era (Internet bubble) where stock prices of some dot-com companies rose more than 1,000% (Cassidy, 2009). The second cryptocurrency bubble was more intense than the first (Figure 5), extremely high price valuations (i.e. the market cap peaked at circa \$2.9 trillion in November 2021 before shedding \$2 trillion by June 2022). This silent crash again proved Newton's law of universal gravitation ("what goes up must come

down”) that prices of cryptocurrencies (or assets) can’t continue their upward trajectory forever. This notion in mind, jittery investors decided to engage in profit-taking before it was too late, which led to a heavy sell-off and a crash as profit-taking turned into panic that spiraled through the broader market (for bubbles and crashes, see Abreu & Brunnermeier, 2003; Kindleberger, 1978).

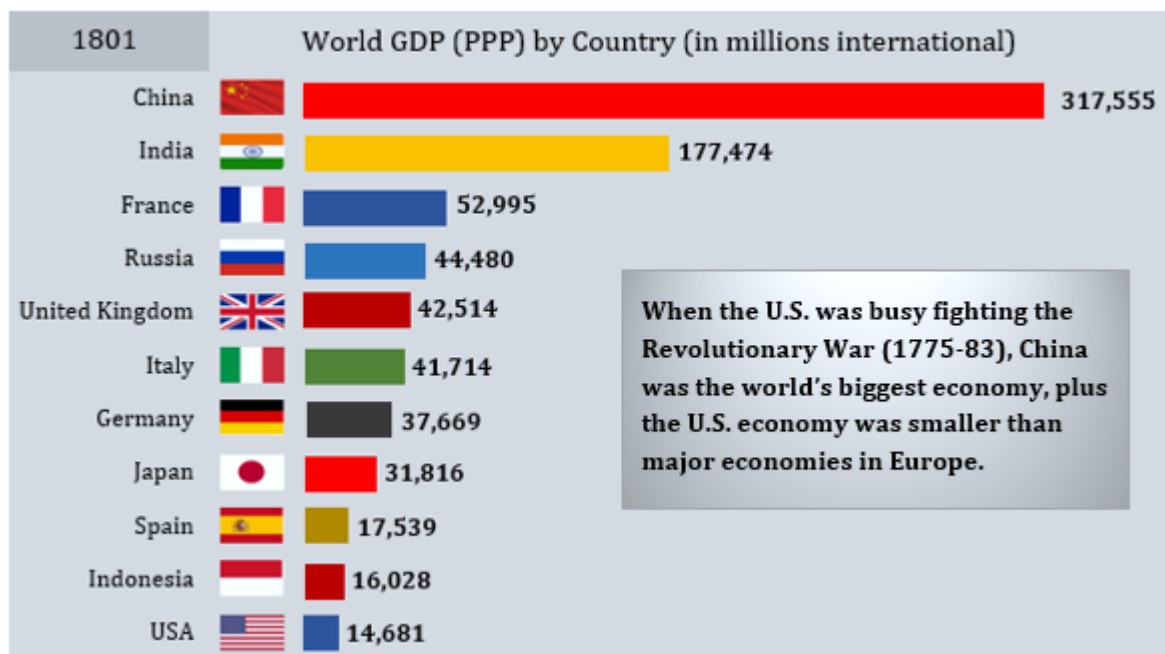


Source of data: Statista; <https://www.statista.com/statistics/377382/bitcoin-market-capitalization/>

Figure 5: Number of cryptocurrencies worldwide from 2013 to 2021

Throughout history, the evolution of money points to an alternated shift between commodity money and credit money (King, 1999; Goodhart, 2000). Since the late 1990s, contemporaneous crises have collectively cost the world’s economies over \$30 trillion; after each crisis followed by a public outcry, the same old debate has kept reemerging that is a new monetary order with a bigger role for gold. In addition to that, Bitcoin advocates and investors strongly infer that Bitcoin in fact is “Gold 2.0”. If both regulatory and political backlash to Bitcoin is alleviated or stopped altogether, Bitcoin can be, as its creator Satoshi Nakamoto originally stated, “electronic cash” version of the gold standard. Unlike the dollar centric monetary system (i.e. politicized, weaponized, and overly hegemonic), a Bitcoin-based monetary system (digital gold standard) should not be compared with the gold standard of the late 19th and early 20th centuries because the antecedent was very brief and had serious problems (i.e. scalability and cost). There have been compelling arguments that Bitcoin is ready and capable of assuming the role as a global reserve currency that the Gold 1.0 failed to do so, only if governments, central banks, law makers, and regulators stop raging a war on Bitcoin and altcoins.

Many economists along with historians argue that Bitcoin has emerged due to instability-inflicting policies created by the centralized systems using state-controlled fiat currencies (i.e. intrinsically useless paper), values of which fluctuate based on a confluence of domestic and macroeconomic factors (see Bernanke, 2013; Bordo, 2007; Davies, 2000, 2002; Kasper, 2017). In stark contrast to the U.S. accepting Bitcoin in 2013¹⁵ and classifying it as a commodity in 2015, the US Fed, regulators and law makers have waged a war on non-central bank issued digital coins; furthermore, Trump slammed Bitcoin by tweeting “I am not a fan of Bitcoin and other cryptocurrencies, which are not money, and whose value is highly volatile and based on thin air”. To remind people of the American primacy and dollar’s monetary hegemony (i.e. China was the biggest economy in 1800), Trump tweeted, “We have only one real currency in the USA, and it is stronger than ever, both dependable and reliable. It is by far the most dominant currency anywhere in the world, and it will always stay that way. It is called the United States Dollar!”¹⁶ Besides Bitcoin, Trump also criticized Facebook’s Libra coin by tweeting that Libra will have “little standing or dependability”.¹⁷ The US government (the Fed) may have won the war against Libra (rebranded as Diem in 2020), but Bitcoin is still going very strong.



Source of data: <https://www.youtube.com/watch?v=4-2nqd6-ZXg> (modified by author)

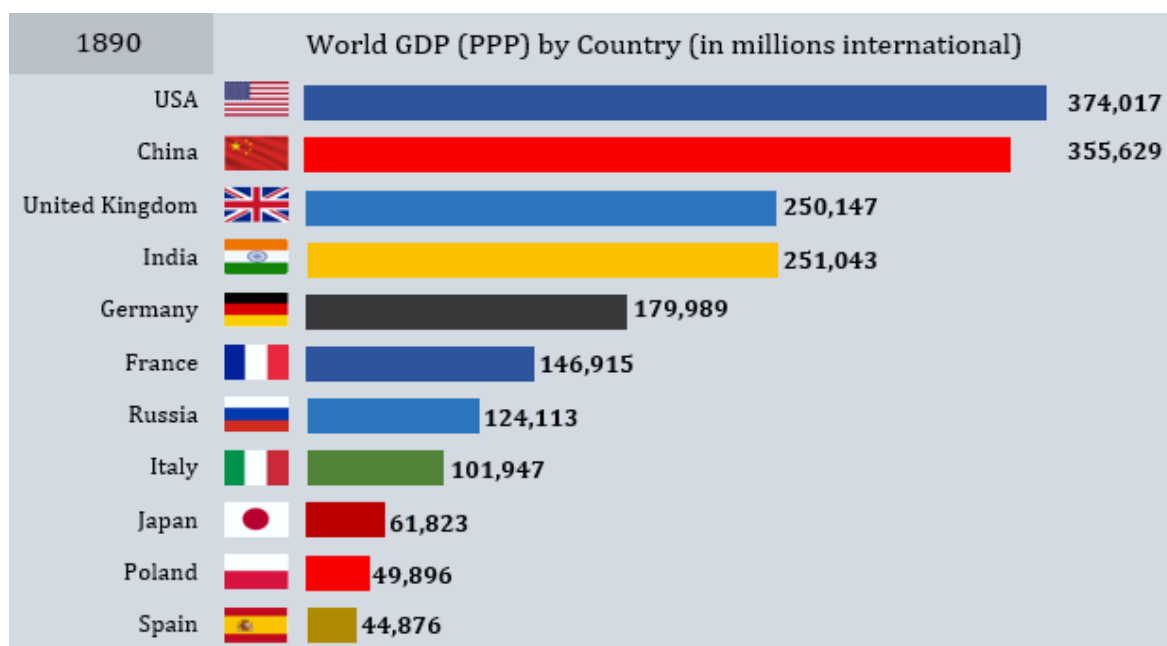
Figure 6: World GDP (PPP) by country in 1800

¹⁵ It was concluded at the hearings held by the U.S. Senate Committee on Homeland Security and Governmental Affairs that “Bitcoin had the potential to play useful roles in the commercial payment system.” For the actual hearing document, see U.S. Senate Committee on Homeland Security and Governmental Affairs, *Beyond Silk Road: Potential Risks, Threats, and Promises of Virtual Currencies*, Hearings held November 18-19, 2013, available at hsgac.senate.gov/hearings/beyond-silk-road-potential-risks-threats-and-promises-of-virtual-currencies.

¹⁶ He tweeted (3:15 AM – Jul 12, 2019); <https://www.foxbusiness.com/markets/bitcoin-price-tumbles-trumps-criticism>

¹⁷ <http://thenextweb.com/hardfork/2019/07/12/donald-trumps-critique-of-bitcoin-is-antiquated-and-irrelevant/>

Since the onset of the 18th century, the world's biggest economy status has rotated between the U.S. and China for every 100 years or so (Figure 7), but the concentration of power in the hands of the U.S. throughout the 20th century has been rather disruptive for world peace and global economy which along with the world's scarce resources are controlled by the core countries (imperial powers before the 16th century, succeeded by European powers up to the 20th century and later by the industrial capitalists) that enjoy the benefits, consisting the United States, Canada, most of Western Europe, Japan, Australia, Russia and China. Conversely, periphery and semi-periphery states (larger part of the world) for their shares of the resources are at the mercy of the core countries (Abernethy, 2000; Cell, 1997). Two Great Wars (WWI and WWII) culminated Europe's hegemony, which was decimated by the end of WWI and the emergence of United States as a major power. The defeat of Nazi Germany in WWII caused another power shuffle in Europe, this time Britain was sidelined (i.e. John Maynard Keynes, the influential English economist and leader of the British delegation at the 1944 Bretton Woods Conference, ridded the UK of its imperial grandeur and the reserve currency) and the United States along with the Soviet Union (USSR) became two superpowers, which also meant a step towards a bipolar world order, i.e. the Western bloc against the Eastern bloc during the Cold War era starting in 1947 till the USSR's collapse in 1991 (see Marples, 2004; Shannon, 1996).

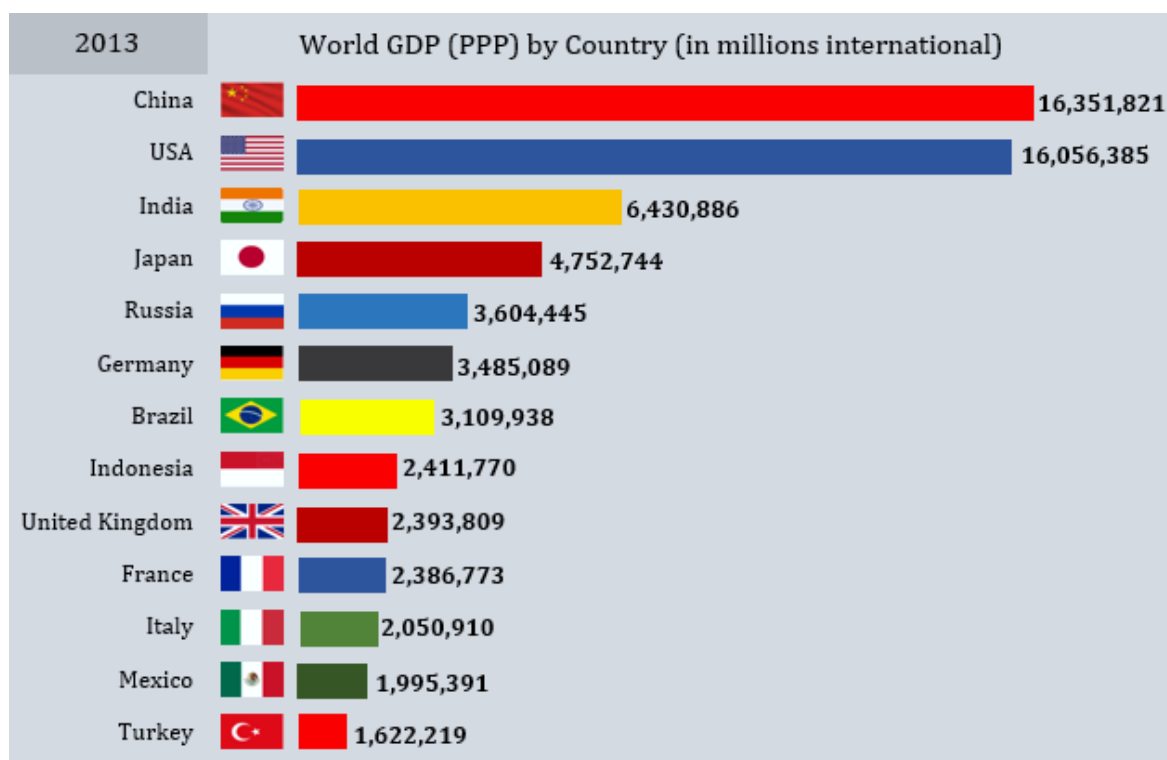


Source of data: <https://www.youtube.com/watch?v=4-2nqd6-ZXg> (modified by author)

Figure 7: World GDP (PPP) by country in 1890

Since the 2008 global financial crisis ("Great Recession"), the worst crisis since the Great Depression, the U.S. government's abuse of sanction power as a foreign policy and its increasing weaponization of dollar (the Trump administration in particular) have forced countries like China, Russia, Venezuela,

Iran, and Turkey to become overly cautious about participating in dollar regimes; for instance, Iran began to use gold and Bitcoin to bypass heavy US embargo/sanctions and the ongoing disruptive U.S. – China trade war prompted China to launch its own digital yuan. After China once again passed the U.S. as the biggest economy in the world measured by GDP/PPP (see Figure 8), United States has been keen on using the weaponized dollar (the weapon of mass economic destruction) against any nation (even WWII allies) that is not acting in the best interest of “America First” policy. To preserve the dollar’s “exorbitant privilege” and not to lose its prolonged economic and geopolitical dominance to China, former President Trump has been more aggressive than his predecessors in pushing sanctions as a main foreign policy (i.e. he has imposed a total of 1,474 sanctions on entities).



Source of data: <https://www.youtube.com/watch?v=4-2nqd6-ZXg> (modified by author)

Figure 8: World GDP (PPP) by country in 2013

Cash is no longer “king”¹⁸; the evolution of money has taken several forms since 250 B.C.¹⁹, i.e. metallic money (coins), paper money (bank notes), plastic money (credit, debit, and department store cards), electronic/digital money (e-commerce), and crypto money (i.e. Bitcoin). Even though central banks (alchemist of modern banking) only became part of money’s evolution in the 17th century, the U.S.

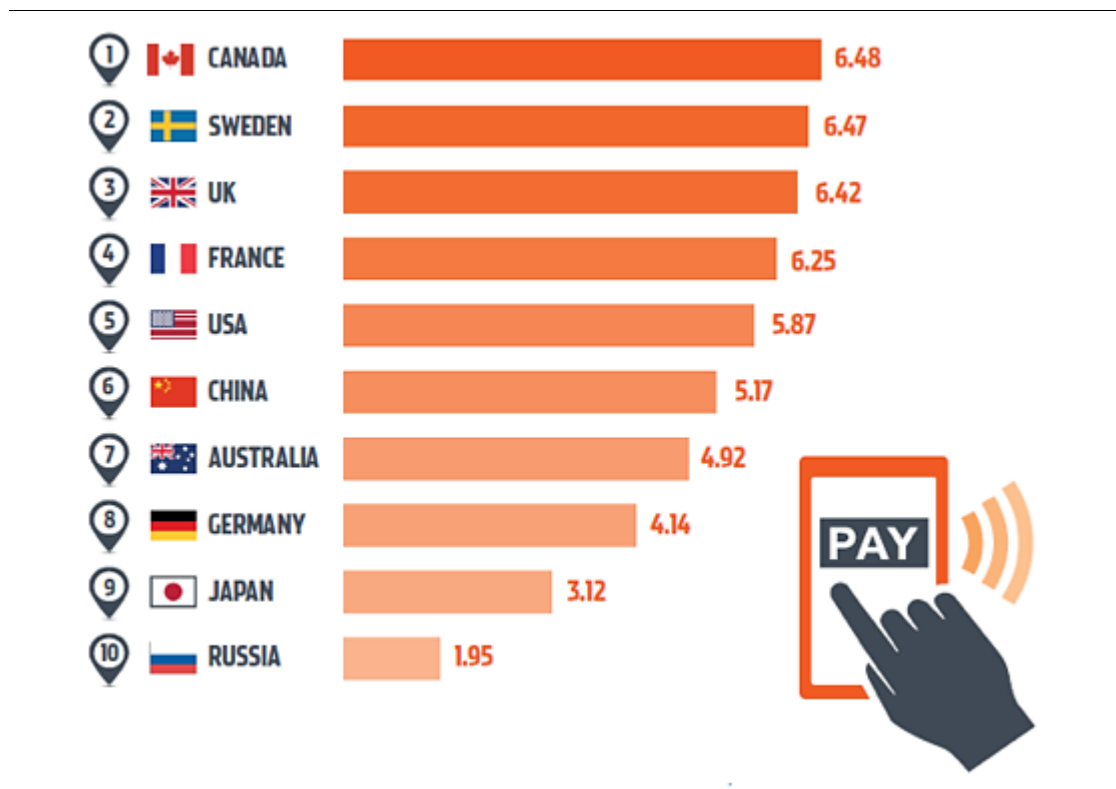
¹⁸ Cash, once the most dominant medium of exchange, accounts less than 10% of all economic transactions in Western European economies (Ferguson, 2012). In America, cash transactions still account 30% of all transactions.

¹⁹ Chief legal advisor Julius Paulus Prudentissimus to the Roman emperor Severus Alexander described then what we know today as the basic functions of money, i.e. unit of account, store value, and medium of exchange (see Bordo & Levin, 2017).

Federal Reserve System (the Fed) belonged to a later wave of central banks established at the turn of the 20th century (the Fed was created on December 23, 1913). Similarly, the U.S. also belongs to a later wave of central bank digital currency (CBDC) adoption in the 21st century.

2.1 The Birth, Growth & Identity Crisis

The old famous phrase “*cash is king*” looks and feels like a distant memory. Advanced economies are moving fast towards a cashless society despite concerns over data breaches, hackers, identity theft, or stolen sensitive financial data, i.e. only 2% of retail transactions in Sweden in 2020 consisted cash (the U.S. is at least a decade behind in this transition, see Figure 9). Although COVID-19 pandemic had farfetched implications in every facet of life, nevertheless the Great Health Crisis (GHC) of the 21st century led to a reduction in cash use and a surge in cashless payment methods (i.e. including U.S., the amount of physical cash in circulation worldwide declined noticeably in 2022).

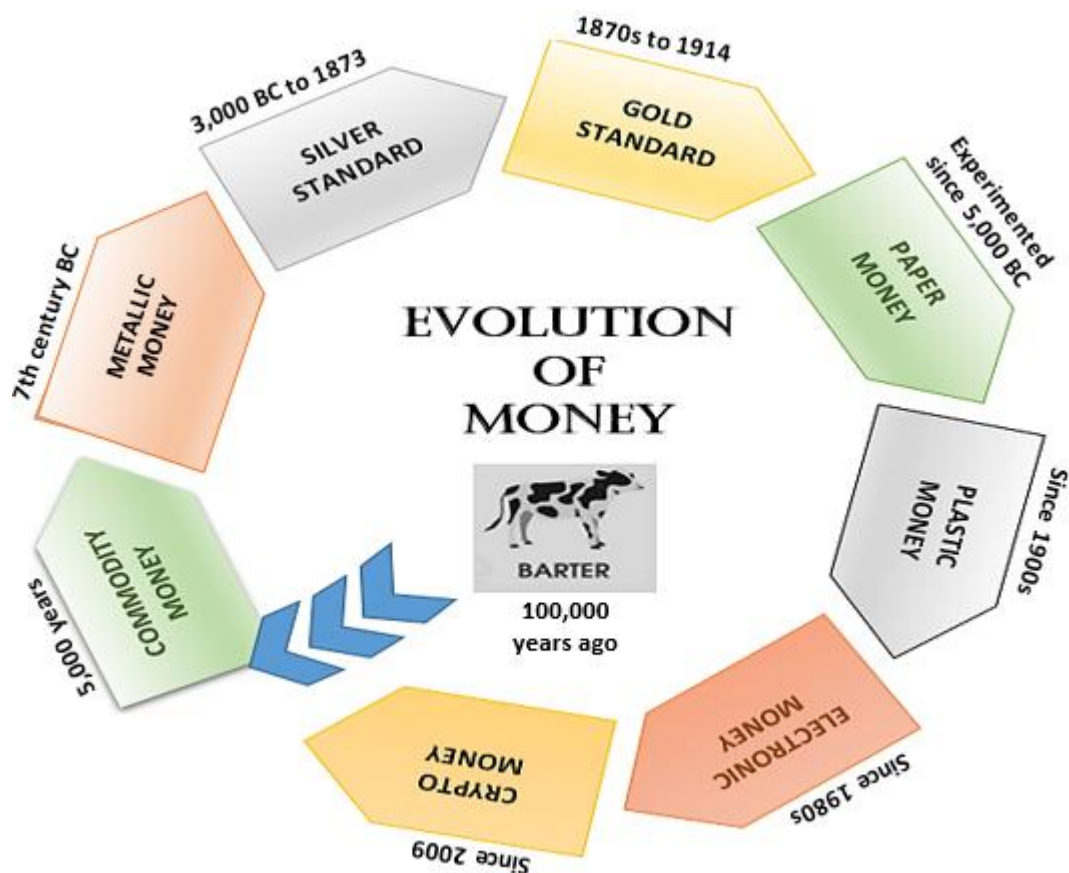


Source: <https://www.fisglobal.com/en-gb/insights/merchant-solutions-worldpay/article/cashless-society-learning-from-global-markets>

Figure 9: The world's most cashless societies

The amount of physical (printed) money (aka M0) in circulation worldwide was roughly \$36 trillion in 2019; as shown in Figure 10, M2 (M0 + M1 plus savings deposits and money market shares) was around \$75 trillion (\$90 trillion in 2022). After peaking at \$6.41 trillion in December 2021, physical cash (M0) in circulation in the U.S. declined over \$600 billion to \$5.5 trillion in June 2022.

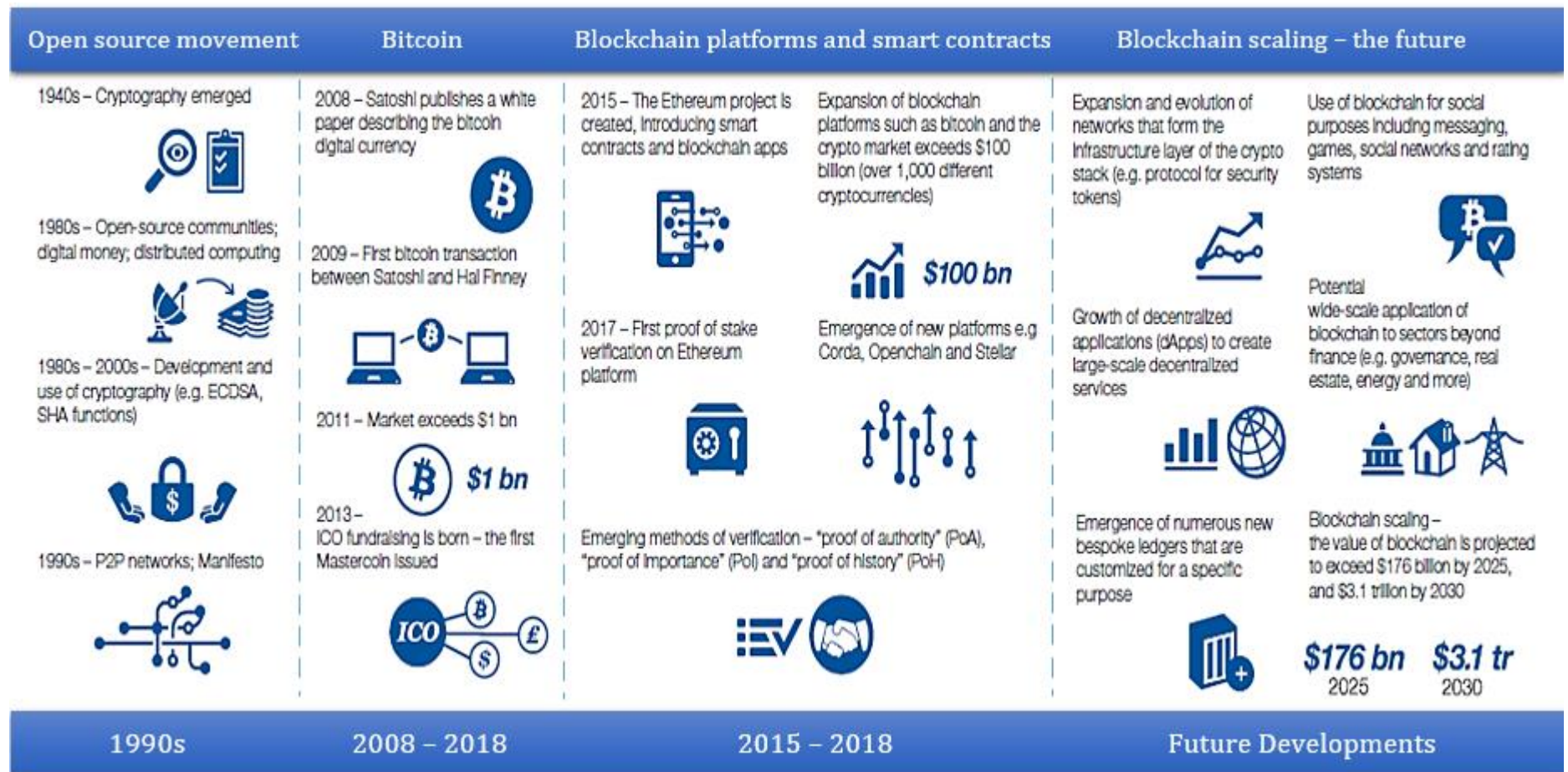
Not just money's evolution (Figure 11), but the fast-paced evolution of web-enabled technologies since the 1990s have played a pivotal role in research, development and emergence of Bitcoin (Figure 12); as such, first modern computer (Z1) was created by Konrad Zuse in the 1930s; in 1950s, computers were only used for scientific/engineering research; personal desktop computers (Programma 101) became available in the 1960s (the Internet also started in this decade but its birthday was in January 1983); in 1969, the Advanced Research Projects Agency (ARPA) of the U.S. Department of Defense awarded contracts for the development of the ARPANET project; packet-switched networks emerged in the 1970s and the advent of microprocessors reduced prices of computers significantly; Macintosh was introduced by Apple in the 1980s (Xerox Alto was its predecessor which had the capability of sending/receiving documents and emails); the adoption of DNS (Domain Name System), IP (Internet Protocol), and TCP (Transmission Control Protocol) internationally contributed to the emergence of the World Wide Web (www) and commercial internet service providers (ISPs) in the late 1980s (i.e. AOL); the release of Windows 95 by Microsoft in 1995 and the iMac G3 by Apple in 1998 along with web-enabled technologies cultivated the internet (web 2.0) or digital revolution.²⁰



Source: Author

Figure 11: Evolution of money

²⁰ For a longer discussion and a historical perspective, see Baran (1964), Kahn (1972), Cerf and Kahn (1974), Kleinrock (1961, 1964, 1976), Licklider and Clark (1962), Roberts (1967), and Roberts and Merrill (1966).



Source: PwC (PricewaterhouseCoopers)

Figure 12: Timeline of blockchain developments

1980s – Cryptography emerged; 1980s – Open-source communities, digital money, distributed computing; 1980s – 2000s – Development and use of cryptography (e.g. ECDSA, SHA functions); 1990s – P2P networks, Manifesto; 2008 – Satoshi publishes a white paper describing the bitcoin digital currency; 2009 – First bitcoin transaction between Satoshi and Hal Finney; 2011 – Bitcoin market cap exceeds \$1 bn; 2013 – ICO fundraising is born – the first Mastercoin issued; 2015 – The Ethereum project is created, introducing smart contracts and blockchain apps (Expansion of blockchain platforms such as bitcoin and the crypto market exceeds \$100 billion with over 1,000 different cryptocurrencies); 2017 – First proof of stake verification on Ethereum platform (Emergence of new platforms e.g. Corda, Openchain and Stellar) and emerging methods of verification – “proof of authority” (PoA), “proof of importance” (PoI) and “proof of history” (PoH).

Blockchain scaling – the future: Expansion and evolution of networks that form the infrastructure layer of the crypto stack (e.g. protocol for security tokens); Use of blockchain for social purposes including messaging, games, social networks and rating systems; Growth of decentralized applications (dApps) to create large-scale decentralized services; Blockchain scaling – the value of blockchain is projected to exceed \$176 billion by 2025, and \$3.1 trillion by 2030.

2.1.1 The Birth with Historic Significance

Although the recorded historical data shows that a systemic banking crisis is infrequent, occurring every 20-25 years with the probability of circa 5% (BIS, 2010); during the period of 1985-2009, G-10 countries²¹ faced a crisis with the probability of 5.2% (Reinhart & Rogoff, 2009) and 4.1% (Laeven & Valencia, 2008). Eight centuries of financial folly (Reinhart & Rogoff, 2009) evidently attest that most financial markets worldwide have undergone repeated economic and financial crises, attributable to a confluence of driving and contributing factors such as policy errors by central banks (for the Fed's untimely expansive and contractionary monetary policies, see Friedman & Schwartz, 1963; Bernanke, 2013; Eichengreen, 2002); highly leveraged financial systems and sovereign states (Blundell-Wignall et al., 2014; BCBS, 2004; Saurina & Trucharte, 2007); too-big-to-fail banks (i.e. moral dilemma); financial openness - interconnected banking sectors (contagion risk); excessive indebtedness both by private and household sectors; lax credit environment and predatory lending during benign economic times; procyclical deleveraging under menacing market conditions (credit crunch); heavy reliance on rating by the ECAs²² played a crisis-intensifier role prior to and during the GFC (BCBS, 2004).

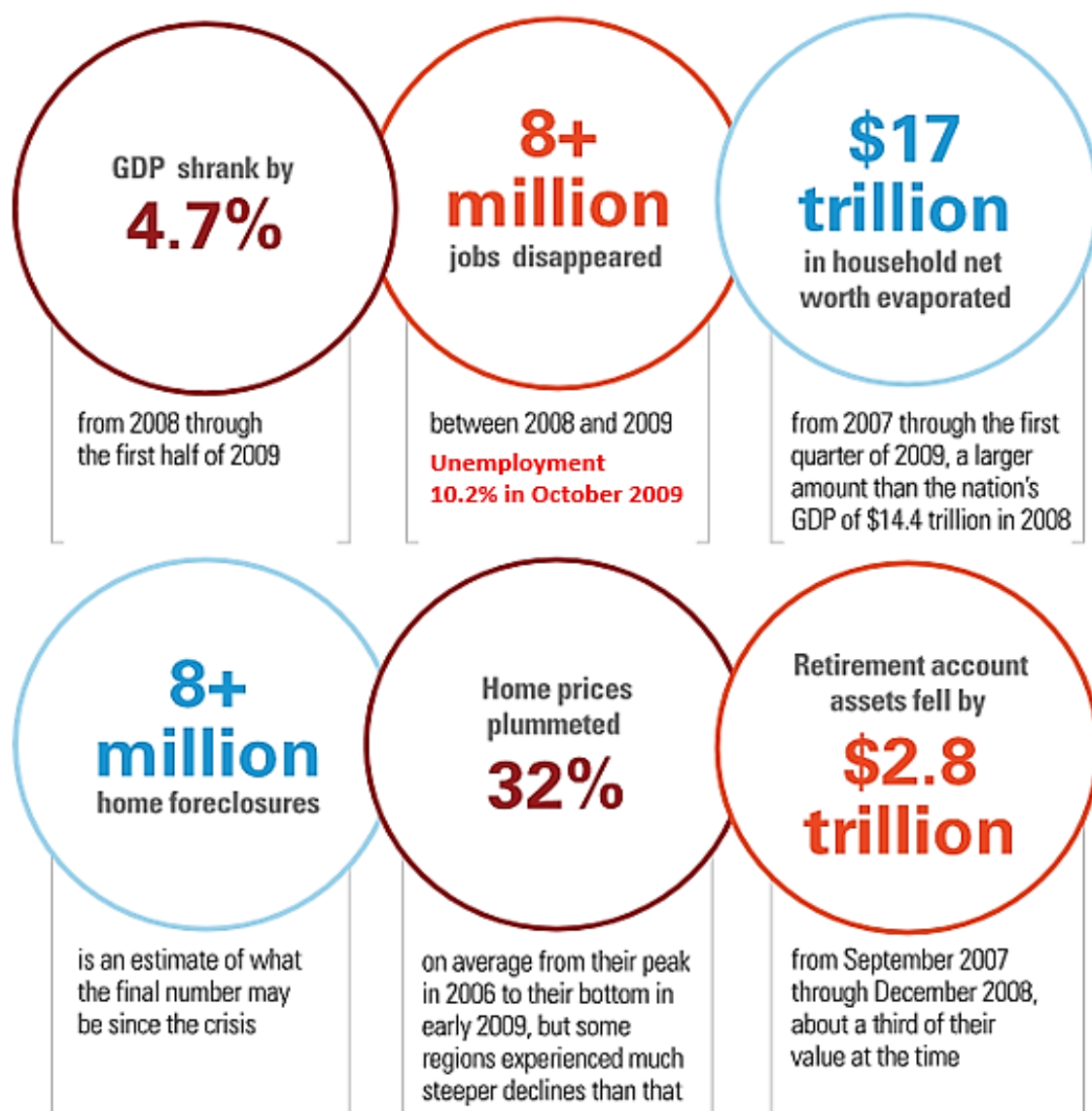
Out of several hundred economic, financial and currency crises in the course of two millennia²³, the farfetched impact of two events in particular had shattered public trust towards financial systems and central governments globally; first was the Great Depression of the 1930s where the young Fed (created on December 23, 1913) with a disaster record was at risk of losing its independence and legitimacy as the public outcry and its inquisitorial turned into a loss of confidence in the American financial system (Bernanke, 1983; Bernanke & Gertler, 1995; BIS, 2011; Bordo, 2007; Eichengreen, 2003); and the second unprecedented event was the Great Recession (Great Contraction) of 2007-08, the severest shock in the post-WWII era, which made the level of distrust of the US financial system and the Fed peak in 2009 (Atkinson et al., 2013; Bernanke, 2012; Boulding, 1971; Cochrane, 2012). The latter financial mayhem (see Figure 13) also paved the road to the historic birth of Bitcoin – the world's first successful cryptocurrency (digital coin), created by a mysterious engineer/programmer under the alias Satoshi Nakamoto (a pseudonym) who supposedly worked two years for writing the

²¹ G-10 refers to the following group of countries: Belgium, Canada, France, Germany, Italy, Japan, Netherlands, Sweden, Switzerland, the United Kingdom (UK) and the United States (US). Even though the group consists of 11 countries, the name has remained as G-10.

²² The use of credit ratings provided by the external credit assessment institutions (ECAs) created three adverse incentives for banks; (1) banks felt no real urgency to develop their own internal risk-assessment frameworks; (2) ECAs misused the rating process to issue artificially-inflated ratings to clients from whom they earned lucrative service fees; (3) banks' overreliance on ECAs resulted in a cliff effect in capital requirements (BCBS, 2004).

²³ Exception of rare conformances, crises do occur in waves; for instance, when Laeven and Valencia (2008, 2012) have analyzed the frequency of starting month of banking crises, they have found that September has the highest frequency (averaging 25 crises), the rest of the months as follow; January (4), February (1 or 2), March (2 or 3), April (2), May (1), June (1), July (4), August (7), October (1), November (6), and December (7).

codes of Bitcoin. Once the writing of codes was completed, he/she/it registered the domain name *bitcoin.org* on 18 August 2008 and published the White Paper on 31 October 2008, “*Bitcoin: A peer-to-peer Electronic Cash System*”. In the following month, he kicked off Bitcoin project on 9 November 2008. On 3 January 2009, Nakamoto (2008) created the genesis block (first block) and a week later on 9 January 2009, Bitcoin v0.1 was released (i.e. the backbone of the future cryptocurrency market). As a symbolic first peer-to-peer transaction involving bitcoin took place on 12 January 2009 when 10 BTC was sent by Satoshi to a computer programmer by the name of Hal Finley.



Source: Vanguard²⁴

Figure 13: Consequences of the GFC of 2008

²⁴ <https://www.vanguard.com.au/adviser/en/article/markets-economy/the-global-financial-crisis>

The farfetched impact of the GFC of 2008 provided a perfect ground for Satoshi Nakamoto to launch Bitcoin (Nakamoto, 2008). In the pre-Bitcoin world, traditional online transactions relied on trusted third parties (banks and credit card companies) to perform three essential tasks; (i) to validate entire transactions; (ii) to ensure secured execution of transactions; and (iii) to maintain a chronology of transaction history. However in the Bitcoin world, trusted third-party intermediation is replaced by a trusted machine (i.e. blockchain with a distributed ledger); furthermore, there is no Fed-like central authority to set the value of digital currency or to control/manipulate the aggregate money supply by monetary policies involving a quantitative easing (credit expansion) or tight-money (contractionary) stance. Bitcoin with uppercase “B” denotes the network of electronic cash and bitcoin with lowercase “b” denotes a unit of account to facilitate exchange online where bitcoin is accepted.

Table 5: Blockchain segmented by permission model

Blockchain Types	Type		Read	Write	Commit	Example
	Open	Public permissionless	Open to anyone	Anyone	Anyone	Bitcoin, Ethereum
		Public permissioned	Open to anyone	Authorized participants	All or subset of authorized participants	Supply chain ledger for retail brand viewable by public
	Closed	Consortium	Restricted to an authorized set of participants	Authorized participants	All or subset of authorized participants	Multiple banks operating a shared ledger
		Private permissioned “enterprise”	Fully private or restricted to a limited set of authorized nodes	Network operator only	Network operator only	External bank ledger shared between parent company and subsidiaries

Source: Hileman and Rauchs (2017)

Although blockchain and Bitcoin are used interchangeably, they are distinctly different; as illustrated in Table 5, blockchain is an underlying infrastructure technology on which Bitcoin (alpha coin) and all altcoins (i.e. Ethereum) run; the latter however is a network of electronic cash (e.g. Dwyer, 2014). Based on needs and features of a digital coin, a blockchain can be designed as an open (permissioned or permissionless) or a closed (consortium or private permissioned) system which can either operate centralized or decentralized with a public distributed ledger technology - DLT (Berryhill et al., 2018; Cascarilla, 2015; Duffie, 2019; Dwyer, 2014; Pike, 2018). As outlined by Demirors (2017), blockchains have three key layers; an application layer (applications used by participants), a networking layer (protocol rules), and a protocol layer (programming language and computational rules). A blockchain is made up of two parts; blocks, like a storage unit, contain data and keep a chronology of transactions,

whereas a chain can be metaphorically viewed as a string that holds blocks together via a consensus algorithm based on proof-of-work (PoW), proof-of-stake (PoS), or other algorithms (see Baek & Elbeck, 2015; Bartos, 2015; Berentsen & Schär, 2018; Catalini et al., 2019; Chiu & Wong, 2014).

Bitcoin by far is the most popular and valuable cryptocurrency (i.e. peaked \$68,000 and market cap of over \$1.2 trillion in November 2021) the world has seen today, but its arduous journey to stardom has not been instantaneous at all. The birth of Bitcoin should not be taken as a direct consequence of the GFC of 2008 (i.e. funding freeze and the resultant severest credit crunch) because the previous attempts to create electronic cash actually began in the 1980s, accelerated in the 1990s with advent of the Internet and increased e-commerce activity, and finally became a reality in the new millennium. Nakamoto certainly had knowledge of the antecedent research and appreciated Chaum's "*untraceable payments*", Merkle's trees (Merkle, 1980, 1987), the proof-of-work (PoW) in the early 1990s, and Back's Hashcash (Back, 2002; Woodford, 2000; Camenisch et al., 2005; Okamoto & Ohta, 1992).

Since the late 1990s (starting with the Asian financial crisis of 1997-98), contemporaneous crises²⁵ have collectively cost the world's economies over \$30 trillion. Repeated financial shocks have, directly or indirectly, caused a spike in the acceleration of money's evolution into cryptocurrency which has occurred in three waves. The first wave of research began in the early 1980s with Chaum's proposal of "*untraceable payments*" (Chaum, 1982). The second wave of research was in the 1990s, consisting variations and extensions of the antecedent research in the previous decade, but the 1990s proved to be a decade of great transformation that took a leap of faith that internet along with a flock of startups of web-enabled technologies would make the dream of electronic cash a reality. Unfortunately, nearly all of the applications developed in the 1980s and 1990s never saw a widespread deployment; as such, combating email spam (Dwork & Naor, 1992; reasons of failure, see Laurie & Clayton, 2004), internet-based payment systems (Sirbu & Tygar, 1995) and minting digital coins (Rivest & Shamir, 1997). Furthermore, growing skepticism among risk-averse investors caused attempts by DigiCash (Chaum et al., 1998; Schoenmakers, 1998) and Peppercoin (Rivest, 2004) to end up failing; as a result, DigiCash (1990) filed for bankruptcy in 1998. Nevertheless, a decade before the birth of Bitcoin, two attempts at creating a decentralized digital currency emerged in the late 1990s; "b-money"²⁶ by Wei Dai (Dai, 1998) and Bitgold by blockchain pioneer Nick Szabo²⁷ (see Goldschlag & Stubblebine, 1998; Vishnumurthy et al., 2003; Okamoto & Ohta, 1992; Kocherlakota, 1998; Sander & Ta-Shma, 1999).

²⁵ A study by the Government Accountability Office (GAO) shows the 2008 global financial crisis (GFC) in systemic nature cost the U.S. economy alone over \$22 trillion (Blinder, 2013; Laeven & Valencia, 2008; Gorton, 2008). The IMF estimated the cost of the GFC as \$11.9 trillion (\$10.2 trillion in the U.S. and \$1.7 trillion in developing countries).
<https://www.telegraph.co.uk/finance/newsbysector/banksandfinance/5995810/IMF-puts-total-cost-of-crisis->

²⁶ <http://www.weidai.com/bmoney.txt>

²⁷ Szabo (2005), "Bit Gold" available at: <http://unenumerated.blogspot.rs/2005/12/bit-gold.html>

The last of the Mohicans, e-gold (established in 1996) after reaching several million users was also shut down by the U.S. government in 2008 citing legal issues. Although more than a decade has passed since Satoshi launched Bitcoin, the regulatory hurdle is still a major issue for cryptocurrencies.

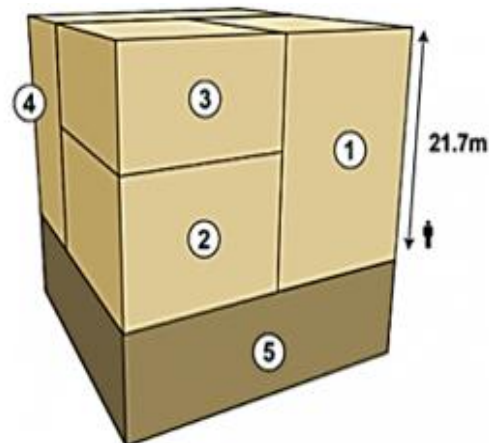
2.1.2 The Growth in Spite of Unending Obstacles and Backlash

Bitcoin, the world's first successful cryptocurrency, is not backed by gold²⁸ or any other tangible and intangible assets²⁹ of intrinsic value; its value literally comes from a combination of trust, acceptance, and a degree of speculation. Satoshi Nakamoto (a pseudonym) was probably fascinated by gold³⁰ and wanted to create something close to it or better. Bitcoin and gold have so much in common; as gold, Bitcoin is not infinite; on the contrary, the maximum number of mineable Bitcoin is fixed at 21 million, strikingly all the gold that has been mined to date can fit a cube of 21 meters on each side (Figure 14) and is worth \$12.54 trillion based on gold price of \$1,800/ounce on 10 September 2021 (to match that, the price of Bitcoin should be circa \$600K). At first, price of Bitcoin was secondary to engineers and enthusiasts who were more captivated by the unique design and its mysterious creator, but they also believed Bitcoin had an economic value due to expensive mining and other operational costs.

Total above-ground stocks (end-2019): 197,576 tones

1. Jewelry 92,947 tones, 47%
2. Private investment 42,619 tones, 21.6%
3. Official holdings 33,919 tones, 17.2%
4. Other 28,090 tones, 14.2%
5. Below ground reserves 54,000 tones

Source: Metals Focus, GFMS, Thomson Reuters, US Geological Survey, World Gold Council



Source: World Gold Council, <https://www.gold.org/about-gold/gold-supply/gold-mining/how-much-gold>

Figure 14: Total above ground stocks of gold mined throughout history

²⁸ The monometallic (gold only) standard was officially inaugurated in 1834 by the British Empire which marked the new beginning in trade where British pound sterling was fully backed by gold. By 1908, most countries adopted the gold standard. US President Franklin D. Roosevelt abandoned the half a century long practice (since 1789) in 1933 (due to the Great Depression) and in 1971 President Richard Nixon completely cut the link between the U.S. dollar and gold.

²⁹ Before the invention of money in the form of coins and bank notes, throughout history sea salt, large stones, clam shells, rice, blankets, cowrie shells, and peppercorns were used which are examples of commodity money.

³⁰ Gold meant wealth and power in many ancient civilizations including Incas, Aztecs, Roman, Persian, Greek, Mayan, Egyptian, and Mesopotamian; moreover, gold was used as a symbol to separate emperors as well as the members of royal family and high-ranked religious figures (priests) from the general public, and the rich from the poor. Gold was even metaphorically linked to Gods, i.e. ancient Greek lyric poet Pindar in 5th B.C. described gold as "a child of Zeus, neither moth nor rust devoureth it, but the mind of man is devoured by this supreme possession" (Bernstein, 2000).

Another great similarity between Bitcoin and gold is that both are mineable; nonetheless, the concept of “mining” through solving a mathematical problem (i.e. double spending) is not an arcane approach (Dwork & Naor, 1993; Back, 2002); as illustrated in Table 6, e-commerce on internet using various cryptographic payment systems is even much older (Wenninger, 2000).³¹

Table 6: Earlier electronic payments systems & proposals

ACC	CommerceSTAGE	Hashcash	Mini-Pay	PayMe	Secure Courier
Agora	Cybank	HINDE	Minitix	PayNet	Semopo
AIMP	CyberCash	iBill	MobileMoney	PayPal	SET
Allopass	CyberCoin	IMB-MP	Mojo	PaySafeCard	SET2GO
n-money	CyberGold	InterCoin	Mollie	PayTrust	SubScrip
BankNet	DigiGold	Ipin	Mondex	PayWord	Trivnet
Bitbit	Silk Road	Javien	MPTP	Peppercoin	TUB
Bitgold	E-Gold	Karma	Net900	PhoneTicks	Twitpay
Bitpass	ECash	LotteryTickets	NetCard	Playspan	VeriFone
C-SET	eCharge	Lucre	NetCash	Polling	VisaCash
CheckFree	eCoin	MagicMoney	NetCheque	Proton	Wallie
ClickShare	First Virtual	MicroMint	NetFare	Redi-Charge	Way2Pay
CommerceNet	FSTC	Micromoney	No3rd	S/PAY	WorldPay
CommercePOINT	Globe Left	MilliCent	One Click Charge	Sandia Lab E-Cash	X-PAY

Source: Narayanan et al (2016)

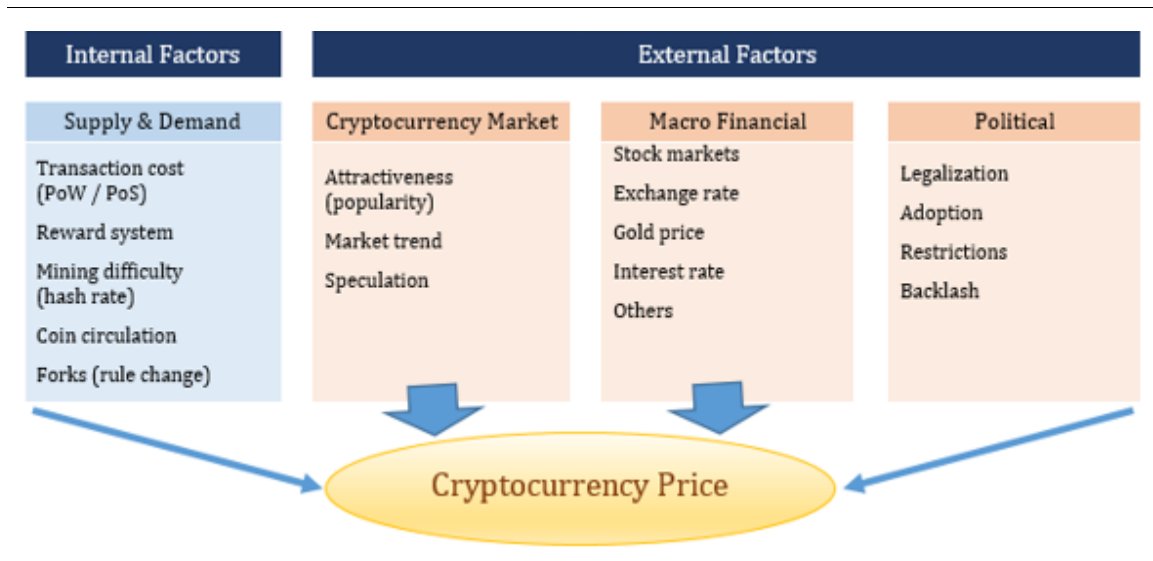
Bitcoin’s arduous price journey began with practically \$0.00 in the absence of an official exchange³²; except the symbolic transaction of 10 BTC sent on 12 January 2009 by Satoshi to a computer engineer Hal Finley (Nakamoto, 2008), the first real financial transaction involving Bitcoin took place on 22 May 2010 when Laszlo Haynek in Florida bought two Papa John's pizzas (about \$25) and paid 10,000 BTC, making the value of each Bitcoin at 0.0025 (i.e. after some exchanges opened in 2010, i.e. Mt. Gox, the price of Bitcoin crossed the \$1 threshold in April 2011). In the early days (Jan. 2009 – May 2012), Bitcoin did not have a very positive image due to claims of its association with the dark web (i.e. Silk Road) and its alleged involvement in unlawful activities such as terrorism-financing, money-laundering, illegal drugs and human trafficking (Bonneau et al., 2015; Gudgeon et al., 2019; Christin, 2013), all of which put a tight lid on Bitcoin’s true potential of a price surge. Furthermore, collapsed exchanges (Moore & Christin, 2013; for Ponzi schemes and Botnets³³, see Huang et al., 2014), hacked

³¹ Since advent of the Internet, a long list of cryptographic payment systems have come and gone (the most popular and still currently being used is PayPal. The following earlier examples paved the road for the birth of Bitcoin and so many other cryptocurrencies (over 2000 today). Ecash by David Chaum (1982), Hashcash by Adam Back (1997), B-money by Wei Dai (1998), Bit gold by Nick Szabo, Liberty Dollar (1998-2009), E-gold (1996-2007), and Reusable P-o-W by Hal Finney.

³² About a year after Bitcoin’s debut, a portal called Bitcoin Market was created by a Bitcointalk user named “dwdollar”, which was associated with frauds and PayPal stopped supporting it in June 2011. Mt. Gox arrived (July 18, 2010) which quickly became a popular exchange, but its creator Jed McCaleb sold it on 6 March 2011 to Mark Karpeles who later strangely disappeared with all the users’ bitcoins. About 2-3 weeks after this incident, three new non-dollar fiat currency pairs were created, i.e. Bitcoin (BTC)/pounds (GBP), Bitcoin (BTC)/reals (BRL), and Bitcoin (BTC)/euros (EUR).

³³ A Botnets uses Trojan viruses to control users’ computers and Ponzi schemes are fraudulent investing scam s which are similar to a pyramid scheme where funds from new investors are used to pay the earlier investors.

accounts, thefts³⁴ of cryptocurrencies and a greater scrutiny placed on Bitcoin by governments only made the matter get a lot worse. Since no Fed-like central authority sets the price of Bitcoin, it is therefore influenced by a confluence of both internal and external factors (Figure 15); for this reason, the price of Bitcoin could be anything in its eco system as long as people are willing to pay for it.



Source: Author; Poyser (2017); Sovbetov (2018)

Figure 15: Key characteristics of Bitcoin blockchain and price influencers

In spite of criticism and Bitcoin bashing, a noticeable surge in Bitcoin price coincided with two key developments; the number of exchanges has increased dramatically throughout the world, see Table 7 for key exchanges (as of 13 October 2022, 21,317 cryptocurrencies are traded in 526 exchanges, their total market cap is \$941 billion³⁵, but peaked at over \$2.7 trillion in November 2021, the market cap of Bitcoin alone reached \$1.2 trillion); and there was a shift in the mindset of advanced nations' governments from outright banning Bitcoin to accepting it as a security, asset, or commodity for tax purposes, see Table 8 (El Salvador is the only country that accepts Bitcoin as a currency).

Table 7: Key exchanges

United States	Europe	Asia
Kraken Coinbase Bittrex Poloniex	Bitsamp LocationBitcoin CEX.io	Bitfinex Huoban of Okex bitflyer Binance

Source: Author

³⁴ Gox (thousands of bitcoins were stolen, usernames and email accounts were compromised); Bitomat (Polish exchange, 17.000 BTC are stolen from users); MyBitcoin (120.000 bitcoins were stolen from users); TradeHill (\$100.000 was lost); Bitcoin (18.000 bitcoins were stolen. The market value of stolen and lost bitcoins is over \$5 billion.

³⁵ <https://coinmarketcap.com/>

To this date, categorization of Bitcoin has been a hotly debated topic; for instance in the U.S. alone, for tax purpose the Internal Revenue Service (IRS) has classified Bitcoin as a property, the Securities and Exchange Commission (SEC) has categorized cryptocurrencies as securities, and the Commodities Futures Trading Commission (CFTC) has recognized Bitcoin as a commodity (see Table 8).

Table 8: Cryptocurrency regulations in US, EU, and Japan

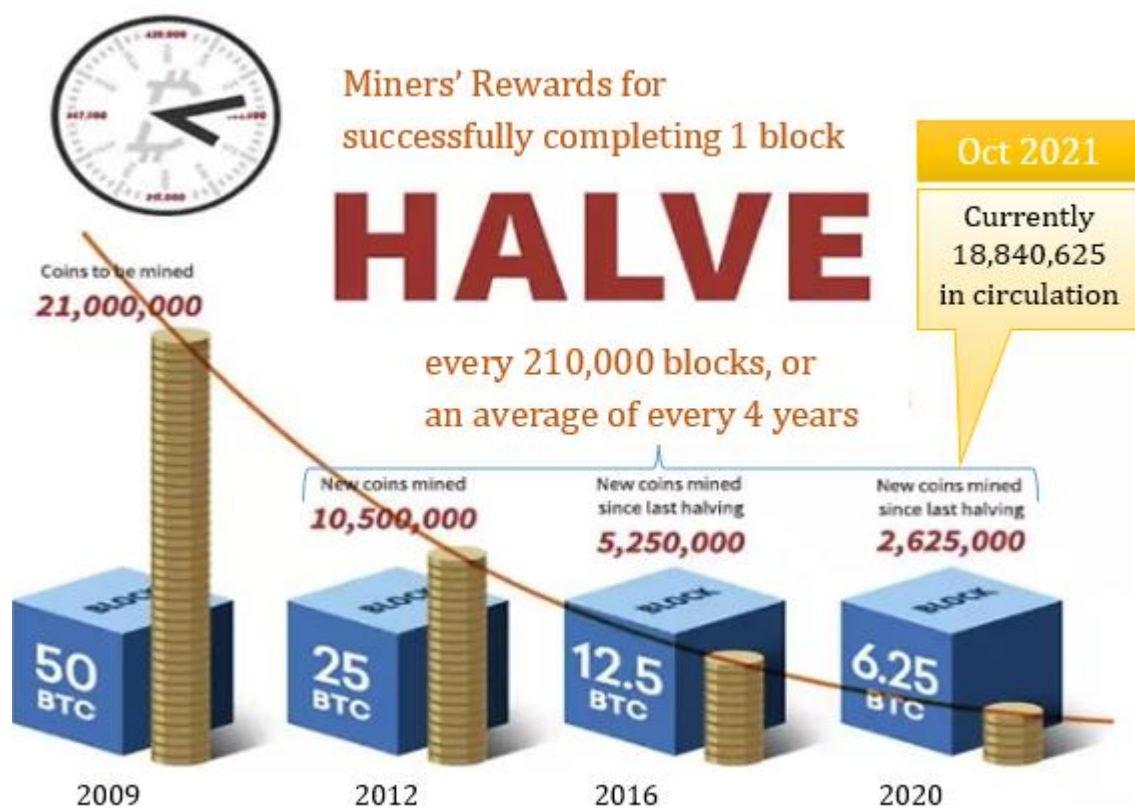
Country	Crypto-friendly norms	Regulations on illicit use of cryptocurrencies
US	U.S. accepted Bitcoin in 2013 and classified it a commodity in 2015, but cryptocurrencies are not considered legal tender but exchanges are legal in the United States under the regulatory scope of the Bank Secrecy Act (BSA). However, cryptocurrency exchanges are regulated differently by states, nonetheless they must obtain a license from the Financial Crimes Enforcement Network - FinCEN which considers them money transmitters.	US Securities and Exchange Commission (SEC) considers cryptocurrencies as securities, the same securities laws apply to digital wallets. On the other hand, the Commodities Futures Trading Commission (CFTC) has recognized Bitcoin and Ethereum as commodities. Under the guidelines published by FATF in June 2019, FinCEN expects crypto exchanges to comply with record-keeping requirements and the AML and CFT laws.
EU	Unlike the U.S., cryptocurrencies are legal across the EU, but regulations and taxation may vary by member-states. Cryptocurrency exchanges must comply with European Banking Authority (EBA), European Commission (EC), European Central Bank (ECB), European Insurance & Pension (EIOPA), and European Supervisory Authority for Securities (ESMA). In 2015, the EU Court of Justice ruled that cryptocurrencies should be exempt from value-added tax (VAT). In January 2020, EU's 5th Anti-Money Laundering Directive (5AMLD) came into effect. In December 2020, 6AMLD came into effect, making cryptocurrency compliance more stringent (cybercrime added).	Cryptocurrencies are legal across the EU and are classified as qualified financial instruments (QFI's). Exchanges are regulated at a regional level under QFI licenses; in certain EU states, they have registration requirements with their respective regulators, i.e. France's Autorité des Marchés Financiers (AMF), Germany's Financial Supervisory Authority (BaFin), Italy's Ministry of Finance. Authorizations and licenses granted by these regulators can be <i>passport</i> exchanges, allowing them to operate under a single regime across the entire bloc. Following 5AMLD and 6AMLD, cryptocurrency exchanges are liable for money laundering offences to legal persons as well as individuals (i.e. greater oversight).
Japan	Japan currently has the world's most progressive regulatory climate; Bitcoin and other coins are recognized as legal property under the Payment Services Act (PSA). Following those regulations, crypto exchanges in Japan are required to be registered and comply with traditional AML and CFT rules. The National Tax Agency ruled that gains on cryptocurrencies are taxed. The recent regulations include amendments to the PSA and FIEA effective in May 2020, which place greater restrictions on cryptocurrency custody service providers that fall under the scope of the PSA while cryptocurrency derivatives businesses fall under the scope of the FIEA.	Under the PSA, only businesses with a competent local Financial Bureau are allowed to operate as a cryptocurrency exchange; foreign cryptocurrency exchanges are permitted to register where they can comply with equivalent registration standard in their host country. Due to a series of high profile hacks, crypto regulations have become an urgent national concern, therefore the Financial Services Agency (FSA) has stepped up efforts to regulate trading and exchanges: amendments to the PSA require cryptocurrency exchanges to register and comply with cybersecurity and AML/CFT laws. In 2016 and 2019, another amendment updated this requirement to include checking customer ID and to cover custodian services providers.

Source: Mondaq³⁶; Comply Advantage³⁷

³⁶ <https://www.mondaq.com/india/fin-tech/1044546/global-cryptocurrency-regulatory-landscape>

³⁷ <https://complyadvantage.com/blog/cryptocurrency-regulations-around-world/>

With the absence of trusted third-part intermediators, miners (nodes) validate all transactions based on proof-of-work (PoW) and protect the blockchain ledger from all types of attacks. To incentivize miners to validate transactions and induce competition among them, Nakamoto created a very unique reward scheme that pays miners (validators) for every successfully completed block of transactions. As illustrated in Figure 16, the block reward was first set at 50 bitcoins (BTC) and has been halved for every 210,000 validated blocks (roughly, once every four years), currently at 6.25 BTC. Miners also make money from transaction fees, which were practically set to zero at the time of Bitcoin's debut in January 2009. Over a decade later, with the price of each bitcoin over \$67,600 in November 2021, the validation fees of transactions now make up a larger portion of the incentive scheme.



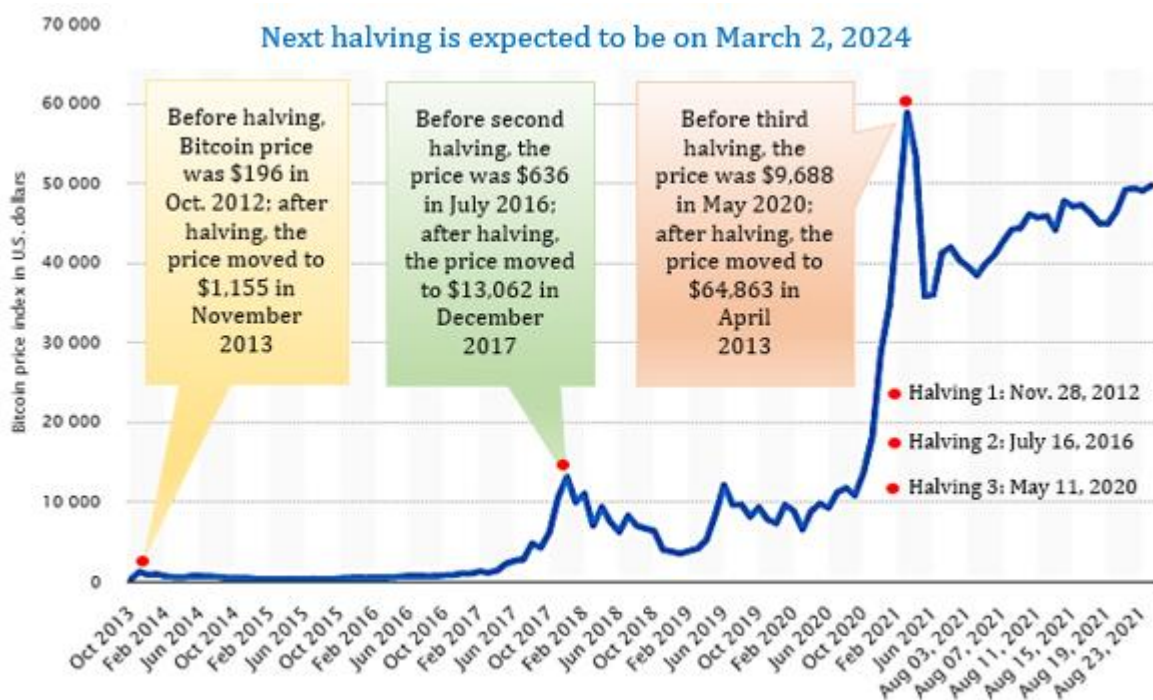
Source: <https://www.investopedia.com/terms/b/bitcoin-mining.asp>

Figure 16: Bitcoin incentive proof-of-work scheme

Miners solve the double-spending problem based on proof-of-work consensus protocol, and in return, randomly selected winner (miner) is awarded the set block reward (i.e. 6.25 BTC after the May 2020 halving) plus any corresponding transaction fees. A block is completed roughly every 10 minutes, meaning miners get bitcoin rewards roughly six times per hour; one with a larger computational power has a better chance of winning the rewards. During each 10-minute time interval, confirmed

and unconfirmed transactions along with the block header from the previous block are hashed into a new block using SHA-256 algorithm. Miners guessing the right nonce (a number) are able to generate an output below the target output threshold. Then the block solution is broadcast to all nodes, if 51% of miners reach a consensus on the solution to the problem, this as a new block is then added to the blockchain and the block reward goes to its successful owner (19,179,300 bitcoins on Oct. 14, 2022).

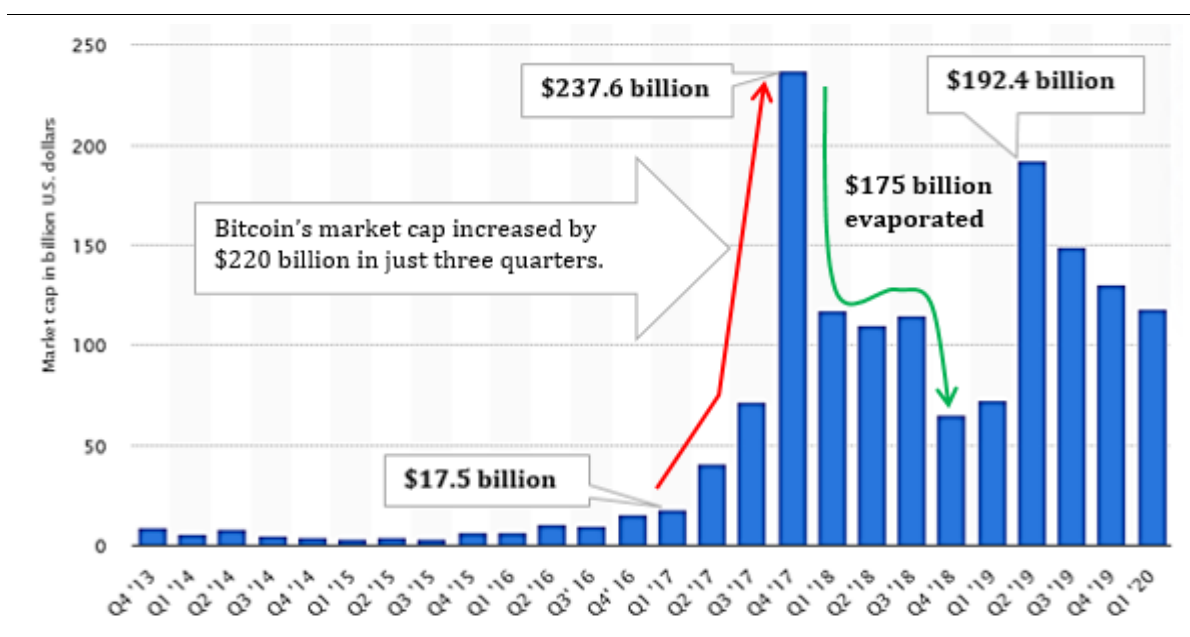
Halving of Bitcoin in about every four years or so has generated unmatched hype through media frenzy which fueled further speculation along with manipulative trading activities; as illustrated in Figure 17, the price of bitcoin set new price records in the aftermath of each halving event. So far, there have been three halving (28 November 2012, 16 July 2016, 11 May 2020, and the fourth halving is expected to happen on 2 March 2024, 509 days from today), three bubbles, and two major crashes. Between 2010 and 2022, Bitcoin (and the cryptocurrency market as a whole) has undergone many crashes, in fact 21 of them (see Table 3 on page 9 for a list of price corrections of 30% or higher), but the last Great Silent Crash (GSC) was more severe than all previous crashes combined. The bursting of the first major Bitcoin bubble was in December 2017 after its price moved to mindboggling \$19,783 (intraday high of \$20,089); the damage was, \$700 billion of value just evaporated (Bitcoin's market cap alone shrank by \$300 billion). Bursting of the second bubble was a lot more severe, after the price of Bitcoin saw \$67,566, the cryptocurrency market crashed with a loss of close to \$2 trillion.



Source: Statista, <https://www.statista.com/statistics/326707/bitcoin-price-index/>

Figure 17: Bitcoin halving and price movements

Bitcoin mania is also often compared with the Dutch tulipmania of the 17th century. Although the levels of speculation were more severe in Bitcoin than tulipmania, but their sudden collapse and the duration it took were almost identical, i.e. extraordinary price rises in each event occurred in the last month of the year (December 1636 and December 2017 respectively) and both crashes were followed by sharp price falls. In a matter of several days in December 2017, the price of bitcoin doubled from \$10,000 to intraday high of \$20,089 on December 17, 2017; at its peak, Bitcoin's market capitalization stood at \$237.6 billion (\$348 billion at the intraday high price of \$20,089). With the arrival of the New Year (January 2020), Bitcoin underwent one of the biggest price corrections; in just one year, the price of bitcoin dropped below \$4,000 and its market cap to \$65 billion (Figure 18).

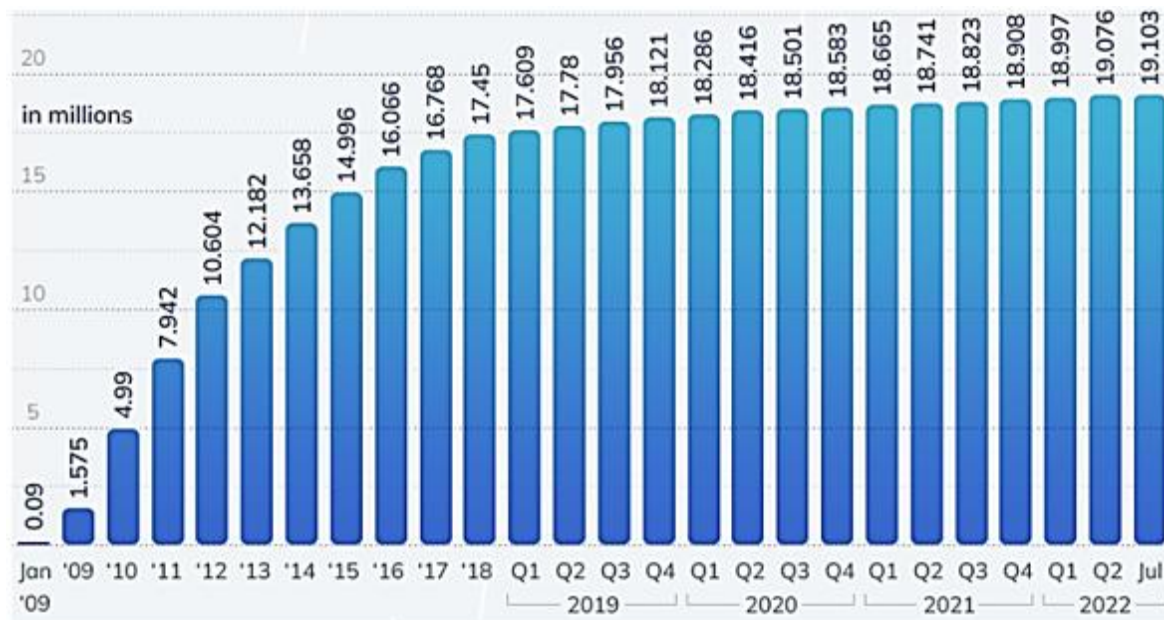


Source: Statista; <https://www.statista.com/statistics/377382/bitcoin-market-capitalization/>

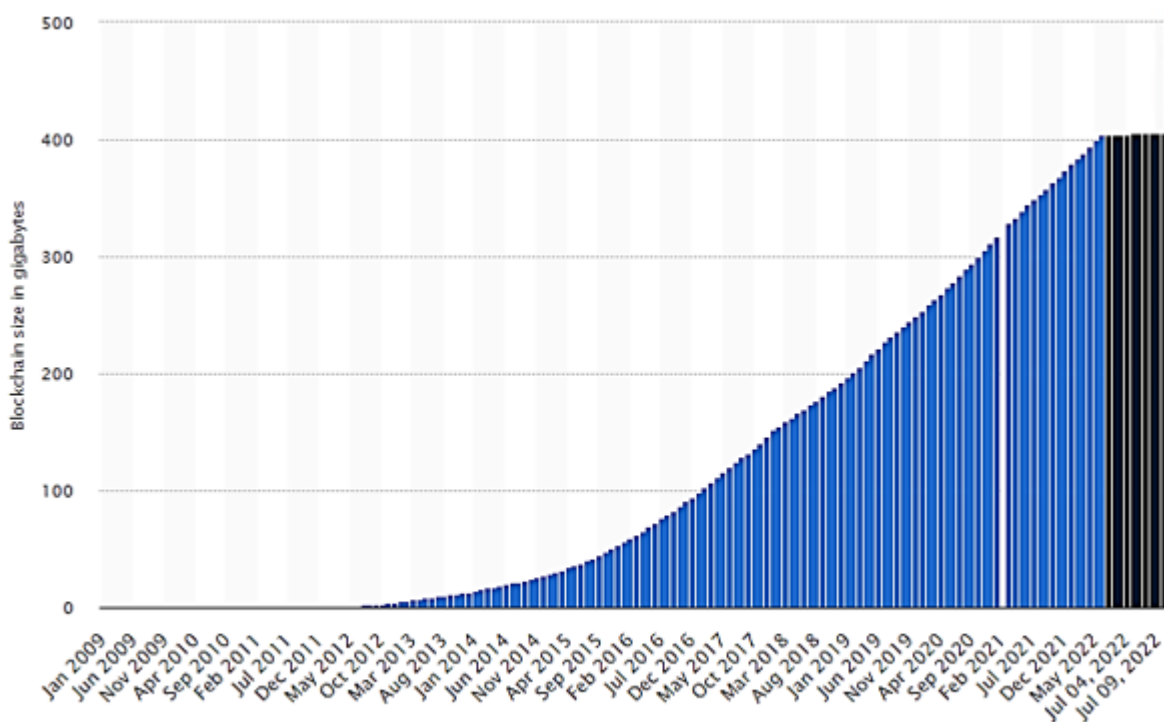
Figure 18: Market capitalization of Bitcoin from 4th quarter 2013 to 1st quarter 2020

Bitcoin is no fluke, it is real and has been with us over a decade. It is true and well documented that Bitcoin has certain inherent weaknesses and flaws, but its extreme volatility is due to uncertainty stemming from increased regulatory efforts in the U.S., Japan and Europe to curb the true potential of Bitcoin and other cryptocurrencies. For instance, the U.S. will do everything in its power to defend the dollar's hegemony and keep enjoying its "exorbitant privilege". If two dozens of major shocks have been unsuccessful to kill off Bitcoin, it is certain that both Bitcoin and altcoins will continue to forge ahead unabated regardless of ever more backlash (the Fed and the ECB in particular) and ongoing ill-efforts by doubters, pessimists, doomsayers, skeptics, disbelievers, and short-sighted and blindfolded politicians waiting on the lookout to put another nail in the Bitcoin coffin. Not dead yet, as the U.S. government (i.e. former president Trump) has waged war against Bitcoin, more people have become

supportive of Bitcoin which broke new records during May-November 2021 in terms of price (peaked \$67,566, see Figure 20), blockchain size (406.05 gigabytes in July 2022) and wallet users.



Source: <https://moneytransfers.com/news/content/how-many-bitcoins-are-there>

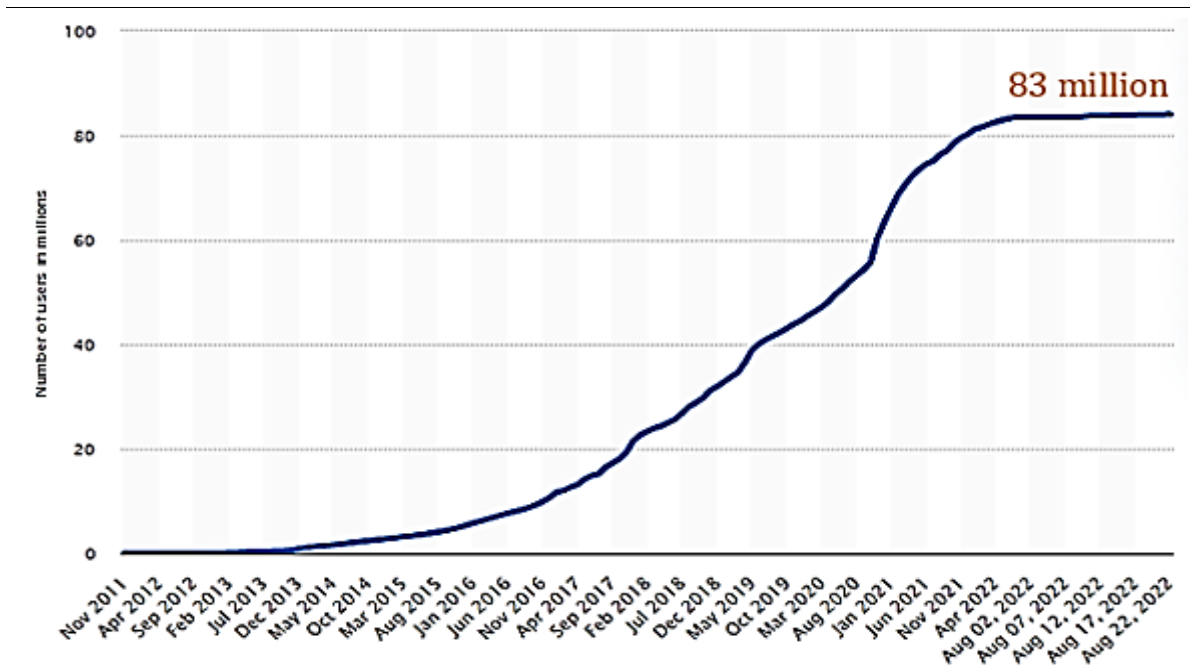


Source: <https://www.statista.com/statistics/647523/worldwide-bitcoin-blockchain-size/>

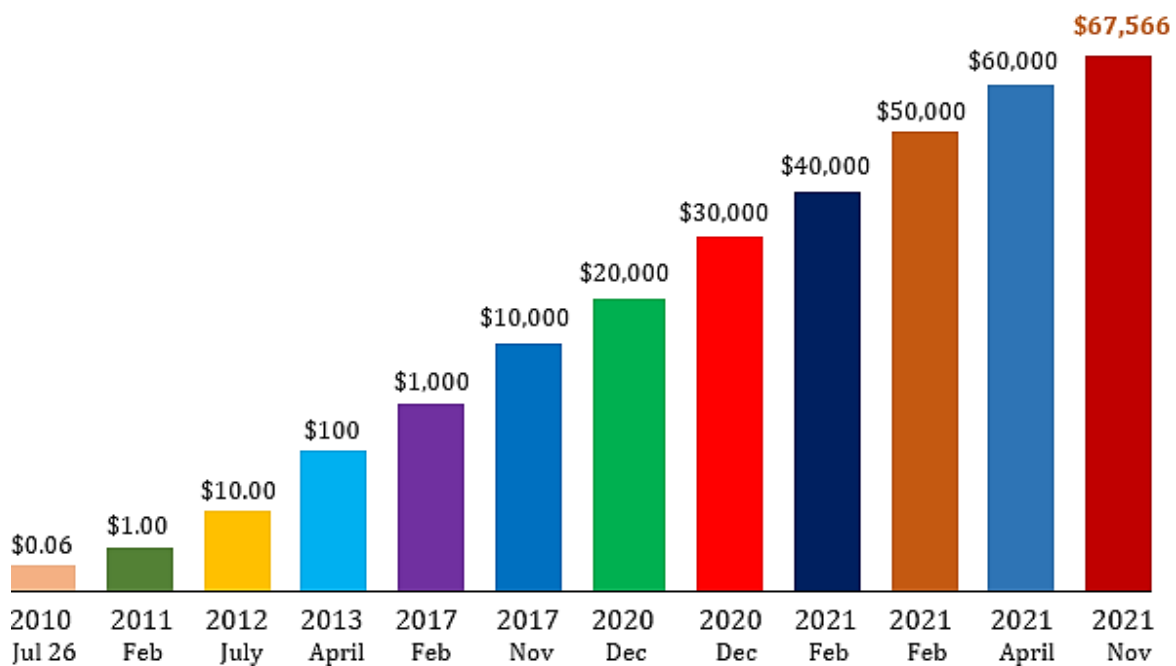
Source: Moneytransfers.com; Statista

Figure 19: Number of Bitcoins mined and Bitcoin blockchain size

Bitcoin has undeniably become a household name; for instance, nearly 90% of adults in America have heard of Bitcoin, and in just less than three years, the number of unique Bitcoin wallet users has more than doubled, i.e. from about 40 million in 2019 to over 80 million as of July 2022 (Figure 20).



Source: Statista, <https://www.statista.com/statistics/647374/worldwide-blockchain-wallet-users/>



Source: Author

Source: Statista; Author (the height of bars are visual and do not indicate actual price levels)

Figure 20: Bitcoin wallet users and Bitcoin price records

2.3 Factors that Hinder Bitcoin's Potential to Become a Global Reserve Currency

The Bitcoin literature covers a variety of factors why Bitcoin (as anticipated) has not become a global reserve (supranational) currency. This paper will mainly discuss five of those factors:

2.3.1 Intertwined American Imperialism and Dollar's Hegemony

American imperialism (dollar's hegemony) is a show-stopper for cryptocurrencies. European settlers arrived in North America in the 16th century as part of several colonization programs (i.e. the first colony was founded at Jamestown, Virginia, in 1607), launched by the British Empire, France, Spanish Empire, and the Dutch Republic. Before American imperialism was born (already in its DNA), global trade and the world's scarce resources were largely controlled by the imperial powers³⁸ before the 16th century. When the Continental Congress (i.e. the government of 13 original American colonies, and the U.S. 1774 and 1789) declared independence from the Great Britain on 4 July 1776, it had no currency of its own (i.e. British pound and Spanish peso were granted legal tender for an indefinite period of time). However, George Washington as the first U.S. President enacted the Coinage Act of 1792 into law, making the dollar as a unit of account and "\$" as its symbol (see Bogart, 1930; Friedman & Schwartz, 1963; Bense, 2000). To finance the Revolution War (1775-83), the Continental Congress authorized the issuance of paper notes called "*continental*" in 1775 (IOUs³⁹). However, due to a lack of trust and hyperinflation by 1778, it was only worth 1/7 of the original face value; after another devaluation in 1780 (at 1/40), continentals (currency) were out of circulation permanently in 1781 (for more history, see Bogart, 1930; Friedman & Schwartz, 1963; Bense, 2000).

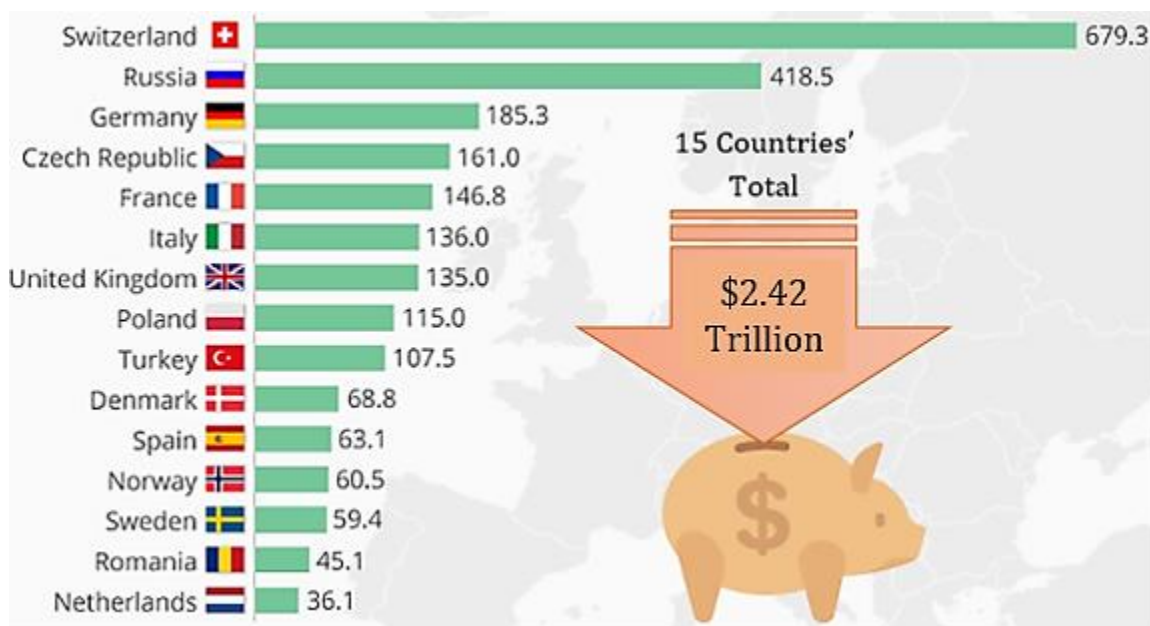
Imperial powers were succeeded by European powers (who were also chief enemies of each other) up to the 20th century and later dominated by the industrial capitalists in the leadership of the United States (e.g. Abernethy, 2000; Cell, 1997). After acquisitions of new territories in western Pacific and Latin America such as Puerto Rico, Philippines, and Guam (see Corbin & Levitsky, 2003) and winning insular possessions in the Caribbean and Hawaii, the United States craved more and its next mission was to restructure the power-sharing in Europe. Two Great Wars (WWI and WWII) were enough to culminate Europe's hegemony, which was decimated by the end of WWI and the U.S. emerged as a super power. The defeat of Nazi Germany in WWII caused another power shuffle in Europe, this time Britain was sidelined (i.e. John Maynard Keynes, the influential English economist and leader of the British delegation at the 1944 Bretton Woods Conference, ridded the UK of its imperial grandeur and

³⁸ Pre-13th century: Persian, Indian, Roman empires, the Muslim Caliphates, the Chinese and Egyptian dynasties; Mongol Empire (13th – 15th century), Ottoman Empire (13th – 16th century), and European hegemony (18th – 20th century).

³⁹ IOU stands for "I owe you".

the reserve currency). The United States and the Soviet Union (USSR) became two superpowers, but only one main reserve currency (the dollar-USD). In order to contain Soviet influence and to prevent the expansion of communism, President Harry S. Truman announced the Truman Doctrine on March 12, 1947 which prompted the Cold War (1947-1991) that meant a step towards a bipolar world order, the Western Bloc⁴⁰ against the Eastern Bloc⁴¹ (Marples, 2004; Shannon, 1996).

Competition between the U.S. and Europe increased noticeably with the creation of the European Economic Community⁴² (EC) in 1957 that became the European Union (EU) in 1993. The idea of one large common market enticed U.S. companies to invest in Western Europe, this prompted EU nations to stash dollars in foreign reserves (Figure 21). Technology was at the forefront of an accelerated transformation in the 1950s and 1960s; for instance, first commercial telecommunications satellite was launched in 1965, fax machines were available in the 1970s, and containerized shipping reduced the cost of moving goods across the world; as a result, world trade grew exponentially.



Source: Statista, <https://www.statista.com/chart/14850/foreign-reserves-by-european-country/>

Figure 21: European countries foreign reserves (2016-17, in \$billion)

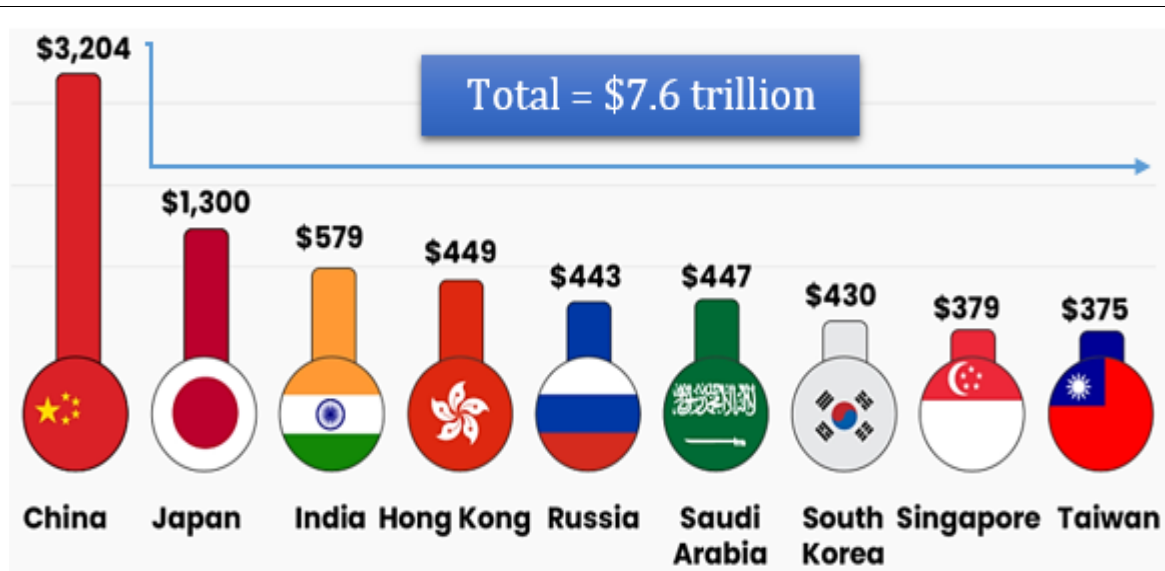
United States finances trillions of its debt through peripheries. In the post-WWII era, developing and emerging economies have become increasingly dollarized. The triad instability (i.e. political, financial, and currency) has been the usual motive behind massive dollarization in developing and low-income

⁴⁰ Other terms used to describe: the Free Bloc, the Capitalist Bloc, the American Bloc, and the NATO Bloc,

⁴¹ Other terms used to describe: the Communist Bloc, the Socialist Bloc and the Soviet Bloc.

⁴² The Economic Community was formed by Belgium, France, Italy, Luxembourg, the Netherlands and West Germany.

countries (LICs). In spite of the great strides toward macroeconomic and political stability in some countries, dollarization has continued to rise; in fact, foreign currency (mainly dollar) has exceeded 50% of national money stock in a number of countries (see Berg & Borensztein, 2000; Chang, 2000; Galindo & Leiderman, 2005). Besides Panama and Liberia (fully dollarized), de facto dollarization has been prevalent in Asia, Latin America and some parts of Europe (i.e. 20% in Greece, 45% in Turkey over \$200 billion, and over 80% in Bolivia). Interestingly, only one-third of all dollar currency is held within the United States and the remaining two-thirds is held outside of the U.S. (Figure 22).



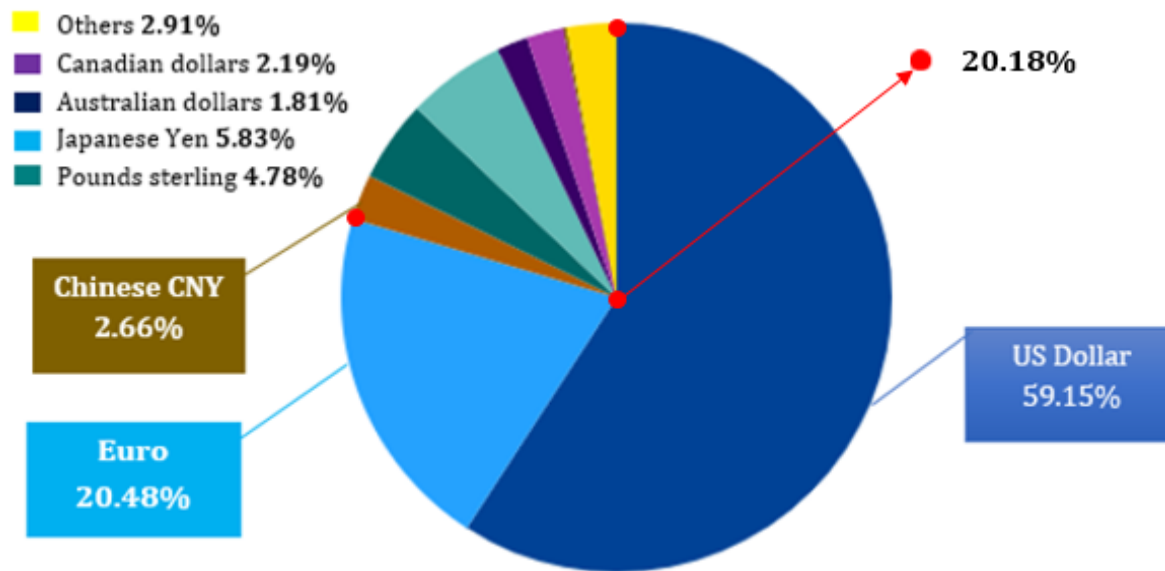
Source: Research and Rank, <https://twitter.com/researchandrank/status/1394935544203726855/photo/1>

Figure 22: Dollar reserves of top 9 countries (\$ billion)

Today, virtually all economies of the world operate under the artifice of a hegemon (the U.S.) which assumes the role of acting as a stabilizer and the lender of last resort to avoid a panic-induced funding freeze either due to a contraction (i.e. banks may hoard capital) or a severe case of countercyclical deleveraging in an acute financial distress (see Dooley et al., 2004; Garber, 1993; Kirshner, 2008). The utmost responsibility of the hegemon was to use its currency (dollar) as an anchor to provide global liquidity with the objective of promoting a hegemonic stability underpinned by unrestricted access to capital markets, free cross-border flow of goods and capital, and expansive monetary policies (e.g. Kindleberger, 1973). This grand strategy was self-defeating as it had serious operational side effects; the 'exorbitant privilege' has turned the U.S. into an 'anarchic Hobbesian' whose hegemonic status is backed by its military might and sanction power (Eichengreen, 2011; Hudson, 1972).

The U.S. dollar faces no real competition to affect its role as the world's main reserve currency (Figure 23), but the repeated financial crises and the resultant massive losses since the late 1990s put a scuff

on its unblemished reputation of capital flight to safety. Even former Fed Chairman Alan Greenspan⁴³ along with Chinn and Frankel (2006) claimed that by 2020 euro could dethrone the U.S. dollar as the world's main reserve currency. However, there are mixed views regarding the dollar's stabilizer role; advocates argue that dollar will continue to dominate global economy due to network externalities (contagion or network effects), but the opponents (i.e. the influential British economist John Maynard Keynes) refute the idea of a single dominant reserve currency (Côté, 1994; Rose, 1999).



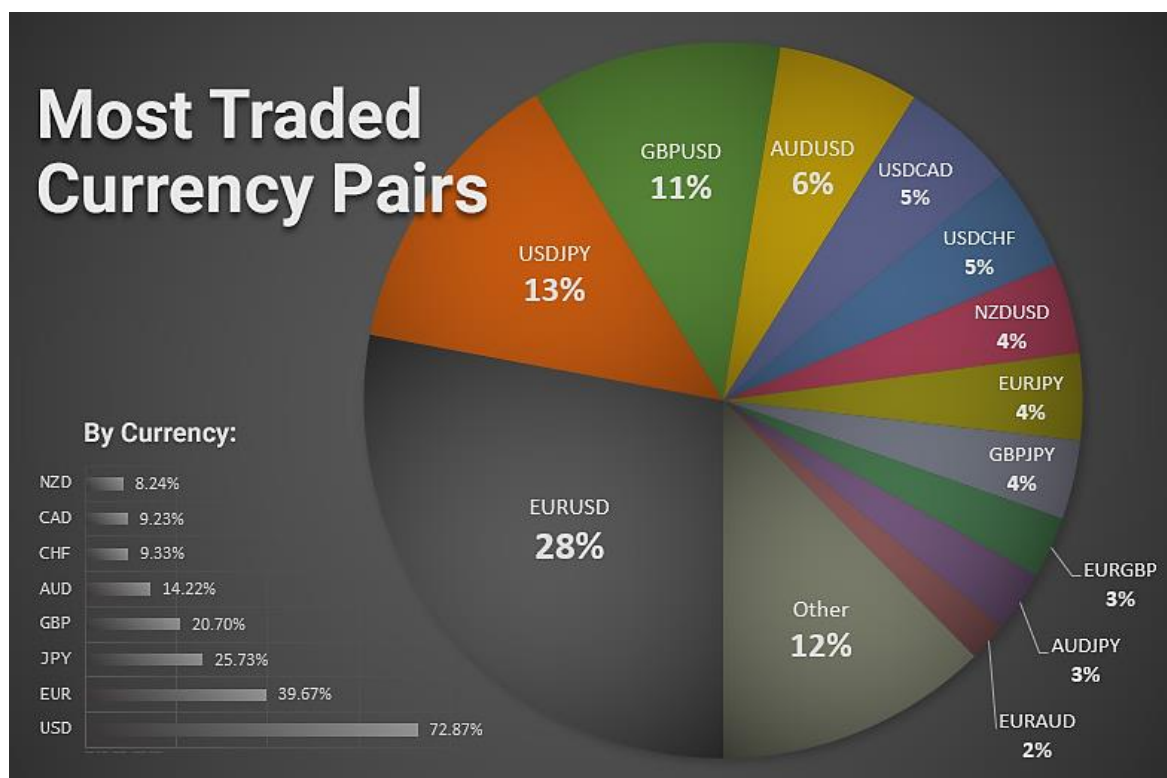
Source: IMF; <http://data.imf.org/?sk=E6A5F467-C14B-4AA8-9F6D-5A09EC4E62A4>

Figure 23: Currencies percent of total foreign exchange reserves (2021)

No country other than the United States (although China is closing the gap) has the right power mix to defend its national currency. Therefore, American imperialism and the dollar's hegemonic status are interlinked through four pillars; 1) economic power; 2) military might; 3) superior technological power; and 4) geopolitical power. U.S. presidents and various government branches of the United States rather have a forceful influence on other countries and their politicians. Anytime the U.S. can't seem to get what she wants or the dollar's hegemony is questioned, the US-influenced (extension of its political power) multilateral organizations such as the IMF, UN, NATO, WTO, World Bank and many others contribute to the amount of international pressure for a satisfactory outcome for the U.S. To preserve the dollar's hegemony and maintain its exorbitant privilege, former president Trump had been very aggressive in weaponizing the dollar and pushing sanctions as a foreign policy, i.e. Trump imposed a total of 1,474 sanctions on entities – far more than any U.S. presidents.

⁴³ <https://www.reuters.com/article/greenspan-euro/euro-could-replace-dollar-as-top-currency-greenspan-idUSL1771147920070917>

Despite repeated high-magnitude financial crises and the ongoing search for an alternative medium of exchange (i.e. yen, euro, and the latest Bitcoin and yuan have failed even to make a dent on dollar's damaging power), close to a billion people worldwide still use U.S. dollar as a main currency to engage in economic activity; moreover, a good number of sovereign states and territories use dollar as their official currency⁴⁴, plus in many countries dollar is used along with local currencies.⁴⁵ Because most of the key commodities (i.e. oil, gold, coffee, sugar, pork, etc.) are priced in dollars, the U.S. dominates both foreign exchange markets and the international trade (i.e. over 72% of the daily forex trading volume involves U.S. dollar, see Figure 24). Although there have been many efforts to kill off, deflate, or end the dollar's smile, diminishing value of the mighty dollar could have a noticeable impact on the world's 185 currencies that maintain a tight trading range to the dollar. Nevertheless, fall of the dollar may benefit central banks that invest in U.S. Treasury or other high interest yielding bonds. Due to the U.S. bond markets being the largest and most liquid (over \$21 trillion), virtually all central banks choose to stash over \$6 trillion dollars of reserves in the U.S. bond market to earn interest.



Source: <https://fxssi.com/the-most-traded-currency-pairs>

Figure 24: 10 Most traded currency pairs in 2022

⁴⁴ Ecuador (since 2000), El Salvador, Marshall Islands, Micronesia, Palau, Timor-Leste, Zimbabwe (after 2000), the U.S. territories (American Samoa, American Virgin Islands, Guam, Northern Mariana Islands, Puerto Rico), territories of the United Kingdom (British Virgin Islands, Turks and Caicos).

⁴⁵ Bahamas, Belize, Barbados, Costa Rica, Nicaragua, Panama, Myanmar, Cambodia, Liberia, St. Kitts and Nevis, territories of the Netherlands (Aruba, Bonaire, Curacao, St. Maarten, St. Eustatius and Saba), territories of the United Kingdom (Bermuda, Cayman Islands).

The United States dollar is still the world's main reserve currency by a large margin, but its power has somewhat diminished as China has passed the U.S. in 2014 as the world's largest economy in terms of GDP/PPP (purchasing power parity). The gap between two economies widens in 2030 (Table 9); and by 2050, the U.S. drops to third place overtaken by India; strikingly, Brazil moves to 4th, Japan to 7th and the UK to 11th positions. In light of these developments plus China's unwillingness to store billions in liabilities of the U.S., gold is assured to play a more vital role in the future monetary order. With macro events such as hard Brexit, China-US trade war, the U.S. abuse of sanction power as a foreign policy, war in Syria, OVID-19 (coronavirus) pandemic, Ukraine-Russia war; central banks across the world (peripheries) are mainly concerned with preserving value of their reserves and have been accumulating gold to hedge against the dollar's use as a weapon of mass economic destruction.

Table 9: GDP at PPP ranking
\$USD, Billion (2014)

PPP Rank	Country	GDP at PPP 2014 (\$bn)	Country	GDP at PPP 2030 (\$bn)	Country	GDP at PPP 2050 (\$bn)
1	China	17,632	China	36,112	China	61,079
2	United States	17,416	United States	25,451	India	42,205
3	India	7,277	India	17,138	United States	41,384
4	Japan	4,788	Japan	6,006	Indonesia	12,210
5	Germany	3,621	Indonesia	5,486	Brazil	9,164
6	Russia	3,559	Brazil	4,996	Mexico	8,014
7	Brazil	3,073	Russia	4,854	Japan	7,914
8	France	2,587	Germany	4,590	Russia	7,575
9	Indonesia	2,554	Mexico	3,985	Nigeria	7,345
10	United Kingdom	2,435	United Kingdom	3,586	Germany	6,338
11	Mexico	2,143	France	3,418	United Kingdom	5,744
12	Italy	2,066	Saudi Arabia	3,212	Saudi Arabia	5,488
13	South Korea	1,790	South Korea	2,818	France	5,207
14	Saudi Arabia	1,652	Turkey	2,714	Turkey	5,102
15	Canada	1,579	Italy	2,591	Pakistan	4,253
16	Spain	1,534	Nigeria	2,566	Egypt	4,239
17	Turkey	1,512	Canada	2,219	South Korea	4,142
18	Iran	1,284	Spain	2,175	Italy	3,617
19	Australia	1,100	Iran	1,914	Canada	3,583
20	Nigeria	1,058	Egypt	1,854	Philippines	3,516

Source: PricewaterhouseCoopers (PwC); Taskinsoy (2019a)

Today, competition is fierce in every sense; people, entities, and even sovereign states are prepared to do everything in their power to protect what they see as value. In that line of logic, former U.S. president Trump showed the world how he feels about the United States dollar (most valuable asset and weapon of the U.S.) by tweeting "We have only one real currency in the USA, and it is stronger than ever, both dependable and reliable. It is by far the most dominant currency anywhere in the world, and it will always stay that way. It is called the United States Dollar!" The U.S. military might is a big fan of the dollar, which reserves the right to intervene whenever necessary (Table 10).

Table 10: Military strength ranking (2017)^a

Country	World Rank	Power Rating	Country Population	Military Personnel	Total Aircraft	Fighter Aircraft	Combat Tanks	Naval Assets*	Budget \$billion	Credit Suisse Report (2018) ^b		
										Country	World Rank	Budget \$billion
United States	1	0.0857	323,995,528	2,363,675	13,762	2,296	5,884	415	587.8	United States	1	601.0
Russia	2	0.0929	142,355,415	3,371,027	3,794	806	20,216	352	44.6	Russia	2	84.5
China	3	0.0945	1,373,541,278	3,712,500	2,955	1,271	6,457	714	161.7	China	3	216
India	4	0.1593	1,266,883,598	4,207,250	2,102	676	4,426	295	51.0	Japan	4	41.6
France	5	0.1914	66,836,154	387,635	1,305	296	406	118	35.0	India	5	50.0
United Kingdom	6	0.2131	64,430,428	232,675	856	88	249	76	45.7	France	6	62.3
Japan	7	0.2137	126,702,133	311,875	1,594	288	700	131	43.8	South Korea	7	62.3
Turkey	8	0.2491	80,274,604	743,415	1,018	207	2,445	194	8.2	Italy	8	34.0
Germany	9	0.2609	80,722,792	210,000	698	92	543	81	39.2	United Kingdom	9	60.5
Egypt	10	0.2676	94,666,993	1,329,250	1,132	337	4,110	319	4.4	Turkey	10	18.2
Italy	11	0.2694	62,007,540	267,500	822	79	200	143	34.0	Pakistan	11	7.0
South Korea	12	0.2741	50,924,172	5,829,750	1,477	406	2,654	166	43.8	Egypt	12	4.4
Pakistan	13	0.3287	201,995,540	919,000	951	301	2,924	197	7.0	Taiwan	13	10.7
Indonesia	14	0.3347	258,316,051	975,950	441	39	418	221	6.9	Israel	14	17.0
Israel	15	0.3476	8,174,527	718,250	652	243	2,620	65	15.5	Australia	15	26.1
Vietnam	16	0.3587	95,261,021	5,488,500	278	76	1,545	65	3.4	Thailand	16	5.4
Brazil	17	0.3654	205,823,665	1,987,000	697	43	469	110	24.5	Poland	17	9.4
Taiwan	18	0.3765	23,464,787	1,932,500	850	286	2,005	87	10.7	Germany	18	40.2
Poland	19	0.3831	38,523,261	184,650	465	99	1,065	83	9.4	Indonesia	19	6.9
Thailand	20	0.3892	68,200,824	627,425	555	76	737	81	5.4	Canada	20	15.7
Iran	21	0.3933	82,801,633	934,000	477	137	1,616	398	6.3	----	----	----
Australia	22	0.4072	22,992,654	81,000	465	78	59	47	24.1	----	----	----
North Korea	23	0.4218	25,115,311	6,445,000	944	458	5,025	967	7.5	----	----	----
Saudi Arabia	24	0.4302	28,160,273	256,000	790	177	1,142	55	56.7	----	----	----
Algeria	25	0.4366	40,263,711	792,350	502	89	2,405	85	10.6	----	----	----

Source: <https://www.businessinsider.com/most-powerful-militaries-in-the-world-ranked-2018-2>

Notes: ^a Global Firepower's ranking pays particular attention to the manpower; ^b The Credit Suisse Report calculates the military strength based on the following weights of total score: active personnel (5%), tanks (10%), attack helicopters (15%), aircraft (20%), aircraft carriers (25%), and submarines (25%).

* Number of aircraft carriers out of total naval assets; United States (19), Russia (1), China (1), India (3), France (4), United Kingdom (2), Japan (4), Egypt (2).

The United States has more aircraft carriers than all top 24 countries combined; 19 compared to 17. Also the military spending of the 20 countries in the aggregate is \$1.32 trillion whereas the United States' military budget alone is \$601 billion which amounts to 45.5\$.

The United States' military power is unmatched; it has more aircrafts (i.e. 13,762) than 24 countries are combined and spends nearly three times more than China's \$216 billion defense budget (see Table 10). Therefore, the U.S. government will use its military power to do the expected and the unthinkable to protect its national currency (and its *exorbitant privilege*), even if it means fighting multiple wars at home and abroad. In recent years, some developments (i.e. rising China, cryptocurrencies, Russia as a game-changer etc.) have made the U.S. government (the Fed) feel agitated; for this reason, it has reiterated its position that it will not allow anything to diminish the dollar's power.

2.3.2 Massive Dollarization: A Weapon of Mass Economic Destruction

Persistent dollarization and the United States' weaponization of dollar as a foreign policy continues to dampen any potential upward trajectory in Bitcoin price and the cryptocurrency market. The post-war monetary order, which was drafted with only American view in mind by the U.S. senior economic advisor Harry Dexter White and was overwhelmingly accepted at the Bretton Woods Conference by the delegates from the 44 wartime allies. The two watchdog institutions (i.e. IMF and World Bank) also created in 1944 serve the American imperialism (i.e. modern version of the Roman Empire). The influential British economist John Maynard Keynes' UK plan was also rejected by the same delegates who accepted the US plan; Keynes had serious doubts about the new order's ability to solve interwar problems, those from the Great Depression, and unforeseeable ones in the future. Unfortunately, this inflationary monetary order's main nutrition is debt victims (i.e. excessive indebtedness by people, entities, and sovereign states), and its survival relies on a reload of new peripheries (Table 11).

Table 11: List of periphery countries

 Bangladesh	 Benin	 Bolivia	 Burkina Faso	 Burundi
 C. African Republic	 Chad	 Chile	 China	 Congo
 Gambia	 Ghana	 Guinea-Bissau	 Haiti	 Honduras
 India	 Indonesia	 Kenya	 Lesotho	 Madagascar
 Malawi	 Mauritania	 Nepal	 Niger	 Nigeria
 Papua New Guinea	 Philippines	 Rwanda	 Senegal	 Sierra Leone
 Solomon Islands	 Sri Lanka	 Sudan	 Togo	 Zambia

Source: Babones (2005), https://en.wikipedia.org/wiki/Periphery_countries

Unfavorable macro events both in the 1970s (i.e. Yom Kippur aka Arab-Israeli War of 1973; forced liquidation of Bank Herstatt in 1974; OPEC oil shock/energy crises in 1973-74) and the 1980s (*lost decade*⁴⁶) were enough to ignite the sparks of fear, accelerate the pace of dollarization, and prompt

⁴⁶ Latin American debt crisis; savings and loan (S&L) crisis in the U.S.; US stock market crash of 1987, and more.

the formation of currency-unions in the 1990s (a bipolar view). With the WWII peripheries (i.e. Japan, Germany in particular) graduating to center, the new peripheries' (developing nations and emerging economies) gradual integration into international trade hubs gained momentum (Rose, 2000; Frankel & Rose, 1996). Consequently, their elongated addiction to dollar assets and dollar denominated short term debt triggered a huge spike in dollar's share of global trade and international reserves. Following the Asian crisis of 1997-98 (first systemic crisis), dot.com crisis (2001), sub-prime mortgage debacle (2006) and the GFC of 2008, financially weak countries began piling up dollars because strong foreign reserve position was perceived as a positive indication of financial stability; furthermore, it was a sign of country's ability to repay its debt obligations both in the short-term and long-term.

Table 12: List of semi-periphery countries

 Algeria	 Argentina	 Brazil	 China	 Egypt	 Finland	 Greece
 India	 Indonesia	 Israel	 Iran	 Italy	 Malaysia	 Mexico
 Norway	 Poland	 Portugal	 S. Arabia	 Singapore	 S. Korea	 Spain
 Nigeria	 Taiwan	 Turkey	 Venezuela	 Zaire		

Source: Wallerstein (1976), https://en.wikipedia.org/wiki/Semi-periphery_countries

In addition to multilateral⁴⁷ watchdog institutions implicitly supporting American imperialism and dollar's hegemony, strategic commodities are priced in dollars after the 1945 agreement between the U.S. and Saudi Arabia created the petrodollar (President Franklin D. Roosevelt initiated the alliance with Saudi King Abd al-Aziz). To support dollar's global reserve status, the U.S. has more gold than any nation, valued at \$470 billion with the price of \$1,800 per ounce of gold (Table 13).

Table 13: World gold mine production survey 2018

	Gold Mine Production 2016 (tons)	Gold Mine Production 2017 (tons)	Gold Reserves (tons)	Total Reserves (US\$ bn)	Gold as a Share of Reserves %
United States	222.0	230.0	8,134	452.8	75.2
Germany	-----	-----	3,374	200.6	70.4
IMF	-----	-----	2,814	-----	-----
Italy	-----	-----	2,452	151.6	67.7
France	-----	-----	2,436	156.8	65.1
Russian Federation	253.6	270.7	1,910	433.1	17.8
China Mainland	453.5	426.1	1,843	3,236.0	2.4
Switzerland	-----	-----	1,040	811.2	5.4
Japan	6.5	6.1	765	1,264.3	2.5
Turkey	26.0	26.1	565	107.8	21.9

Source: Taskinsoy (2019a); GFMS, Thomson Reuters

⁴⁷ The IMF and the World Bank were established in 1944 (began operations in 1945), United Nations (UN) in 1945, the General Agreement on Tariffs and Trade (GATT) in 1947 (replaced by the World Trade Organization – WTO in 1995), and the North Atlantic Treaty Organization (NATO) in 1949

Reasons for holding dollars may vary for individuals, companies, or sovereign states; nevertheless, virtually all countries throughout the world are forced to hold dollars as reserve assets either to hedge against macroeconomic instability or to pay for imports of certain commodities such as livestock and meat, energy, industrial and precious metals (Table 14).

Table 14: List of traded key commodities

Commodity	Contract Size	Currency	Main Exchange
Energy			
WTI Crude Oil	1000 bbl. (42,000 U.S. gal)	USD \$	NYMEX, ICE
Brent Crude	1000 bbl. (42,000 U.S. gal)	USD \$	ICE
Ethanol	29,000 U.S. gal	USD \$	CBOT
Natural gas	10,000 million BTU	USD \$	NYMEX
Natural gas	1,000 therms	USD \$	ICE
Heating Oil	1000 bbl. (42,000 U.S. gal)	USD \$	NYMEX
Gulf Coast Gasoline	1000 bbl. (42,000 U.S. gal)	USD \$	NYMEX
Propane	1000 bbl. (42,000 U.S. gal)	USD \$	NYMEX, ICE
Industrial Metals			
LME Copper	Metric Ton	USD \$	London Metal Exchange, New York
Lead	Metric Ton	USD \$	London Metal Exchange, New York
Zinc	Metric Ton	USD \$	London Metal Exchange
Tin	Metric Ton	USD \$	London Metal Exchange
Aluminum	Metric Ton	USD \$	London Metal Exchange
Aluminum alloy	Metric Ton	USD \$	London Metal Exchange, New York
LME Nickel	Metric Ton	USD \$	London Metal Exchange
Cobalt	Metric Ton	USD \$	London Metal Exchange
Molybdenum	Metric Ton	USD \$	London Metal Exchange
Precious Metals			
Gold	100 troy ounces	USD \$	COMEX
Platinum	50 troy ounces	USD \$	NYMEX
Palladium	100 troy ounces	USD \$	NYMEX
Silver	5,000 troy ounces	USD \$	COMEX

Source: Wikipedia, https://en.wikipedia.org/wiki/List_of_traded_commodities

Various levels of additions to dollar are linked to instability; whether partially or fully dollarized⁴⁸, or liability dollarization⁴⁹ is the case, economies and governments being addicted to dollar aggravate instability when it is not properly managed. In a stable economy (i.e. advanced nations), dollars may still be used by residents and entities, but not in the form of an addiction that will lead to instability. The levels and reasons for dollar addiction are varied and disparate, here are some examples;

- Dollar is the official currency of the United States and other countries (i.e. Panama and Liberia are fully dollarized); except the U.S., virtually all countries need dollars to pay for imports of most commodities which are priced in dollars (i.e. oil, gold, steel. Aluminum, etc.).

⁴⁸ In a partially dollarized economy, domestic residents hold dollars for personal or business reasons, dollar-denominated loans, assets, and investment instruments are available. In a dollarized economy, unstable national currency may no longer be issued, or replaced by a more stable foreign currency (dollar).

⁴⁹ Excessive borrowing of dollar by the private and public sectors and households.

- The US dollar is the main international reserve currency since the 1944 Bretton Woods Agreement which was ratified by the delegates from 44 Allied countries; as of 2020, most central banks have stashed circa 60% of their foreign reserves in U.S. dollars (i.e. over \$7 trillion in Q3 – 2021).
- As the strongest and most stable currency of the world, a large portion of the international capital flows and foreign borrowing instruments are dollar-denominated⁵⁰, i.e. over \$12 trillion in 2020.
- Developing and emerging economies need a constant flow of foreign capital in order to leap into a higher level of economic performance, and since the majority of international capital flows are in dollar, therefore they need dollars to finance fiscal deficits along with infrastructure projects.
- People, companies, and governments worldwide prefer to invest in the U.S. financial markets due to their size and security aspects; the combined January 2022 market cap of the three largest U.S. equity indices and bond market was nearly \$150 trillion; Dow Jones Industrial Average – DJIA⁵¹ consisting of only 30 companies is the most popular but not the largest (\$11 trillion), Russell 3000 index (\$47.7 trillion), and S&P 500 index⁵² (\$42.4 trillion), and US bond market⁵³ (\$46 trillion).
- The degree of dollarization in developing and emerging economies spread widely in the 1970s and 1980s, but has risen since the late 1990s; on account of inflation and macroeconomic imbalances, households, private and public sectors have heavily relied on dollar's role as a stable medium of exchange and store of value. Following the 2008 global financial crisis, the Basel Committee on Banking Supervision introduced Basel III in December 2010 (fully effective by Jan. 2019), higher capital and liquidity rules of which required banks to put aside more reserves (capital buffers).

Reinhart et al. (2003) classified the dollarized economies into four types (Table 15 and 16);

Table 15: Varieties of dollarization

	Private sector debt accounts for ten percent or more of total external debt.	Private sector debt accounts for less than ten percent of total external debt.
At least ten percent of broad money or of domestic public debt are denominated in or linked to a foreign currency.	Type I	Type II
Less than ten percent of broad money and of domestic public debt are denominated in or linked to a foreign currency.	Type III	Type IV

Source: Reinhart et al. (2003)

⁵⁰ Barron's. "Dollar-Denominated Debt Outside U.S. Hits \$12.6 Trillion, and Two More Numbers to Know." <https://www.barrons.com/articles/dollar-denominated-debt-outside-u-s-hits-12-6-trillion-51596186002>

⁵¹ https://en.wikipedia.org/wiki/Dow_Jones_Industrial_Average

⁵² https://en.wikipedia.org/wiki/S%26P_500

⁵³ <https://www.forbes.com/sites/kevinmcpartland/2018/10/11/understanding-us-bond-market/?sh=1978a31f1caf>

Table 16: Degrees of dollarization: composite scores, 1996-2001

	Composite Index Level	No. of countries in category	Countries
Very High		16	
Type I	25	1	Ecuador
	22	1	Bolivia
	21	1	Uruguay
	20	1	Argentina
	19	1	Bulgaria
	17	2	Lao, Nicaragua
	16	2	Angola, Peru
	15	2	Cambodia, Paraguay
	14	5	Guinea-Bissau, Lebanon, Mozambique, São Tomé & Príncipe, Zambia
High		34	
Type II	13	6	Bosnia & Herzegovina, Ghana, Honduras, Jordan, Tajikistan, Turkey
	12	8	Congo DR, Croatia, Guinea, Indonesia, Malawi, Sierra Leone, Tanzania, Yemen
	11	4	Kyrgyz Republic, Mongolia, Russia, Vietnam
	10	5	Bahrain, Côte d'Ivoire, Jamaica, Moldova, Philippines
	9	11	Armenia, Belarus, Costa Rica, El Salvador, Estonia, Georgia, Hungary, Pakistan, Thailand, Turkmenistan, Uganda
Moderate		32	
Type III	8	9	Egypt, Israel, Latvia, Lithuania, Macedonia, Papua New Guinea, Romania, St. Kitts and Nevis, Ukraine
	7	9	Brazil, Chile, Czech Republic, Guatemala, Haiti, Hong Kong, Kazakhstan, Malaysia, Slovak Republic
	6	6	Azerbaijan, Mauritius, Poland, Trinidad and Tobago, United Arab Emirates, Venezuela
	5	5	Albania, Colombia, Mexico, Solomon Islands, Uzbekistan
	4	3	Saudi Arabia, Slovenia, South Korea
Low		8	
Type IV	3	1	Kuwait
	2	5	China, Fiji, Netherlands Antilles, Singapore, South Africa
	1	1	Taiwan
	0	1	Oman

Source: Reinhart et al. (2003)

2.3.3 Identity Crisis: Currency, Asset, Commodity, or Thin Air

Bitcoin had a clear identity and vision given by its creator Satoshi Nakamoto (2008) who designed Bitcoin with a supranational currency in mind (once suggested by Keynes at the 1944 Bretton Woods Conference), i.e. stateless (no government), no trusted third-party intermediation (removing banks and credit card companies), and no de facto Fed-like authority (central bank) to set its value, control and manipulate it by the use of repeated tight money (contractionary) and expansive (quantitative

easing) monetary policies. The identity crisis came into play when the opponents in consort with politicians, financial authorities, law makers, and regulators stoked by fears that Bitcoin's increasing popularity (i.e. its craze or mania) would impact the dollar-centric financial system and the control the Fed asserts over money. Although the U.S. dollar has enjoyed its hegemonic status (exorbitant privilege) for more than half a century, Bitcoin supporters began to believe that the dollar's days were numbered and Bitcoin (highly secure, permissionless – decentralized network of electronic cash) could actually become a viable alternative to the U.S. dollar (Cermak, 2017).

In less than a decade, Bitcoin became a household name (Wallace, 2011), “a cultural and financial phenomenon” (Wolla, 2018) has been on the super-highway in terms of both technological evolution and digital transformation. The CEO of Coinbase claims that Bitcoin will replace the U.S. dollar as the world's main reserve currency.⁵⁴ Despite Bitcoin's vast popularity, some traditional economists still find it difficult to get a grasp on the Bitcoin phenomenon (see Sovbetov, 2018). According to Franco (2014), Bitcoin may never become a real currency for everyday use because special knowledge is prerequisite to gain an understanding of three elements about Bitcoin; software (cryptography), hardware (computer network), and the underlying infrastructure technology (blockchain). Yermack (2013) contends that Bitcoin is a currency, while Berentsen and Schär (2018) argue that Bitcoin has applications suitable to become an important asset class, but not a currency. Baek and Elbeck (2015) investigate if Bitcoin is an investment asset or a speculative vehicle; along the same line, a Goldman Sachs executive feels Bitcoin is more of an asset than a currency.⁵⁵

San Francisco Fed President John Williams (2011-18) asserted that Bitcoin fails three basic functions of money such as medium of exchange, store of value, and unit of account (Bitcoin as a real currency, see David, 2014; Davies, 2002 for history of money).⁵⁶ Bitcoins are used as a medium of exchange for goods and services purchased online (e.g. Dwyer, 2014; Eichengreen, 2009; Kelly, 2014). It is true that currently Bitcoin is not used in everyday transactions or accepted by everyone like fiat money; however, the count of merchants and nations accepting Bitcoin as a currency is increasing rather fast. Bitcoin serves as a store of value (for the main drivers of Bitcoin price, see Kristoufek, 2015); unlike fiat currencies, Bitcoin is a decentralized crypto (or digital) currency, therefore its value is not set by a central bank (i.e. Fed). Bitcoin's value comes from trust, acceptance, and a degree of speculation; just like gold or any other commodity, its value appreciates or depreciates based on demand and what people are willing to pay for it. However, extreme volatility can have a damaging effect in Bitcoin's store of value aspect (Katsiampa, 2017; Souza et al., 2017; Wallace, 2011; Wolla, 2018). Bitcoin with

⁵⁴ <https://99bitcoins.com/coinbase-ceo-claims-bitcoin-will-replace-dollar-as-world-reserve-currency/>

⁵⁵ <https://www.technologyreview.com/s/610783/bitcoin-would-be-a-calamity-not-an-economy/>

⁵⁶ <http://fortune.com/2018/04/20/bitcoin-new-york-fed-central-bank-jack-dorsey/>

lower case “b” (bitcoin) is a unit of account; mysterious creator⁵⁷ Nakamoto (2008), in his white paper titled “*Bitcoin: A Peer-to-Peer Electronic Cash System*”⁵⁸, describes bitcoins as a digital cash unit of the peer-to-peer network. Like dollars (unit of the dollar system), bitcoin is divisible (eight decimals) and its smallest denomination is called a *Satoshi* (i.e. 0.00000001 BTC).

Over the past several decades, there have been numerous attempts to kill off the dollar’s smile, but these efforts so far have been a failure. Except euro (20.26% of foreign reserves, see Table 17), other attempts have only made a trifling dent on dollar’s hegemonic status that it has enjoyed since WWII. Bitcoin can certainly try to deflate the dollar’s imperious status as the world’s main reserve currency, but the list of contenders is growing larger with the newly arrived Chinese yuan (2016) as a reserve currency (1.84% of reserves).⁵⁹ Unless backlash to Bitcoin stops, this paper share the view that it is impossible for a fiat or a crypto-currency to become a viable alternative to the U.S. dollar; besides, due to its extreme volatility (erratic price fluctuations), no central banks will appear to be overeager to hold Bitcoin in their reserves. Not omitting Bitcoin’s record-setting performance; unfortunately, Bitcoin can’t become a global reserve currency based on just significant price appreciation.

Table 17: Currency composition of official foreign exchange reserves (COFER)

Category	Q3 2020	Q4 2020	Q1 2021	Q2 2021	Q3 2021
Total Foreign Exchange Reserves	12,250.97	12,705.33	12,583.07	12,813.74	12,827.45
Allocated Reserves	11,459.20	11,864.53	11,730.64	11,946.27	11,973.21
Claims in U.S. dollars	6,927.23	6,990.97	6,971.79	7,065.05	7,081.39
Claims in euro	2,359.64	2,526.41	2,404.80	2,446.82	2,452.10
Claims in Chinese renminbi	247.44	271.60	293.32	314.24	318.99
Claims in Japanese yen	668.19	715.35	686.30	688.11	697.35
Claims in pounds sterling	523.64	561.39	554.28	571.55	572.76
Claims in Australian dollars	199.51	216.87	214.89	220.77	217.19
Claims in Canadian dollars	231.10	246.57	250.01	267.81	262.17
Claims in Swiss francs	19.30	20.74	19.44	20.25	20.42
Claims in other currencies	283.15	314.63	335.82	351.69	348.87
Unallocated Reserves	791.77	840.80	852.43	867.47	856.22

Source: IMF, <https://data.imf.org/?sk=E6A5F467-C14B-4AA8-9F6D-5A09EC4E62A4>

Although Bitcoin has overcome many obstacles and challenges since its debut in January 2009, it is still heavily criticized. Most of the criticism, assertions, claims and concerns fall under four categories;

⁵⁷ Controversial entrepreneur Craig Wright claims to be bitcoin inventor Satoshi Nakamoto.

⁵⁸ See Nakamoto (2008) for helpful references, available online: <https://bitcoin.org/bitcoin.pdf>, also see www.bitcoin.org. Satoshi describes a mathematical problem of double spending with finite number of solutions, in fact, 21 million (which is also the total number of bitcoins arbitrarily set to be mined until the year 2140). This aspect of bitcoin (i.e. for a free-floating cryptocurrency) meets the scarcity element that is required to create real value (i.e. gold and diamonds).

⁵⁹ The IMF added Chinese yuan to the SDR basket effective October 1, 2016. Based on this update, the SDR basket consists the following amounts of each currency: U.S. dollar (0.58252), Euro (0.38671), Chinese yuan (1.0174), Japanese yen (11.900), Pound sterling (0.085946). <https://www.imf.org/en/News/Articles/2016/09/30/AM16-PR16440-IMF-Launches-New-SDR-Basket-Including-Chinese-Renminbi> (accessed on December 11, 2018).

decentralization, price volatility, energy use and illicit activities. The latter earned Bitcoin a bad image and prompted government authorities in many countries to call for a greater scrutiny of Bitcoin on terrorism-financing, money-laundering, illegal drugs (Silk Road⁶⁰) and human trafficking (Bonneau et al., 2015; Gudgeon et al., 2019; Christin, 2013). As of 2018, 24 countries have issued a ban on Bitcoin and other cryptocurrencies, but mainly on Bitcoin (i.e. 15 implicit and 9 absolute).

When blockchain pioneer and cryptographer Nick Szabo spoke at the Israel Bitcoin Summit⁶¹ at Tel Aviv University on 8 January 2019, he said that “the use of censorship-resistant cryptocurrencies will rise in countries sanctioned from trade and economies suffering from failed monetary planning”. Conversely, former U.S. President Donald Trump claims that not only Bitcoin’s value is based on “thin air”, but it has “little standing or dependability”.⁶² However, advocates of Bitcoin believe that Trump and those who share his short-sighted view of cryptocurrencies fear Bitcoin because they do not have a clue of the inner workings of money’s natural evolution. Szabo also argued that the many of the world’s central banks will eventually turn to cryptocurrency reserves in near future in order to supplement national gold reserves”. He also asserted that “even central banks may attempt to hoard Bitcoin reserves over gold”. For a longer discussion, see Andolfatto (2018); Chiu and Wong (2014); Duffie (2019); Fung et al (2014); Kahn and Roberds (2009); Kocherlakota (1998). Reitman (2019) argues that running headlong into backlash to Bitcoin and other cryptocurrencies⁶³ can backfire and cause inadvertent negative effects on the developments of future innovative technologies.

Today, Bitcoin is a legal tender only in El Salvador; the country’s president Nayib Bukele made Bitcoin a national currency on 7 September 2021 alongside the U.S. dollar. Although many consider Bukele’s decision one of his boldest moves, this is actually what Bitcoin needed because its identity crisis will only continue if no presidents other than Bukele overcome their fears of the U.S. and its weaponized dollar and make a bold move. Unfortunately, backlash to Bitcoin is not just common to the Trump administration (2016-20), here are criticisms by prominent industry experts⁶⁴; JP Morgan CEO Jamie Dimon said Bitcoin fits perfect illicit activities and “it is worse than tulip bulb”; Jack Bogle (Vanguard founder) advised people to “avoid Bitcoin like a plague”; Andreas Treichi (Erste Group Bank AG)

⁶⁰ On October 1, 2013 the FBI shut down the black market website Silk Road and seized its assets including 26,000 bitcoins.

⁶¹ Redman (2019), <https://news.bitcoin.com/nick-szabo-central-banks-may-turn-to-cryptocurrency-reserves-over-gold/> (January 9, 2019), Economics (Accessed August 14, 2019).

⁶² <http://thenextweb.com/hardfork/2019/07/12/donald-trumps-critique-of-bitcoin-is-antiquated-and-irrelevant/>

⁶³ Facebook announced in 2019 that its stablecoin project dubbed Libra would be launched in June 2020 as a permissioned (not permissionless such as Bitcoin) where billions of transactions would be governed by the 28 heavyweight validator-firms (publicly-traded companies), each of which is a member of the Libra Association. But due to the intensifying war (backlash) against Libra, first Facebook rebranded Libra as Diem in 2020 and even it has changed its own company name from Facebook to Meta, but these efforts (three-year odyssey) were not enough to stop lawmakers and regulators in the United States (Trump and his Fed) and Europe (ECB). Sadly, shortsighted politicians and their blindfolded followers stoked by fears won the war against cryptocurrencies by derailing the Libra (or Diem) project.

⁶⁴ <https://coinpedia.org/information/experts-criticisms-bitcoin-cryptocurrencies/>

argues that Bitcoin “will make central banks lose control”; Jim Kramer (CNBC) points to great resemblance between Bitcoin and monopoly money; Tony Robbins (self-made millionaire) feels Bitcoin is “very iffy” and “I look at that as it’s like going to Vegas” David Gledhill (CIO at DBS) believes Bitcoin is a “Ponzi scheme”; Howard Marks (co-chairman of Oaktree Capital) called it “an unfounded fad (or perhaps even a pyramid scheme)”; and Michael Novogratz (former Fortress hedge fund manager) said that Bitcoin mania (bubble) is fueled by “a lot of froth and fraud”. According to San Francisco Fed President John Williams, Bitcoin does not meet functions of money such as medium of exchange, store of value, and unit of account (also see Kristoufek, 2015).

Bitcoin’s acceptance as an asset, commodity, property, or security contributes negatively to the status of Bitcoin; furthermore, treating Bitcoin as such for tax purposes (see Table 18) only aggravates its identity crisis. Is there a simple solution? The answer to the question differs greatly depending on who you ask; for Bitcoin advocates, the answer is “yes” but unfortunately, Bitcoin is still banned in countries like Algeria, Bolivia, Ecuador, Bangladesh, Nepal, and Macedonia (see Figure 25).

Table 18: Developments & acceptance of cryptocurrencies in countries

Key Developments	Cryptocurrencies Declared as Legal		
Japan legalized cryptocurrencies and set up a Payment Services Act.	Morocco	Iran	Slovakia
	Nigeria	Lebanon	Switzerland
	Namibia	Turkey**	United Kingdom
US government accepted Bitcoin in 2013 and classified it as a commodity in 2015.	South Africa	India*	Australia
	Zimbabwe	Pakistan	New Zealand
	Canada*	Japan	Ireland
	Mexico	Hong Kong	Netherlands
Germany legalized Bitcoin and its citizens are allowed to trade and transact in this coin.	United States	Taiwan	Belgium
	Costa Rica	South Korea	France
	Nicaragua	Indonesia	Luxembourg
	Trinidad and Tobago	Philippines	Greece
France legalized the operation of virtual currencies in July 2014.	Jamaica	Cambodia	Italy
	Brazil	Malaysia	Bulgaria
	Argentina	Thailand*	Bosnia and Herzegovina
Malta accepts Bitcoin and other cryptocurrencies as a legal mode for digital transactions.	Colombia	Singapore	Malta
	Chile	Vietnam**	Spain
	Kyrgyzstan	Croatia	Portugal
	Cyprus	Germany	Sweden
Canada legalized cryptocurrency in August 2017.	Israel	Poland	Iceland
	UAE	Austria	Norway
	India	Czech Republic	Denmark
Belarus legalized cryptocurrency in March 28, 2019.	Jordan	Romania	Ukraine
	Saudi Arabia	Slovenia	

Source: Yahoo Finance⁶⁵; Investopedia⁶⁶; Wikipedia⁶⁷

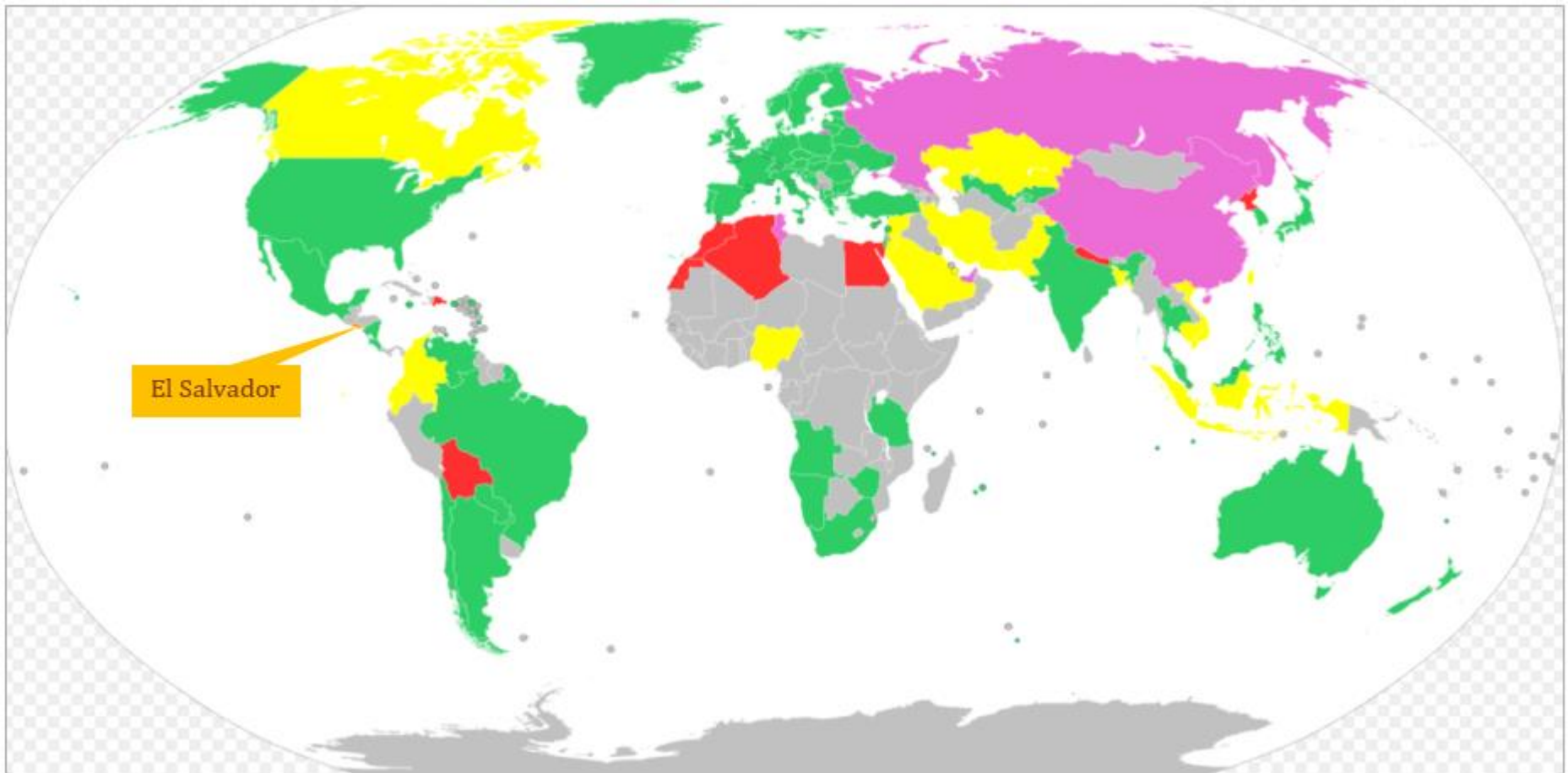
Notes: * banned by banks; ** not allowed as a payment tool

Following countries have banned cryptocurrencies: Algeria, Bolivia, Ecuador, Bangladesh, Nepal, Macedonia

⁶⁵ <https://finance.yahoo.com/news/countries-banned-cryptocurrencies-why-210500613.html>

⁶⁶ <https://www.investopedia.com/articles/forex/041515/countries-where-bitcoin-legal-illegal.asp>

⁶⁷ https://en.wikipedia.org/wiki/Legality_of_bitcoin_by_country_or_territory



Source: Wikipedia, https://en.wikipedia.org/wiki/Legality_of_bitcoin_by_country_or_territory#/media/File:Legal_status_of_bitcoin.svg

Figure 25: World map of Bitcoin status

- Legal tender: El Salvador
- Permissive (legal to use bitcoin): European Union and G7, Albania, Angola, Argentina, Austria, Australia, Bangladesh, Belgium, Belarus, Brazil, Brunei, Bulgaria, Cambodia, Canada, Chile, China, Colombia, Costa Rica, Croatia, Cyprus, Czech Republic, Denmark, Ecuador, El Salvador, Estonia, Finland, France, Germany, Greece, Hong Kong, Hungary, Iceland, India, Iran, Ireland, Italy, Jamaica, Japan, Jordan, Indonesia, Israel, Kyrgyzstan, Lebanon, Luxembourg, Malaysia, Malta, Mauritius, Mexico, New Zealand, Namibia, Netherland, Nicaragua, Nigeria, Norway, Pakistan, Philippines, Poland, Portugal, Qatar, Romania, Russia, Saudi Arabia, Singapore, Slovakia, Slovenia, South Africa, South Korea, Spain, Sweden, Switzerland, Taiwan, Tanzania, Thailand, Trinidad and Tobago, Turkey, United Kingdom, United States, Uzbekistan, Venezuela, Vietnam, and Zimbabwe.
- Contentious (some legal restrictions – banking ban): Bangladesh, Canada, China, Colombia, Cambodia, Ecuador, Indonesia, Iran, Jordan, Nigeria, Qatar, Russia, Saudi Arabia, Taiwan, Turkey (prohibited as a payment tool), and Vietnam.
- Contentious (interpretation of old laws, bitcoin is not illegal): Russia
- Hostile (full or partial prohibition): Algeria, Bolivia, Egypt, Morocco, and Nepal.

2.3.4 Inherent Flaws – Deficiencies of Bitcoin Blockchain

When mysterious creator Satoshi Nakamoto launched Bitcoin in January 2009 as the first successful cryptocurrency (alpha coin), this also meant the start of a new phase in money's evolution. At first, Bitcoin seemed to have a superior technology (blockchain) and no inherent flaws or deficiencies; but as time has passed, altcoins (Bitcoin's prodigies) with better designed blockchains and a lot faster transaction speeds have sprouted like wild mushrooms (21,542). While Bitcoin blockchain processes 7 transactions per second (TPS), its competitor altcoins are a lot faster; as such, Ethereum (20), Dash (48), Litecoin (56), Bitcoin Cash (60), Ripple (1,500), OES (3,000), and Stellar (3,000), only Solana is capable of processing 50,000 TPS which is twice more than the credit card giant Visa that handles 24,000 TPS (i.e. Bitcoin blockchain would take 119.04 and 57.14 minutes to process that volume).

In the early days, Bitcoin and blockchain were misunderstood; therefore, people often used these two terms interchangeably, like the chicken and egg problem (it is hard to decide whether blockchain gave life to Bitcoin or vice versa). Regardless of mounting criticism and claimed flaws, i.e. low scalability (rigorous PoW algorithm via a mining process), high latency (a block is completed about every ten minutes), and high energy cost, Bitcoin blockchain is still considered to be a technology revolution which made Bitcoin become the most popular and dominant cryptocurrency in the world.

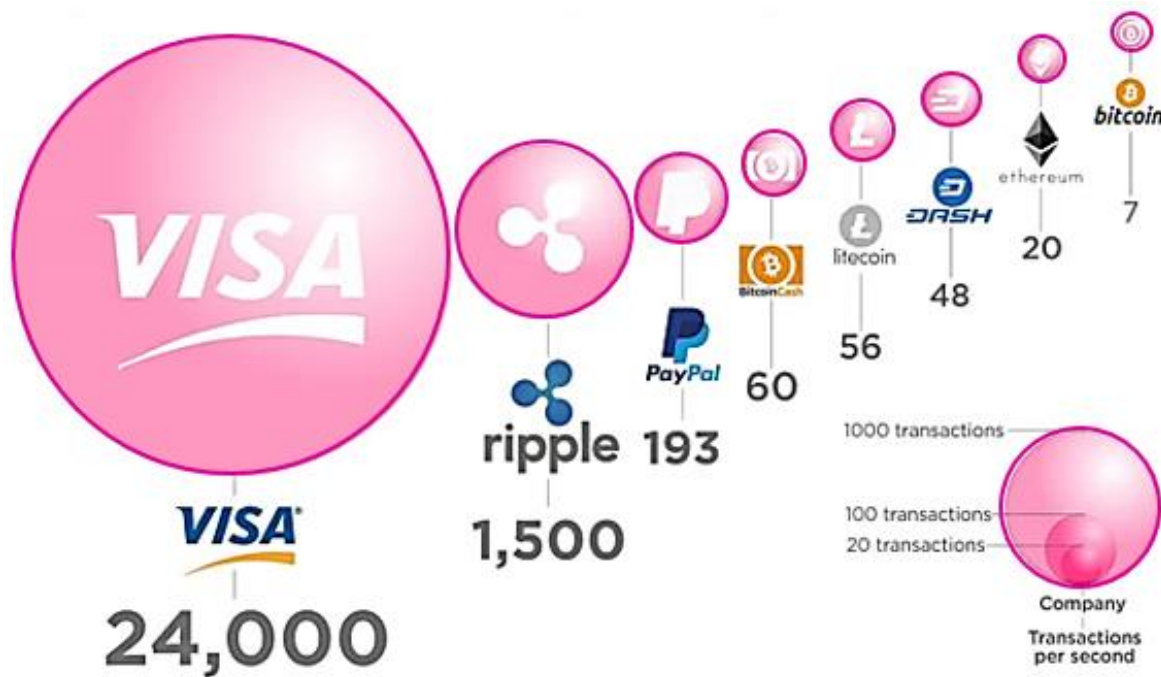
Low Scalability of Bitcoin Blockchain

Bitcoin blockchain, a decentralized – permissionless network of electronic cash, heavily relies on PoW (proof-of-work) using cryptography, this is the main cause of Bitcoin's low-scalability which makes per transaction cost higher. PoW uses a consensus approach, meaning all nodes in the network must reach a consensus as to what transactions are valid. While there is no issue of Bitcoin scaling to mass users (i.e. millions of participants), its blockchain cannot simply scale to thousands of transactions per second (TPS). Of course, this is not a problem without solutions; Bitcoin blockchain can be scaled via hard fork (in 2017, Bitcoin was split into two separate chains, Bitcoin and Bitcoin Cash, while Bitcoin handles 7 TPS (see Figure 26), Bitcoin Cash's scalability improved to 60 TPS). Another solution to Bitcoin's scalability problem is to move transactions off the main blockchain.

In terms of fast scalability, there are better cryptocurrencies than Bitcoin and Ethereum; for example, Solana (SOL) is capable of processing circa 50,000 TPS, this amounts to an improvement of 7,142-fold over Bitcoin, 3,125-fold over Ethereum and twice more than Visa's 24,000 TPS⁶⁸. A larger number of transactions means a lower cost per transaction; currently, no cryptocurrency can match Solana's

⁶⁸ <https://www.fool.com/investing/2022/01/19/5-of-the-fastest-scaling-cryptocurrencies/>

\$0.00025 per transaction fee. In terms of payment-focused networks, no better cryptocurrency than Stellar (XLM), although it's 3,000 TPS is lower than Solana, but XLM's average transaction fee is only \$0.00001 which means a user can execute about 350,000 transactions for \$1 in fees compared with SOL's 4,000 transactions per \$1. On Stellar network (blockchain), sending/receiving money and the validation (settlement) take less than 5 seconds, as opposed to days in the current financial system.⁶⁹



Source: Seeking Alpha⁷⁰

Figure 26: Cryptocurrencies transaction speeds compared with Visa and PayPal

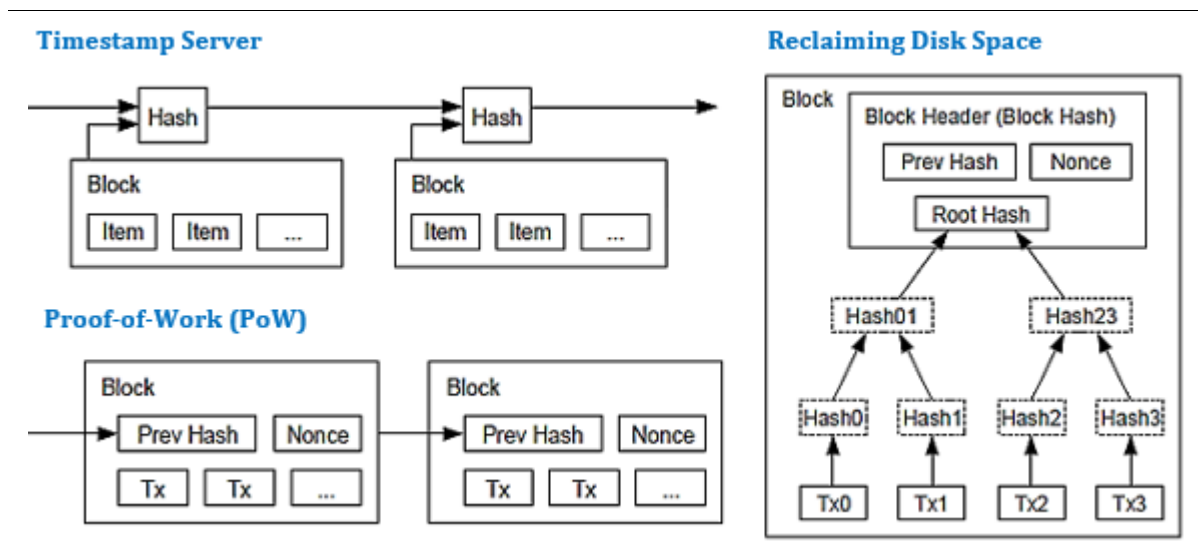
Bitcoin blockchain uses a consensus approach to ensure high security (temper-proof and resistant to attacks, i.e. Bitcoin blockchain can function even if 50% of the network is compromised, but 51% attack is a problem). In the proof-of-work (PoW) hashing scheme which was derived from Adam Back's Hashcash (Back, 2002) using SHA-256 hash function, transactions in a chronological order are timestamped and hashed in a Merkle tree (Merkle, 1980, 1987; Eastlake & Hansen, 2011) that verifies and validates the accuracy of the hash value in the block header (see Massias et al., 1999; Haber & Stornetta, 1991). When the hashed data is confirmed as valid and added to a block (chained), from this point on forward, the data is inalterable and can never be changed, modified, or deleted (Ben-Sasson et al., 2014; Buchman et al., 2018; Yin et al., 2018).

⁶⁹ Top five fast-scalable cryptocurrencies are: Solana - SOL (50,000 TPS), Avalanche - AVAX (4,500 TPS), Stellar - XLM (3,000 TPS), IOTA - MIOTA (1,000 TPS), and Ripple - XRP (1,500 TPS).

⁷⁰ https://static.seekingalpha.com/uploads/2021/5/1/saupload_crypto-transactions-compared-ce3b_thumb1.jpg

High-Latency of Bitcoin Blockchain

By design, high latency (inherent flaw) is an Achilles heel in Bitcoin blockchain; on average, verifying (i.e. conforming and validating) transactions takes around 8 minutes (low of 5 and high of 20 minutes) which does not resonate with requirements of the modern financial system and makes Bitcoin unfit to compete against other very fast-scalable cryptocurrencies (covered in the previous section) that also have extremely low latency. The latency problem was not a software bug that has surfaced after Bitcoin was launched, it was in fact the proof-of-work (PoW) that Satoshi Nakamoto made it central (heart) of Bitcoin design (Figure 27) where the level of difficulty of the hashing algorithm is constantly adjusted so that a new block can only be created every 10 minutes. Unfortunately, Bitcoin's very high latency cannot be resolved without making foundational changes to its blockchain; for instance, one solution may be moving away from both hideous and time-consuming PoW consensus approach; another solution could be taking the validation process off the blockchain so that Bitcoin's TPS can be improved significantly, nevertheless these solution may still not be enough to compete with the transaction speed of Solana (50,000 TPS) or XML's average transaction fee of \$0.00001.



Source: Nakamoto (2008)

Figure 27: The overview of Bitcoin protocol

Finding the correct hash is easier said than done⁷¹, i.e. guessing the right 67-digit number (below the threshold target) among possibilities of all 67-digit numbers is about 45,000 times more difficult than winning the lottery jackpot with a single ticket. Furthermore, the difficulty level gets adjusted (i.e. easier or harder depending on the number of miners in the network) for every two weeks (or after a completion of 2,016 blocks, whichever comes first).

⁷¹ <https://www.investopedia.com/terms/b/bitcoin-mining.asp> JULIUS MANSA

Different methods⁷² are used under POS⁷³ to prove ownership of cryptocurrency (i.e. stake), and the argument in support of POS is that the likelihood of an attack by those with higher stakes (i.e. owning a large amount of coins) is less likely. POS is certainly better and more economical than PoW, but POS may lead to unfairness and domination by a single person or entity with the largest holding of coins; to avoid this, POS uses random selection along with stake size and coin age (i.e. older coins with larger stake), which was employed by Blackcoin (see Vasin, 2014) and Peercoin respectively (King & Nadal, 2012; Vukolić, 2015). Another consensus algorithm used by blockchains is Byzantine Fault Tolerant, which is based on military combat situation where generals must decide whether attack or retreat (Lamport et al., 1982). Variations of BFT have been developed; as such, a replication of BFT as Practical Byzantine Fault Tolerance (PBFT)⁷⁴, Stellar Consensus Protocol (SCP), Delegated Byzantine Fault Tolerance (dBFT); for a longer discussion, see Mazieres (2015), Miguel and Barbara (1999).

High Energy Cost of Bitcoin Blockchain

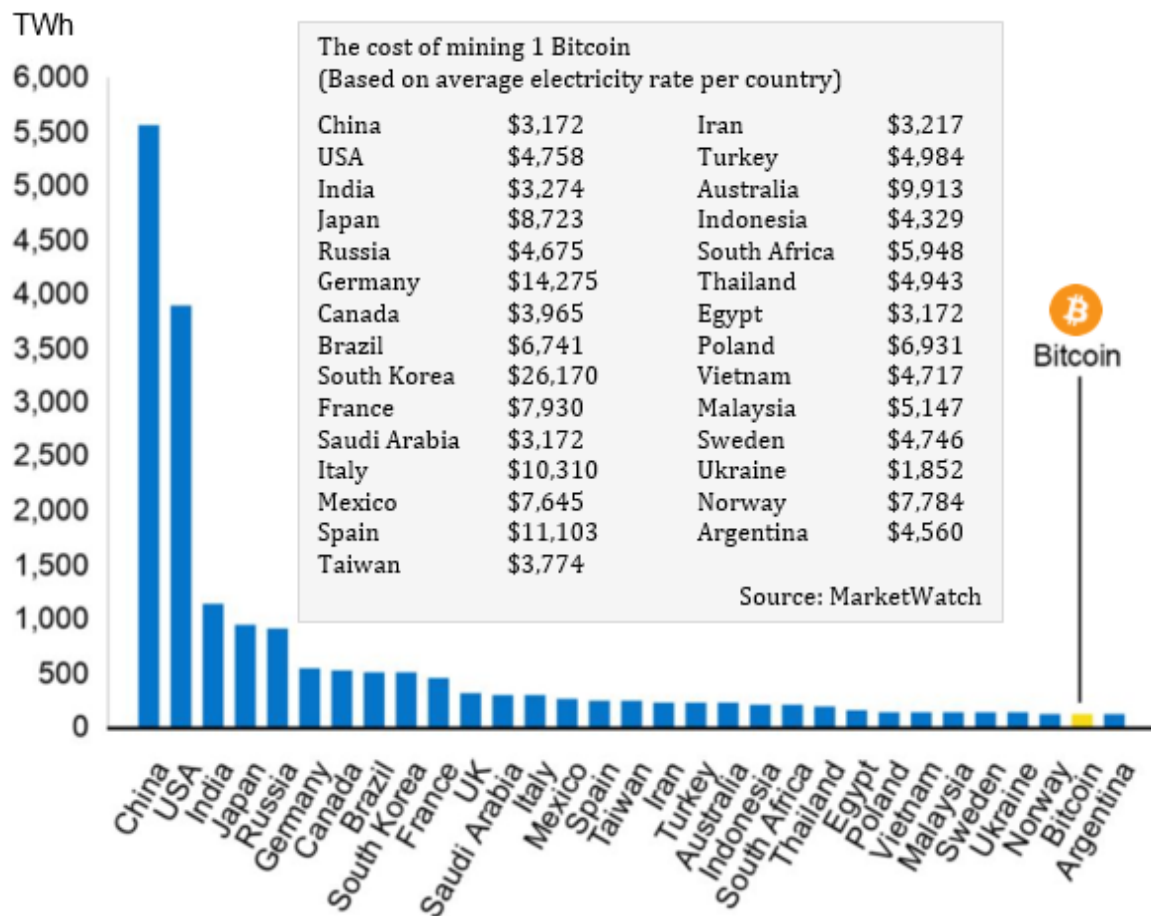
Electric cost for mining (minting) a single bitcoin varies significantly among countries; for example, the highest is in South Korea (\$26,170) and lowest is in Ukraine (\$1,852), but on average \$6,191. As illustrated in Figure 28, Bitcoin mining uses around 160 billion kilowatt (or 160 TWh) of electricity annually, which exceeds the power consumption of many countries including Argentina. During the years of 2016-19, about 150 bitcoins were mined per hour (circa 3,000 per day) but almost half of the mined bitcoins were sold to cover huge energy cost. The four pillars of the Bitcoin blockchain design such as the Proof-of-Work (PoW) consensus protocol, cryptography, mining (minting) process and the distributed ledger technology (DLT) are the sources of Bitcoin's massive energy consumption.

Bitcoin and Ethereum account for more than 95% of total global crypto-asset electricity usage, 68% and 28% respectively, which leaves only about 2% consumed by other 20,000 cryptocurrencies that are not based on PoW consensus mechanism. Again these two cryptocurrencies' electricity usage (2/3 by Bitcoin and 1/3 by Ethereum) equals to about 1.4% of total U.S. annual electricity consumption, and this is enough to light up about 140 million homes or power up that many computers in the U.S.

⁷² Also see the Ripple Protocol Consensus Algorithm, here subnetworks (i.e. server for consensus and client for the transfer of funds). Each server has a unique node list (UNL), for consensus purposes, UNLs with 80% or higher are included in the ledger and UNLs with less than 20% are considered to be faulty (see Schwartz et al., 2014). Although Bitcoin is mineable, out of the top-100 list by market capitalization, about two-thirds or 64 cryptocurrencies are non-mineable such as Ripple XRP, Binance Coin, Tether, EOS, Stellar, UNUS SED LEO, Chainlink, Tezos, NEO, and IOTA. Consensus protocol without mining used by Tendermint involves three stages; pre-vote step, pre-commit step, and the final commit step; after rounds of voting, a block that gets two-thirds or more of commits is accepted and published as a new block (see Kwon, 2014).

⁷³ Delegated proof of stake (DPOS) is a close variation of POS; the difference between the two is while in POS the selection is random whereas in DPOS blocks are generated and validated by nodes selected by stakeholders like a corporation. After the first proof of stake verification on Ethereum platform different methods of verification have emerged; "proof of authority" (PoA), "proof of importance" (PoI) and "proof of history" (PoH).

⁷⁴ PBFT protocol was used in Hyperledger project Available: <https://www.hyperledger.org/>



Source: University of Cambridge Bitcoin Electricity Consumption Index⁷⁵; slightly modified

Figure 28: Bitcoin uses more electricity than Finland and Argentina

Environmental Hazard of Bitcoin Blockchain

Global warming phenomena is not a fluke, a science-fiction, or something that is taking place in a distant future, it is with us now and we must deal with it before it is too late. Global warming's most emblematic impact on climate change (i.e. melting glaciers and polar ice) is clearly visible wherever we look (Oerlemans, 2005). The exponential growth in crypto-assets since the debut of Bitcoin in January 2009 and their massive electricity usage (95% by Bitcoin and Ethereum) makes the global only get worse, which could result in significant environmental impacts (hazards).

As illustrated in Figure 29, the Earth warms up as greenhouse gases trap the heat mainly caused by human-induced emissions from burning fossil fuels, livestock farming, and deforestation (Barnett et al., 2005). As the Earth's surface and lower atmosphere warm up further (Hassol, 2004; Hansen et al.,

⁷⁵ <https://www.bbc.com/news/technology-56012952>

2005), the global warming's adverse impacts on climate change are increasingly more visible⁷⁶ (Stott et al., 2004). The average global temperature has increased by between 0.6 °C and 0.8 °C since 1900s (Industrial Revolution); sea level rose by 10 to 20 cm; Arctic ice thickness decreased by 40%; Mount Kenya and Mount Kilimanjaro lost more than 80% of their ice mass (Kaser et al., 2004; Oerlemans, 2005); and 40% to 60% of total available water in basins of Niger, Lake Chad and Senegal evaporated (McCarthy et al., 2001; Pounds et al., 1999). For more details on global warming's visible impacts, see IPCC (2001), NCDC (2004), O'Neill and Oppenheimer (2002), UNEP (2003), and Wigley (2005).



Source: Climate Atlas of Canada; <https://climateatlas.ca/greenhouse-gases>

Figure 29: The greenhouse gas effect

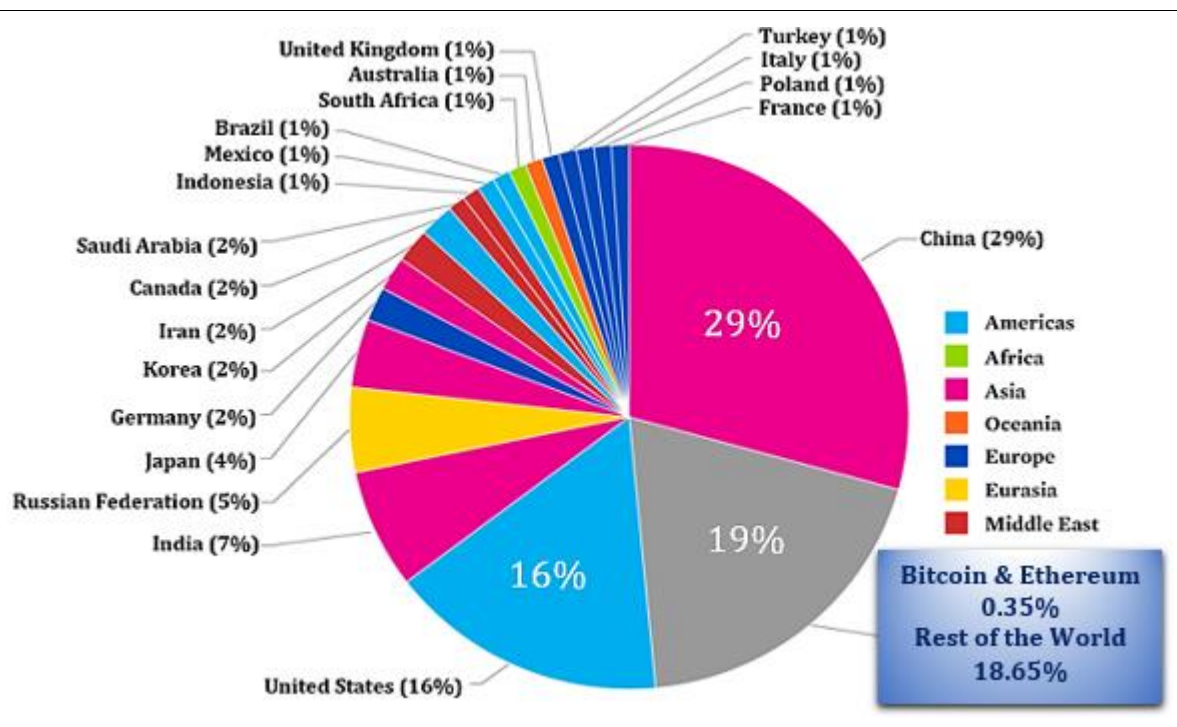
Weather events in 12 millennia, i.e. from 10,000 BC to 1700, pretty much followed seasonal patterns, therefore they were predictable; however, increased human-induced greenhouse gas emissions due to the burning of vast amounts of fossil fuels along with crypto-assets (Table 19 and Figure 30) have pushed the Earth's natural systems (balanced carbon cycle) out of balance causing extreme climate changes in the new millennium. For the world population to reach the first billion (1803) took literally thousands of years, but getting to 2 billion took only 127 years (1803-1927); and after 1960, 1 billion was added to the world population in every 13-14 years (i.e. world population increased 7-fold in 216 years). If global warming continues at its current accelerated pace (Figure 31) and no precautions are taken in the next decade (2030s), scientists warn that by 2050 climate change due to global warming could result in severe consequences in every facet of human life.

⁷⁶ All time high temperatures broke records in many countries (ranged between 40 °C and 50 °C) which led to financial losses in sectors like agriculture, fishery, forestry, and electric power plants. If the acceleration of greenhouse gases continue by 2050, the farfetched implications of killer heat waves could increase 100-fold.

Table 19: Emissions by country

Rank	Country	Emissions in 2017 (MtCO ₂)	% of Global Emissions
1	China	10,877	29.34%
2	United States	5,107	13.77%
3	India	2,454	6.62%
4	Russia	1,765	4.76%
5	Japan	1,321	3.56%
6	Germany	797	2.15%
7	Iran	671	1.81%
8	Saudi Arabia	639	1.72%
9	South Korea	673	1.82%
10	Canada	617	1.66%
11	Mexico	507	1.37%
12	Indonesia	511	1.38%
13	Brazil	493	1.33%
14	South Africa	468	1.26%
15	Turkey	430	1.16%
	Bitcoin & Ethereum ⁷⁷	130	0.35%
	Total	27,460	74.06%
	World	37,077	100.00%
	Rest of World	9,617	25.94%

Source: Wikipedia; https://en.wikipedia.org/wiki/List_of_countries_by_carbon_dioxide_emissions

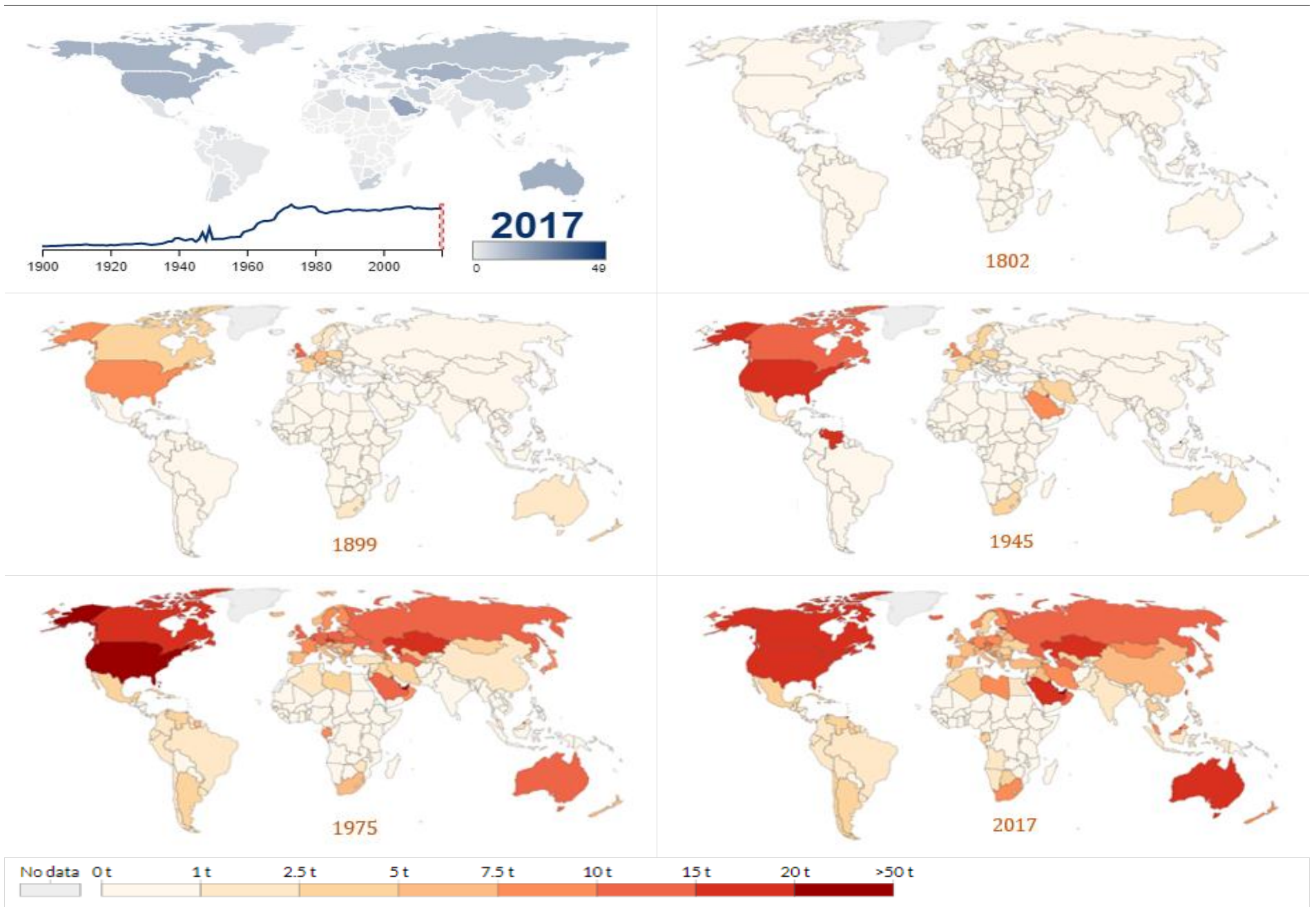


Source of Data: Union of Concerned Scientists⁷⁸

Figure 30: Largest greenhouse gas emitter countries

⁷⁷ <https://www.whitehouse.gov/ostp/news-updates/2022/09/08/fact-sheet-climate-and-energy-implications-of-crypto-assets-in-the-united-states/#:~:text=As%20of%20August%202022%2C%20published,such%20as%20Argentina%20or%20Australia.>

⁷⁸ <https://www.ucsusa.org/resources/each-countrys-share-co2-emissions>



Source: Adapted from OWID based on CDIAC; Global Carbon Project; Gapminder & UN; <https://ourworldindata.org/per-capita-co2>

Figure 31: CO₂ Emissions per capita 1802 – 2017

Bitcoin is more than a cryptocurrency, a digital coin, or a network of electronic cash as its mysterious creator Satoshi Nakamoto described it; it would have been 15th largest as a nation and 5th largest as a corporation (Table 20). Generally, currencies do not have a carbon footprint – a measure of the total greenhouse gas emissions (i.e. primarily carbon dioxide and methane), but Bitcoin does (79.38 MtCO₂ per annum, see Figure 32). Similar to developed nations with higher carbon footprints and higher CO₂ emissions, PoW based Bitcoin and Ethereum networks consume 95% of electricity of all crypto-assets, they are also the biggest GHS-emitters among cryptocurrencies because the vast electricity used for mining and validating transactions is generated by burning large amounts of fossil fuels.

Table 20: Bitcoin as new 17th largest economy

Rank	Country	Nominal GDP	Rank	Company	Market Cap	Unit
1.	United States	21,428	1.	Apple	2,341.2	USD Billion
2.	China	14,343	2.	Microsoft	1,828.8	USD Billion
3.	Japan	5,082	3.	Amazon	1,673.3	USD Billion
4.	Germany	3,846	4.	Alphabet	1,408.0	USD Billion
5.	India	2,875	5.	Bitcoin	1,217.0	USD Billion
6.	United Kingdom	2,827	6.	Tesla	818.4	USD Billion
7.	France	2,716	7.	Facebook	759.1	USD Billion
8.	Italy	2,001	8.	Alibaba	710.5	USD Billion
9.	Brazil	1,840	9.	Taiwan Semi	687.4	USD Billion
10.	Canada	1,736	10.	Berkshire Hathaway	557.3	USD Billion
11.	Russia	1,700	11.	Visa	442.2	USD Billion
12.	South Korea	1,642	12.	Johnson & Johnson	434.2	USD Billion
13.	Spain	1,394	13.	JPMorgan Chase	427.2	USD Billion
14.	Australia	1,393	14.	Walmart	410.3	USD Billion
15.	Bitcoin	1,217*		NVIDIA	357.5	USD Billion
16.	Mexico	1,200	15.	Disney	344.7	USD Billion
17.	Indonesia	1,119	16.	Mastercard	336.0	USD Billion
18.	Netherlands	909	18.	PayPal	330.6	USD Billion
19.	Saudi Arabia	793	19.	Procter & Gamble	318.1	USD Billion
20.	Turkey	754	20.	UnitedHealth	307.7	USD Billion
21.	Switzerland	703	21.	Home Depot	301.5	USD Billion

Source of data: Trading Economics; <https://tradingeconomics.com/country-list/gdp?continent=g2>

Notes: * market capitalization of \$1.217 trillion in November 2021

Bitcoin and Ethereum (or any other cryptocurrencies) cannot continue to exist with their design flaws that are environmentally hazardous (large amounts of electricity usage and CO₂ emissions). Instead of stubbornly insisting on using PoW (and despite its adverse impacts on global warming and climate change), like the competition, both Bitcoin and Ethereum can reduce their energy consumption by more than 90% if they shift from PoW to POS or other approaches for validating transactions.

Annualized Total Bitcoin Footprints	Single Bitcoin Transaction Footprints
Carbon Footprint 79.38 Mt CO2  Comparable to the carbon footprint of Oman.	Carbon Footprint 868.18 kgCO2  Equivalent to the carbon footprint of 1,924,191 VISA transactions or 144,697 hours of watching Youtube.
Electrical Energy 167.11 TWh  Comparable to the power consumption of Poland.	Electrical Energy 1827.75 kWh  Equivalent to the power consumption of an average U.S. household over 62.65 days.
Electronic Waste 23.09 kt  Comparable to the small IT equipment waste of the Netherlands.	Electronic Waste 252.63 grams  Equivalent to the weight of 1.54 iPhones 12 or 0.52 iPads. (Find more info on e-waste here.)

Source: DIGICONOMIST, <https://digiconomist.net/bitcoin-energy-consumption/>

Figure 32: Bitcoin's annual and single transaction footprints

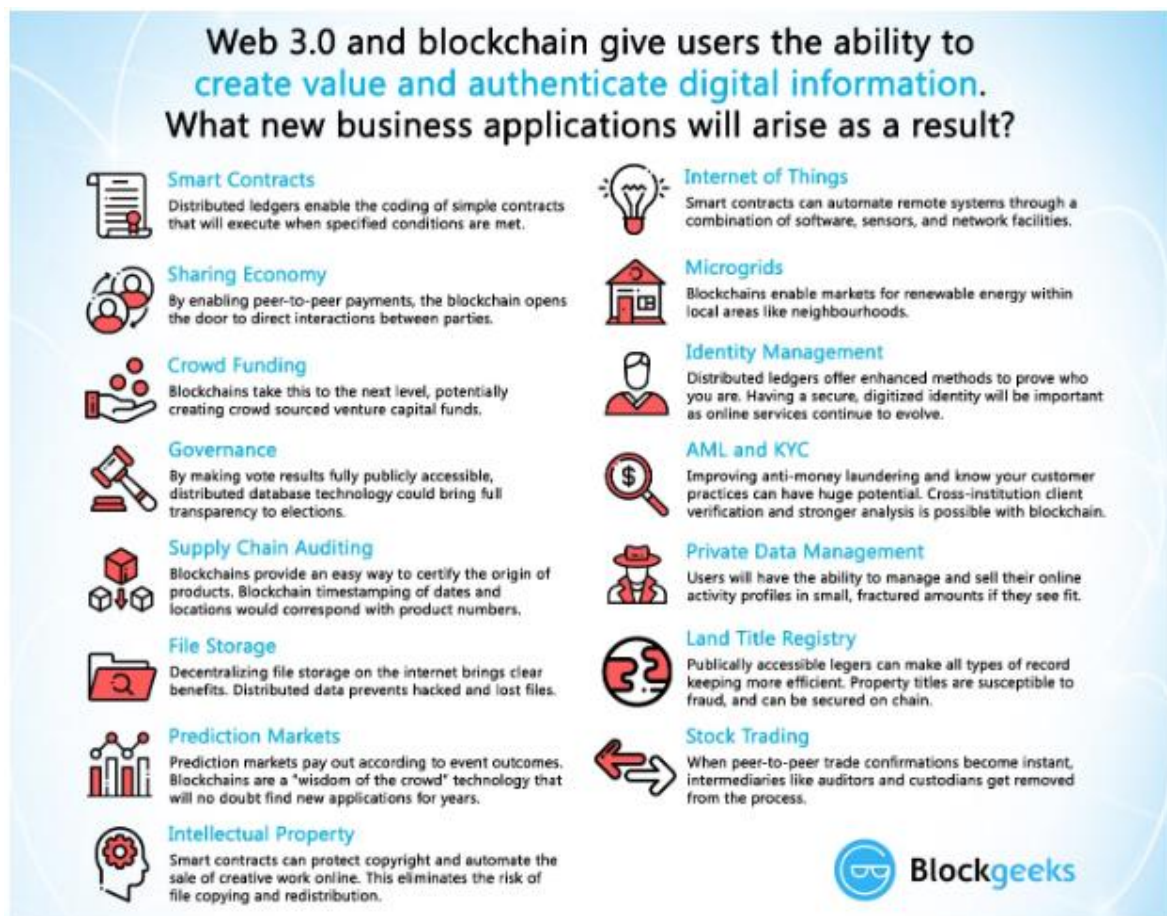
In terms of the rankings of carbon footprints, the combined CO₂ emissions of Bitcoin and Ethereum ranks 38th in the world (Table 21), which means that energy-intensive PoW-based blockchains of these two highly popular crypto-assets emit more CO₂ than each of about 180 countries. As nations have pledged under the Paris Agreement to reduce their carbon footprints, Bitcoin and Ethereum must initiate design related changes (i.e. taking out consensus based validation off their blockchains) to become carbon neutral. While the U.S. and China are the two largest emitters of CO₂ in 2020 (32% and 12.6% respectively), not surprisingly, they also have larger cryptocurrency markets.

Table 21: Country rankings of carbon footprints

Country	2020 CO ₂ Emissions (Mt)	2017 CO ₂ Emissions (Mt)	2020 Emissions Per Capita	2017 Emissions Per Capita	2022 Population
China	11680.42	10877.22	8.2	7.7	1,425,887,337
United States	4535.3	5107.39	13.68	15.7	338,289,857
India	2411.73	2454.77	1.74	1.8	1,417,173,173
Russia	1674.23	1764.87	11.64	12.3	144,713,314
Japan	1061.77	1320.78	8.39	10.4	123,951,692
Iran	690.24	671.45	8.26	8.3	88,550,570
Germany	636.88	796.53	7.72	9.7	83,369,843
South Korea	621.47	673.32	12.07	13.2	51,815,810
Saudi Arabia	588.81	638.76	16.96	19.4	36,408,820
Indonesia	568.27	511.33	2.09	1.9	275,501,339
Canada	542.79	617.3	14.43	16.9	38,454,327
Brazil	451.8	492.79	2.11	2.4	215,313,498
South Africa	435.13	467.65	7.41	8.2	59,893,885
Mexico	407.7	507.18	3.05	3.9	127,504,125
Turkey	405.2	429.56	4.83	5.3	85,341,241
Australia	386.44	402.25	15.22	16.5	26,177,413
Vietnam	321.93	218.73	3.27	2.3	98,186,856
United Kingdom	313.73	379.15	4.66	5.7	67,508,936
Italy	297.35	361.18	5.03	6.1	59,037,474
San Marino	297.35	361.18	5.03	6.1	33,660
Vatican City	297.35	361.18	5.03	6.1	510
Poland	292.56	319.03	7.71	8.4	39,857,145
Taiwan	280.56	279.74	11.78	11.8	23,893,394
France	279.99	338.19	4.26	5.2	64,626,628
Monaco	279.99	338.19	4.26	5.2	36,469
Egypt	269.55	258.67	2.62	2.7	110,990,103
Kazakhstan	267.1	266.21	14.22	14.6	19,397,998
Malaysia	262.17	258.78	7.98	8.2	33,938,221
Thailand	255.46	279.3	3.68	4	71,697,030
Pakistan	217.02	197.3	1.04	1	235,824,862
Spain	214.85	282.36	4.62	6.1	47,558,630
Andorra	214.85	282.36	4.62	6.1	79,824
UAE	203.14	202.8	20.7	21.6	9,441,129
Iraq	191.3	199.3	4.61	5.2	44,496,122
Ukraine	189.3	205.72	4.34	4.7	39,701,739
Argentina	176.51	209.97	3.88	4.7	45,510,318
Algeria	163.47	159.93	3.77	3.9	44,903,225
Netherlands	144.69	174.77	8.42	10.3	17,564,014
Bitcoin/Ethereum	130.00	110.00	6.19	5.24	21,000,000

Source: <https://worldpopulationreview.com/country-rankings/carbon-footprint-by-country>

Before a broader adoption of blockchain for new business applications occur (Figure 33), the negative impacts (i.e. electronic waste) of crypto-asset mining (primarily Bitcoin, and Ethereum as the second largest CO₂ emitter) must be reduced significantly via powering with clean electricity and ultimately becoming carbon neutral before 2030. As the U.S. government is very keen on achieving its National Determined Contribution under the Paris Agreement, a greater scrutiny will be placed on Bitcoin and other cryptocurrencies, but this should not turn into a case of running headlong to political backlash to Bitcoin, this is exactly what happened under the Trump administration (2017-21).



Source: Blockgeeks; <https://blockgeeks.com/guides/what-is-blockchain-technology/>

Figure 33: Blockchain applications

To reach climate goals of the U.S. and the world, and at the same time for a smooth transition of crypto-assets' to clean energy, a collaborative process should be initiated with the cryptocurrency market and various government agencies (at federal, state, and local); furthermore, technical assistance as to impacts on global warming and climate change should be provided by the Environmental Protection Agency (EPA), the Department of Energy (DOE), and other relevant agencies. Bitcoin (and altcoins) is

an inevitable outcome of money's natural and unabated evolution which has accelerated with COVID-19 (coronavirus) pandemic; facts for moving towards a digital global economy⁷⁹;

- 104 million consumers have cards in a digital format.
- 47% of consumers are very interested in using digital cards in stores in the next 12 months.
- 30% of consumers like to access their cards via digital wallets.
- 58% of multinational firms use at least one type of cryptocurrency.
- 10% of financial institutions currently provide access to at least one form of cryptocurrency.
- 73% of financial institutions plan to expand access to cryptocurrencies over the next 12 months.
- 63% of cryptocurrency holders are highly interested in using coins to make online purchases.
- 45% of consumers say cryptocurrencies would make transactions more private.
- 59% of cryptocurrency owners are interested in online purchases resulting in discounts.
- 49% of consumers are highly interested in receiving online banking services.
- 25% of consumers are concerned about security for not using digital-only banking services.
- 62% of consumers want digital-only banks to offer debit cards.
- 28% of international eCommerce merchants use IP recognition technology.
- 6.8 payment methods are supported by eCommerce merchants for cross-border transactions.
- 10.8 currencies eCommerce merchants support for cross-border transactions.
- 73% of big-ticket buyers arranged for their payments offline.
- 55% of millennials are highly interested in buying big-ticket items on digital platforms.
- 59% of millennials would be highly interested in purchasing real estate via digital platforms.
- 68% of payers are willing to pay a fixed fee to offer instant payments for free.
- 67% of microbusinesses say free instant payment options would boost their loyalty as clients.
- 20% of consumers are willing to pay to receive instant nongovernment disbursements.
- 67.6% of consumers were able to pay all their bills on time in Q3 2018.
- 59% of digital application users in large cities or urban areas use digital wallets.
- 92% of U.S. consumers who have placed an online order.
- 60% of consumers (148 million people) want to use contactless payments at the point of sale.
- 32% of U.S. and U.K. businesses plan to automate cross-border receivables in the next 3 years.
- 44.8% of account holders use online and mobile channels for their banking.
- 75% of bank customers plan to maintain digital banking habits they adopted during the pandemic.
- 89% of all 2020 holiday shoppers made at least one purchase online.
- 45% of U.S. shoppers have shifted to digital channels to buy retail products.
- 16% net increase in online use of credit cards since the pandemic's onset.
- \$114 billion is the estimated value of the European mobile wallet market by 2023.
- 4 billion U.K. consumer transactions were powered by open banking in 2020.
- 68% of adults pay for gas with mobile apps and buy it as often as once per week.
- 57% of consumers consider digital payment options when deciding where to shop.
- 24.7% year-over-year increase in consumers shopping and paying for food via digital channels.

⁷⁹ For more details and stats, check out PYMNTS.com at <https://www.pymnts.com/study/>

3.0 Concluding Remarks

Bitcoin is still the world's most popular cryptocurrency (just as dollar is still the world's main reserve currency), but its domination by market cap has been facing the same predicament of the U.S. dollar in terms of its hegemony in international reserves; of course, there are differences. After declaring its independence on 4 July 1776 from Great Britain (13 original colonies) and securing it by winning the American Revolutionary War (April 19, 1775 – September 3, 1783), Congress officially adopted the dollar "\$" sign in 1785, and on 2 April 1792, Congress passed "*The Mint Act*" which established the decimal system for currency. When the U.S. dollar was introduced in the late 18th century onward, there were already widely circulated foreign coins and paper money such as British pound (shillings and pence), Spanish dollar, Portuguese real, and to a lesser extent Mexican peso. On the other hand, when Nakamoto launched Bitcoin in January 2009 as the world's first successful cryptocurrency, altcoins did not exist and the dollar was its biggest competition. Even some digital coins emerged by 2014, Bitcoin's dominance stood at over 96%; as of 8 November 2022 following the 2nd severe crash, it stands at little over 38% which is almost identical to that of the first major crash (Figure 34).



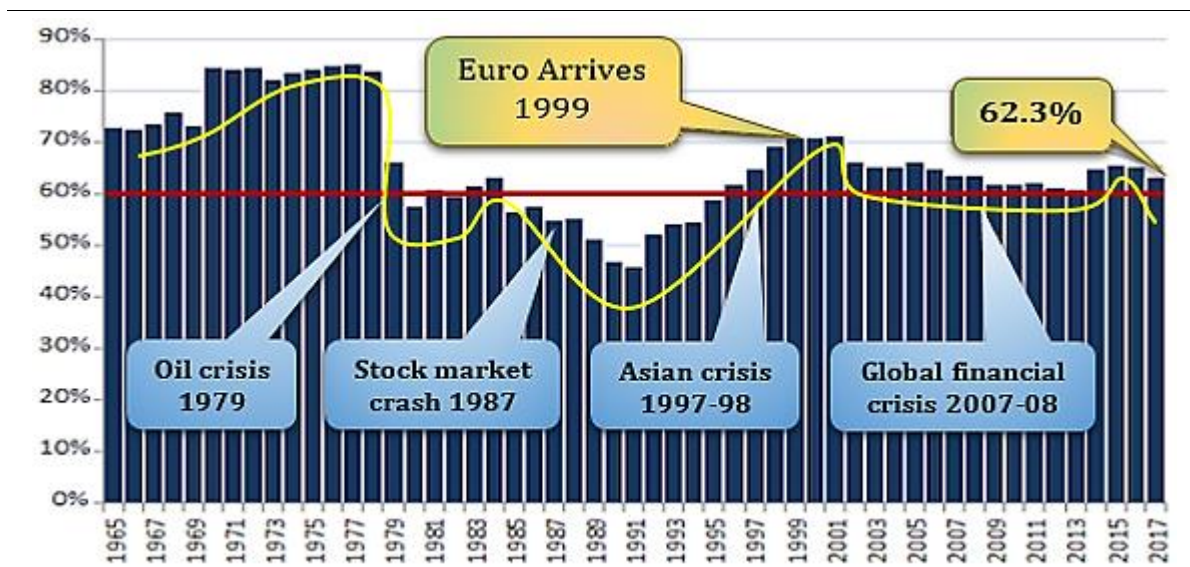
Source: <https://coinmarketcap.com/charts/>

Figure 34: Market capitalization of Bitcoin and other cryptocurrencies

Notes: November 2013, Bitcoin 96.33% (others 4.67%); June 2017, Bitcoin 39.17% (Ethereum 27.47%, others 21.01%); January 2021, Bitcoin 71.89% (Ethereum 10.66%, other 11.56%); November 5, 2022 Bitcoin 38.67% (Ethereum 18.84%, others 18.01%, Tether 6.56%, Binance 5.28%, USD Coin 3.97%, XRP 2.34%, Binance USD 2.13%, Dogecoin 1.56%, Cardano 1.39%, and Solano 1.25%).

As illustrated in Figure 34, the competition among cryptocurrencies has become significantly intense; but in terms of the competition between U.S. dollar and Bitcoin, dollar is a clear winner. Since dollar's hegemony was inaugurated at the 1944 Bretton Woods Conference, and its main reserve currency status in the post-WWII dollar-centric new monetary order was solidified further in May 1971 as US

president Nixon cut the dollar's link to gold which was commonly referred to as the “*Nixon shock*”, the dollar's share of allocated global reserves is still slightly over 62% (Figure 35). Despite unfavorable macroeconomic events, the Fed's policy errors (unjustified expansive and contractionary policies), as well as countless efforts to kill off, deflate, or end the dollar's hegemony; the dollar has been managing it all quite well. Unfortunately, the same conclusion cannot be made for Bitcoin, at least not for now.

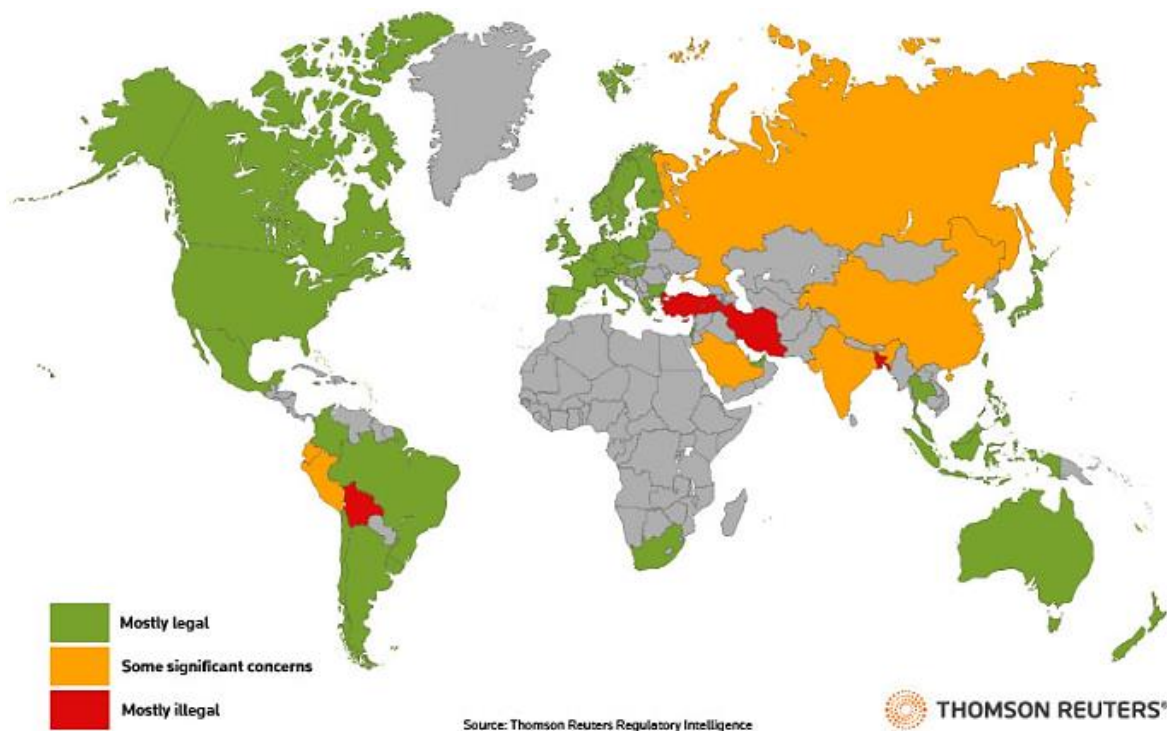


Source: Author; IMF: <http://data.imf.org/?sk=E6A5F467-C14B-4AA8-9F6D-5A09EC4E62A4>

Figure 35: USD share of allocated global official reserves

The launch of euro on 1 January 1999 made only a trivial dent on the dollar's hegemony, i.e. the dollar was affected more from the dot.com crisis (2001-02), sub-prime mortgage debacle (2006), and global financial crises (2007-08) than arrival of the euro. However, the story was different for Bitcoin; just two years after Ethereum was launched on 30 July 2015, it was able to capture 27% of the market share, and this was a major blow to Bitcoin's dominance. Except the regulatory hurdle and political backlash to Bitcoin, the design flaws of Bitcoin blockchain (i.e. PoW-based consensus approach) are at the heart of Bitcoin's meager performance against the new entry of fast-scalable cryptocurrencies with far greater TPS. Although it seems rather extremely difficult, solutions exist to fix Bitcoin's design related problems; for instance, switching from a PoW-based consensus mechanism for mining and validating transactions to a POS or another method could probably reduce high-latency from several minutes to few seconds, turn low-scalability to high-scalability, and increase 7 TPS to maybe 100,000 TPS. Moving away from PoW (or moving validation off blockchain) could also reduce Bitcoin's high electricity usage from 60-70% of crypto-assets to less than 1%, which could enable Bitcoin to become carbon neutral before 2030 and power its blockchain through clean energy; aligning with the U.S. goal (Paris Agreement) may inadvertently take care of the non-technical backlash issue.

More than a decade has passed since Satoshi Nakamoto launched Bitcoin in early January of 2009, but heavy criticism, false claims, tight scrutiny, and regulatory uncertainty have never ended, and neither did the resultant identity crisis and extreme price volatility; furthermore, criticism took the form of bashing and political backlash under the Trump administration, which along with its shortsighted politicians and blindfolded followers (stoked by fears) forced Facebook's stablecoin project dubbed Libra (or rebranded as Diem) to derail. True, early on Bitcoin did not have a very positive image due to claims of its association with the dark web (i.e. Silk Road) and its alleged involvement in unlawful activities such as terrorism-financing, money-laundering, illegal drugs and human trafficking, all of which put a tight lid on Bitcoin's true price potential (i.e. \$100,000 or even \$1,000,000?). Even today, there are a number of countries where Bitcoin is either illegal or highly restricted; as such, Algeria, Bangladesh, Bolivia, China, Colombia, Egypt, Iran, and Nepal (Figure 36).



Source: <https://www.thomsonreuters.com/en-us/posts/investigation-fraud-and-risk/crypto-special-report/>

Figure 36: Bitcoin status in countries

The world's nations are categorized in five different stages; 1) advocates (colored in green, steps are taken to legalize and promote digital coins and drive a parity for them); 2) developing (colored in orange, noticeable progress has been made, but still some barriers exist); 3) fence-sitters (colored in grey, governments and law makers are in the wait-and-see mode, no actions have been taken to ban or regulate cryptocurrencies); 4) banned (colored in red, cryptocurrencies are outlawed in these few

countries, people who get caught are subject to fines, prosecution, or imprisonment. In Table 22, the status of Bitcoin in different regions of the world is shown in more detail.

Table 22: Cryptocurrency regulation by country

North America	
Canada	An early adopter and crypto-friendly. The Canada Revenue Authority (CRA) generally treats Bitcoin and other cryptocurrency like a commodity for the purposes of the Income Tax Act.
Mexico	Has a very crypto-friendly jurisdiction. The Mexican government and the financial authority (CNBV) enacted a new set of fintech laws in March 2018. Its largest crypto exchange, Bitsos, has over 1 million users on its platform. Cryptos are treated as intangible assets where gains would be taxed at 30% for corporations and from 2% to 35% for individuals.
United States	Agencies view cryptos differently; according to the SEC, they are securities, the CFTC calls Bitcoin a commodity, and Treasury calls it a currency. The Internal Revenue Service (IRS) defines them as “a digital representation of value that functions as a medium of exchange, a unit of account, and/or a store of value”.
Central & South America	
Argentina	Investing in cryptos is legal but they are not considered legal currency or tender as they are not issued by the government. No regulations, profits are taxable.
Bolivia	Cryptocurrencies such as Bitcoin were banned in 2014. It is illegal to use any kind of currency not issued and controlled by a government.
Brazil	Unregulated. Bitcoin is viewed an asset and therefore is subject to capital gains taxes. The existing AML laws extend to virtual currencies in a few contexts. The Federal Revenue of Brazil has published a document on cryptocurrency taxes.
Chile	Existing regulations are applicable to cryptocurrencies until a regulatory and oversight framework for cryptocurrencies has been developed. The Chilean Internal Revenue Service (SII) has issued legislation on cryptocurrencies in Notice no 963 (May 14, 2018). The SII requires Tax Form 22 for the declaration from the sale of digital/virtual assets, such as cryptocurrencies (for example, bitcoins).
Colombia	No specific legislation on cryptocurrencies. The Banco de la República, the country’s monetary, exchange and credit authority released statements on cryptos warning they are not legal tender or valid investments for supervised entities, and firms are not authorized to advise or manage them. Virtual currencies such as Bitcoin are not money for legal purposes, but income from them is taxed.
Ecuador	Cryptocurrencies are not regulated. In January 2018, the Central Bank of Ecuador informed citizens that Bitcoin is not a means of payment, the purchase and sale of cryptocurrencies - such as bitcoin - through the internet is not prohibited.
Peru	No specific legislation related to cryptocurrencies and no supervision is provided by the Securities Market Agency (SMV), the Banking, Insurance and Pension Fund Manager Agency (SBS), or the Peruvian Central Reserve Bank (BCRP) which has said that cryptocurrencies are not legal tender. Public warnings have been issued about the potential risks of loss in virtual currencies.
Uruguay	No specific legislation on cryptocurrencies. Viewed as Bitcoin friendly, no regulations specifically banning or permitting the use of cryptocurrencies.
Venezuela	Prior to 2018, law enforcement arrested and seized assets of bitcoin miners but has now declared cryptocurrencies such as Bitcoin legal. The government of Venezuela has also created its own cryptocurrency called the Petro, which is backed by the value of Venezuelan oil.

Source: <https://www.thomsonreuters.com/en-us/posts/investigation-fraud-and-risk/crypto-special-report/>

Table 22: Cryptocurrency regulation by country (continued)

Europe	
Germany	One of the first countries to provide legal certainty to financial institutions allowing them to hold crypto-assets. Firms must be licensed with the German Federal Financial Supervisory Authority (BaFin). Germany has signed up to requirements under the EU Fifth Anti-Money Laundering Directive (AMLD5). Germany has defined crypto-assets as a digital representation of value, not issued or guaranteed by a central bank or public authority, with no legal status of currency or money. Cryptoassets can be used as means of exchange or payment, an investment, and can be transferred, stored, and traded electronically.
France	In April 2019, the French National Assembly adopted the Plan d'Action pour la Croissance et la Transformation de Entreprises (PACTE – Action Plan for Business Growth and Transformation) that will establish a framework for digital asset services providers. France's Financial Market Authority (AMF) has adopted new rules and regulations for cryptocurrency service providers and ICOs, related to the (PACTE). The French Ministry of Finance plans to propose new AML/CFT rules related to digital assets.
Italy	Joined the European Blockchain Partnership (EBP) along with 22 other countries in April 2018. The EBP was established to enable member states to work together with the European Commission on blockchain technology. Cryptocurrencies and blockchain are regulated at the legislative level in Italy under Legislative Act no. 90. Cryptocurrencies are a virtual currency used as a medium of exchange for goods and services. AML regulations are based on EU and FATF recommendations.
Spain	A notable early hot spot for cryptocurrencies among EU members, with merchants accepting payments and bitcoin kiosks in the streets. Despite no formal legal status, virtual currencies in Spain are taxable as income and under VAT. In 2021 the Spanish Securities and Exchange Commission, the Comision Nacional del Mercado de Valores (CNMV) and the Bank of Spain issued a joint statement warning of the risks and volatility.
United Kingdom	The UK Financial Conduct Authority (FCA), HM Treasury, and the Bank of England make up the country's Crypto-assets Taskforce. The FCA has created regulations to cover know your customer (KYC), AML and CFT tailored for crypto-assets. It has also created regulations to cover virtual asset service providers (VASPs). Crypto exchanges must register with the FCA unless they have applied for an e-money license. Cryptocurrencies are not considered legal tender and taxes are levied based on activities. The FCA has banned the trading of cryptocurrency derivatives.
Switzerland	The most cryptocurrency-friendly nation in the world. Switzerland's financial markets regulator, the Swiss Financial Market Supervisory Authority (FINMA) has defined licensing requirements for cryptocurrency businesses of all types including bitcoin kiosk operations, and has created requirements for blockchain companies. Cryptocurrency businesses are subject to AML regulations and licensing requirements under FINMA.
Sweden	The Swedish Financial Supervisory Authority (FSA) and the central bank have publicly declared bitcoin as a legal. From a tax perspective they are viewed as an asset, not a currency or cash. The FSA has warned of the risks associated with cryptos and investment products with cryptos as underlying assets such as exchange traded products (ETPs). Sweden's Central Bank, the Riksbanken, has been a leader in developing a digital central bank currency, the e-krona.
Turkey	Cryptocurrencies are not illegal in Turkey, authorities have demanded user information from crypto trading platforms. Turkey's Central Bank has banned the use of cryptocurrencies, and other such digital assets based on distributed ledger technology cannot be used, directly or indirectly, to pay for goods and services. The central bank has said crypto-assets are "neither subject to any regulation and supervision mechanisms nor a central regulatory authority".

Source: <https://www.thomsonreuters.com/en-us/posts/investigation-fraud-and-risk/crypto-special-report/>

Table 22: Cryptocurrency regulation by country (continued)

Asia, Australia and Others	
Australia	In 2018 new laws for digital currency exchange providers operating in Australia were implemented by the Australian Transaction Reports and Analysis Centre (AUSTRAC). Firms are required to register and implement KYC policies, report suspicious transactions, and comply with AML legislation.
Bangladesh	The Bangladesh Central Bank issued warnings in 2014 and 2017 related to transactions in cryptocurrencies and warned violations could be punishable by up to 12 years in jail under existing AML/CFT laws. Despite prohibitions on the use of cryptocurrencies, Bangladesh has proposed a national “blockchain strategy”.
China	Banned financial institutions from dealing in cryptocurrencies in 2013 and later expanded the bans to crypto exchanges, and ICOs. China has been the epicenter for mining because of low electricity costs. Although a ban on crypto mining was considered, in 2019 the government reconfirmed that it would remain legal. In May 2021, China’s Financial Stability and Development Committee, the financial regulatory agency under Vice-Premier Liu He, said the Chinese government would “crack down on bitcoin mining and trading behavior, and resolutely prevent the transfer of individual risks to the society.” Is not illegal for Chinese to hold or trade bitcoin or other cryptocurrencies. The PBOC has embraced blockchain technology and has been on the forefront of developing the central bank’s digital yuan.
Hong Kong	A FinTech hub. However, the Hong Kong Securities and Futures Commission (SFC) has enacted a strict regulatory framework and licensing requirements for virtual asset service platforms (VASPs). It has also proposed a ban of crypto trading for retail investors. Only professional investors who have over HK\$8 million in assets would be allowed to trade. Bitcoin is defined as a virtual commodity and not legal tender. No capital gains taxes and AML/CFT laws apply to every individual or business in Hong Kong, irrespective of FATF requirements.
Japan	Most progressive/developed regulatory regimes for cryptocurrencies, exchanges must be registered and comply with traditional AML/CFT and other regulations. The Payment Services Act (PSA) defines “cryptocurrency” as a property value and not a legal tender. In December 2017, Japan’s National Tax Agency ruled that gains on cryptocurrencies should be categorized as “miscellaneous income” and taxed accordingly. There have been several new regulations and amendments to the PSA and to the Financial Instruments and Exchange Act (FIEA), introducing the term “crypto-asset,” and regulating crypto derivatives trading. In April 2020, Japan was the first country to create self-regulatory bodies, the Japanese Virtual Currency Exchange Association (JVCEA) and the Japan STO Association.
India	In 2018 the Reserve Bank of India (RBI) banned cryptocurrency trading and prohibited Indian banks from dealing with cryptocurrency exchanges over consumer protection, AML, and market integrity concerns. However, in 2020 the Indian Supreme Court struck down the ban clarifying that no prohibition exists. India has encouraged innovation and the use of blockchain. It has also begun work on a state-backed digital central bank currency, the digital Rupee.
Indonesia	Virtual currencies are not considered legal tender. In 2019 the Indonesian Commodity Futures Trading Regulatory Agency (Bappebti) approved regulation which legally recognizes and regulates bitcoin and other cryptocurrencies as commodities. Derivative transactions, and cryptocurrency exchanges are also subject to regulatory requirements of Bappebti.
Iran	The Iranian Central Bank has authorized banks and currency exchanges to use cryptocurrencies mined by licensed crypto miners in the country. Although mining is legal, Iran takes a heavy-handed approach requiring firms to sell cryptos to the central bank to fund imports. Iran has issued over a thousand licenses to crypto miners and shut down unlicensed firms. Trading outside Iran has been banned, to stop capital flight. The use of cryptos for payments has also been banned.

Source: <https://www.thomsonreuters.com/en-us/posts/investigation-fraud-and-risk/crypto-special-report/>

On Tuesday (October 8, 2022), prices of cryptocurrencies tanked, citing the industry-shaking collapse of FTX⁸⁰, some even dubbed the event as “*Crypto’s Lehman moment*”⁸¹. But erratic price movements is not something new in the crypto industry which has been on a roller-coaster since December of 2021, i.e. after Bitcoin price hit almost \$68,000 and its market cap \$1.24 trillion, jittery investors in a hurry began to cash out their hefty gains (see Figure 37). The inability of FTX’s CEO Sam Bankman-Fried to handle his plan to sell FTX (which was regarded as one of crypto’s “blue chip” companies) to the rival crypto exchange Binance set off a widespread selling panic, cryptocurrency market as a whole shed a mindboggling \$236.7 billion in just two days (Tuesday and Wednesday), which by any standard was insanely bonkers. There may be wishful folks hoping that FTX’s collapse will set off a broader market failure (i.e. Lehman Brothers in 2008), but this will not happen because the cryptocurrency market and its participants (i.e. exchanges) are not as interwoven with the rest of the financial system as Lehman was; besides, Bitcoin (and altcoins) has reeled from even much tougher times and losses.



Source: <https://coinmarketcap.com/charts/>

Figure 37: Cryptocurrency market’s and Bitcoin’s market cap (November 2021-2022)

⁸⁰ After Binance (run by the Chinese-born billionaire Changpeng Zhao), FTX is the second largest cryptocurrency exchange founded by Sam Bankman-Fried and Gary Wang in May 2019, it incorporated in Antigua and Barbuda and headquartered in The Bahamas. Just like Binance.US, FTX also operates a separate exchange FTX.US available to US residents.

⁸¹ <https://www.nytimes.com/2022/11/09/technology/cryptocurrency-binance-ftx.html>

References

- Abernethy, D. (2000). *The Dynamics of Global Dominance, European Overseas Empires 1415—1980*. Yale University Press. p. 85. ISBN 0-300-09314-4.
- Abreu, D. & Brunnermeier, M.K. (2003). Bubbles and Crashes. *Econometrica*, 71(1), 173–204.
- Aizenman, J. & Lee, J. (2005). International Reserves: Precautionary versus Mercantilist Views, Theory and Evidence. NBER Working Paper 11366.
- Andolfatto, D. (2018). Assessing the Impact of Central Bank Digital Currency on Private Banks. Federal Reserve Bank of St. Louis Working Paper 2018–026C.
- Arrow, K.J. (1964). The Role of Securities in the Optimal Allocation of Risk Bearing. *Rev. Econ. Stud.*, Apr. 1964, (2), pp. 91–96.
- Atkinson, T., Luttrell, D. & Rosenblum, H. (2013). How Bad Was It? The Costs and Consequences of the 2007-09 Financial Crisis. Staff Papers No. 20, Federal Reserve Bank of Dallas Federal Reserve Board of Governors, July.
- Babones, S. (2009). *The International Structure of Income: Its Implications for Economic Growth*. Saarbruecken: VDM Verlag Dr Muller.
- Back, A. (2002). Hashcash - A Denial of Service Counter-Measure. <http://www.hashcash.org/papers/hashcash.pdf>.
- Baek, C. & Elbeck, M. (2015). Bitcoins as an Investment or Speculative Vehicle? A First Look. *Applied Economics Letters*, vol. 22, no. 1, pp. 30–34.
- Baran, P. (1964). On Distributed Communications Networks. *IEEE Trans. Comm. Systems*, March.
- Barnett et al. (2005). Penetration of Human-induced Warming into the World's Oceans. *Science*, 309, 284–287. 10.1126/science.1112418. Published online 2 June 2005.
- Bartos, J. (2015). Does Bitcoin Follow the Hypothesis of Efficient Market? *International Journal of Economic Sciences* 4: 10-23.
- Boar, C. & Wehrli, A. (2021). Ready, steady, go? – Results of the third BIS survey on central bank digital currency. BIS Papers No 114. Basel: Bank for International Settlements, January.
- BCBS (Basel Committee on Banking Supervision) (2004). *International Convergence of Capital Measurement and Capital Standards: A Revised Framework*. Basel, June.
- Bech, M.L. & Garatt. R. (2017). Central Bank Cryptocurrencies. *BIS Quarterly Review*.
- Ben-Sasson, E., Chiesa, A., Garman, C., Green, M., Miers, I., Tromer, E. & Virza. M. (2014). Zerocash: Decentralized anonymous payments from Bitcoin. In *IEEE Symposium on Security and Privacy*.
- Bensel, R.F. (2000). *The Political Economy of American Industrialization, 1877–1900*. Cambridge: Cambridge University Press. ISBN 978-0-521-77604-2.
- Berentsen, A. & Schär, F. (2018). A Short Introduction to the World of Cryptocurrencies. Federal Reserve Bank of St. Louis *REVIEW*, First Quarter, 100(1), pp. 1-16.
- Berg, A. & Borensztein, E. (2000). Full Dollarization: The Pros and Cons,” *IMF Economic Issues* No. 24, (Washington: International Monetary Fund).
- Bernanke, B.S. (1983). Nonmonetary Effects of the Financial Crisis in Propagation of the Great Depression. *American Economic Review*, June 1983, vol. 73 (3), pp. 257-76.

- Bernanke, B.S. (2004). Money, gold and the Great Depression. Remarks by Ben S Bernanke, Member of the Board of Governors of the US Federal Reserve System, at the H Parker Willis Lecture in Economic Policy, Washington and Lee University, Lexington, Virginia, 2 March.
- Bernanke, B.S. (2012). Monetary Policy since the Onset of the Crisis. Speech at the Federal Reserve Bank of Kansas City Symposium, Jackson Hole, Wyo., August 31.
- Bernanke, B.S. (2013). Monetary Policy and the Global Economy. At the Department of Economics and STICERD, School of Economics, London, United Kingdom, March 25.
- Bernanke, B.S. & Gertler, M. (1995). Inside the Black Box: The credit channel of monetary policy transmission. *Journal of Economic Perspective*, Vol. 9, No. 4, fall 1995, pp. 27-48.
- Bernanke, B.S., & James, H. (1991). The Gold Standard, Deflation, and Financial Crisis in the Great Depression: An International Comparison. This PDF is a selection from an out-of-print volume from the National Bureau of Economic Research, Volume Title: Financial Markets and Financial Crises, University of Chicago Press, January.
- Bernstein, P.L. (2000). *The Power of Gold: The History of Obsession*. John Wiley and Sons, Inc.
- Berryhill, J., Bourgerly, T. & Hanson, A. (2018). *Blockchains Unchained: Blockchain Technology and its Use in the Public Sector*. OECD Working Papers on Public Governance, No. 28, OECD Publishing, Paris, <https://doi.org/10.1787/3c32c429-en>.
- BIS (Bank for International Settlements) (2010). *An Assessment of the Long-Term Impact of Stronger Capital and Liquidity Requirements*. LEI Report, Basel.
- BIS (Bank for International Settlements) (2011). *Central Bank Governance and Financial Stability: A Report by a Study Group*. Basel: Bank for International Settlements. May.
- Blundell-Wignall, A. (2014). The Bitcoin Question: Currency versus Trust-Less Transfer Technology. *OECD Working Papers on Finance, Insurance and Private Pensions*, (37), 1.
- Blundell-Wignall, A., Atkinson, P.E., & Roulet, C. (2014). Bank Business Models and the Basel System: Complexity and Interconnectedness. *OECD Journal: Financial Market Trends* 2014, 2013 (2), 1-26.
- Bogart, E.L. (1930). *Economic History of the American People*. New York.
- Bonneau, J., Miller, A., Clark, J., Narayanan, A., Kroll, J. A. & Felten, E. W. (2015). Sok: Research Perspectives and Challenges for Bitcoin and Cryptocurrencies. In *Symposium on Security and Privacy*. IEEE, 2015, pp. 104–121.
- Bordo, M.D. (2007). *A Brief History of Central Banks*. Federal Reserve Bank of Cleveland, December (ISSN 0428-1276).
- Bordo, M.D. & Levin, A.T. (2017). *Central Bank Digital Currency and the Future of Monetary Policy*. NBER Working Paper Series, Working Paper 23711, <http://www.nber.org/papers/w23711>.
- Boulding, K.E. (1971). The Legitimacy of Central Banks. *Reappraisal of the Federal Reserve's Discount Mechanism*, Vol. 2, pp.1–13. Washington, D.C.: Board of Governors of the Federal Reserve System.
- Broadbent, B. (2016). *Central Banks and Digital Currencies*. Available at: <http://www.bankofengland.co.uk/publications/Documents/speeches/2016/speech886.pdf>.
- Buchman, E., Kwon, J. & Milosevic, Z. (2018). The Latest Gossip on BFT Consensus. *CoRR*, vol. abs/1807.04938.
- Camenisch, J., Hohenberger, S. & Lysyanskaya, A. (2005). Compact ecash. In *EUROCRYPT*.

- Cascarilla, C.G. (2015). Bitcoin, Blockchain, and the Future of Financial Transactions. In CFA Institute Conference Proceedings Quarterly (Vol. 32, No. 3, pp. 18-24). CFA Institute.
- Cassidy, J. (2009). *Dot.con: How America Lost Its Mind and Its Money in the Internet Era*. Harper Collins. ISBN 9780061841781.
- Catalini, C., Kominers, S.D. & Jagadeesan, R. (2019). Market Design for a Blockchain-Based Financial System. Working paper no. 3396834. Social Science Research Network.
- Cell, J.W. (1997). Europe and the World in an Expanding World Economy, 1700—1850. *Asia in World History: Essays*, p. 445, ISBN 9781563242656.
- Cerf, V.G. & Kahn, R.E. (1974). A protocol for Packet Network Interconnection. *IEEE Trans. Comm. Tech.*, vol. COM-22, V 5, pp. 627-641, May.
- Cermak, V. (2017). Can Bitcoin Become a Viable Alternative to Fiat Currencies? An Empirical Analysis of Bitcoin's Volatility Based on a GARCH Model. *SSRN Electronic Journal*.
- Chang, R. (2000). Dollarization: A Scorecard, *Economic Review*, Federal Reserve Bank of Atlanta, issue Q3, pages 1–12.
- Chaum, D. (1982). Blind Signatures for Untraceable Payments. *CRYPTO*.
- Chaum, D., Fiat, A. & Naor, M. (1998). Untraceable Electronic Cash. *CRYPTO*.
- Chinn, M. & Frankel, J. (2006). Will the Euro Eventually Surpass The Dollar As Leading International Reserve Currency? NBER, January.
- Chiu, J. & Wong, T-N. (2014). E-money: Efficiency, Stability and Optimal Policy. Bank of Canada, Working Paper, 2014–16, April.
- Christin, N. (2013). Traveling the Silk Road: A Measurement Analysis of a Large Anonymous Online Marketplace. In *WWW*, 2013.
- Cipolla, C. (1976). *Before the Industrial Revolution: European Society and Economy, 1000-1700*. New York, NY: W.W. Norton & Company, Inc.
- Cochrane, J.B. (2012). The Federal Reserve: From Central Bank to Central Planner. *The Wall Street Journal*, September 1–2.
- Cook, R. M. (1958). Speculation on the Origins of Coinage. *Historia* 7: 257–62.
- Cooper, R.N. (1982). The Gold Standard: Historical Facts and Future Prospects. *Brookings Papers on Economic Activity*, 1:1982.
- Corbin, M. & Levitsky, O. (2003). Vital Statistics: The U.S. Military. *Defense Monitor*. Volume XXXII, Number 5. November/Dec. 2003. Center for Defense Information. Washington, D.C. <http://www.cdi.org/news/defense-monitor/dm.pdf>.
- Côté, A. (1994). Exchange Rate Volatility and Trade: A Survey. Bank of Canada Working Paper No. 94-5.
- Crawford, C. (2011). The Repeal of the Glass- Steagall Act and the Current Financial Crisis. *Journal of Business & Economics Research*, Vol. 9(1), January.
- Dai, W. (1998). B-money. <http://www.weidai.com/bmoney.txt>.
- Dalton, G. (1967). Primitive Money. In George Dalton, ed. *Tribal and Peasant Economies*. Austin, TX and London, UK: University of Texas Press: 254–81.

- David, Y. (2014). *Is Bitcoin a Real Currency? An Economic Appraisal*. New York University Stern School of Business and National Bureau of Economic Research.
- Davies, G. (2000). The Single Currency in Historical Perspective. *British Numismatic Journal*, 69 (London).
- Davies, G. (2002). *A History of Money from Ancient Times to the Present Day*. 3rd ed. Cardiff: University of Wales Press, 720 pages. Paperback: ISBN 0 7083 1717 0.
- Demirors, M. (2017). Oxford Blockchain Strategy Programme. Said Business School, University of Oxford, UK, <https://www.getsmarter.com/presentations/uk/oxford-said/whatstakeholders-are-involved-in-the-blockchain-strategy-system>.
- Dooley, M.P., Landau, D.F., & Garber, P. (2004). The Revived Bretton Woods System: The Effects of Periphery Intervention and Reserve Management on Interest Rates & Exchange Rates in Center Countries. NBER Working Paper 10332.
- Douglas, M. & Wildavsky, A. (1983). *Risk and Culture*. Berkeley: University of California Press.
- Duffie, D. (2019). *Digital Currencies and Fast Payment Systems*. Mimeo, Stanford University.
- Dunning, M. (2001). Manifest Destiny and the Trans-Mississippi South: Natural Laws and the Extension of Slavery into Mexico. *Journal of Popular Culture*. 35 (2): 111–27.
- Dwork, C. & Naor, M. (1992). Pricing via Processing or Combatting Junk Mail. CRYPTO.
- Dwyer, G.P. (2014). The economics of Bitcoin and Similar Private Digital Currencies. *Journal of Financial Stability*, 333, pp. 1-11.
- Eastlake, D. & Hansen, T. (2011). US Secure Hash Algorithms (SHA and SHA-based HMAC and HKDF). RFC 6234 (Informational), May, available at: <http://www.ietf.org/rfc/rfc6234.txt>.
- Eichengreen, B. (2002). *Financial crises: And what to do about them*. Oxford University Press.
- Eichengreen, B. (2003). *Golden Fetters: The Gold Standard and the Great Depression 1919-1939*. Oxford University Press, Inc. New York, ISBN – 0-19-510113—8.
- Eichengreen, B. (2009). The Dollar Dilemma—the World’s Top Currency Faces Competition in Foreign Affairs. September/October.
- Eichengreen, B. (2011). Exorbitant Privilege: The Rise and Fall of the Dollar. New York: Oxford University press. pp. 44–49. ISBN 0199931097.
- Eichengreen, B. & Flandreau, M. (2009). The Rise and Fall of the Dollar, or When Did the Dollar Replace Sterling as the Leading Reserve Currency?” *European Review of Economic History* 13: 377–411.
- Eichengreen, B. & Irwin, D.A. (1995). Trade Blocs, Currency Blocs, and the Reorientation of World Trade in the 1930s. *Journal of International Economics* 38(1), pp. 1-24.
- Eichengreen, B. & Irwin, D.A. (2010). The Slide to Protectionism in the Great Depression: Who Succumbed and Why? *The Journal of Economic History*, Vol. 70 No. 4, December.
- Epstein, G. & Ferguson, T. (1984). Monetary Policy, Loan Liquidation, and Industrial Conflict: The Federal Reserve and the Open Market Operations of 1932. *Journal of Economic History*, December, pp. 957-83.
- Franco, P. (2014). *Understanding Bitcoin: Cryptography, Engineering and Economics*. John Wiley & Sons.

- Frankel, J. & Rose, A.K. (1996). Currency Crashes in Emerging Markets: An Empirical Treatment. *Journal of International Economics*, Vol. 41, Nos. 3-4, pp. 351-66.
- Friedman, M. & Schwartz, A. (1963). *A Monetary History of the United States 1867-1960*, Princeton: Princeton University Press, 1963.
- Friedman, M. & Schwartz, A. (2008). *The Great Contraction, 1929–1933* (New ed.). Princeton University Press. p. 247. ISBN 978-0691137940.
- Fung, B., Molico, M. & Stuber, G. (2014). *Electronic Money and Payments: Recent Developments and Issues*. Bank of Canada, Discussion Paper, 2014–2, April.
- Galindo, A. & Leiderman, L. (2005). *Living with Dollarization and the Route to Dedollarization*, Inter-American Development Bank Working Paper No. 526, New York.
- Garber, P.M. (1993). *A Retrospective on the Bretton Woods System: Lessons for International Monetary Reform* (Chapter Title: The Collapse of the Bretton Woods Fixed Exchange Rate System). University of Chicago Press, January.
- Goldschlag, D.M. & Stubblebine, S.G. (1998). Publicly Verifiable Lotteries: Applications of Delaying Functions. *Financial Cryptography*.
- Goodhart, C. (2000). Can Central Banking Survive the IT Revolution? *International Finance* 3 (2): 189–209.
- Gudgeon, L., Moreno-Sanchez, P., Ross, S., McCorry, P. & Gervais, A. (2019). Sok: Off the Chain Transactions. *IACR Cryptology ePrint Archive*, vol. 2019, p. 360.
- Haber, S. & Stornetta, W.S. (1991). How to Time-Stamp a Digital Document. In *Journal of Cryptology*, Vol 3(2), pp. 99-111, 1991.
- Hall, T. & Ferguson, J.D. (1998). *The Great Depression, an International Disaster of Perverse Economic Consequences*, Ann Arbor: The University of Michigan Press.
- Hamilton, J.D. (1987). Monetary Factors in the Great Depression. *Journal of Monetary Economics* (March 1987), pp. 145-69.
- Hansen, J., et al. (2005). Earth's Energy Imbalance: Confirmation and Implications. *Science*, 308, 1431–1435. 10.1126/science.1110252. Published online 28 April 2005.
- Hassol, S.J. (2004). *Impacts of a Warming Arctic*. Arctic Climate Impact Assessment (ACIA). Cambridge University Press, UK.
- Hileman, G. & Rauchs, M. (2017). *Global Blockchain Benchmarking Study*, https://www.jbs.cam.ac.uk/fileadmin/user_upload/research/centres/alternative-finance/downloads/2017-09-27-ccaf-globalbchain.pdf.
- Huang, D. Y., Dharmdasani, H., Meiklejohn, S., Dave, V., Grier, C., McCoy, D., Savage, S., Weaver, N., Snoeren, A. C. & Levchenko, K. (2014). *Bitcoin: Monetizing Stolen Cycles*. NDSS.
- Hudson, M. (1972). *Super Imperialism: The Economic Strategy of American Empire*. Holt: Rinehart and Winston.
- Ikenberry, G.J. (2001). American Power and the Empire of Capitalist Democracy. *Review of International Studies* 27: 191-212.
- IPCC (Intergovernmental Panel on Climate Change) (2001). *Climate Change 2001: The Scientific Basis*. Cambridge and New York: Cambridge University Press. (http://www.grida.no/climate/ipcc_tar/wg1/index.htm).

- IPCC (Intergovernmental Panel on Climate Change) (2001). Global warming of 1.5°C. Summary for Policy Makers. Printed October 2018 by the IPCC, Switzerland.
- Irwin, D.A. (1998). The Smoot – Hawley Tariff: A Quantitative Assessment. *The Review of Economics and Statistics*, Vol. 80(2), (May 1998), 326-334.
- Kahn, C. & Roberds, W. (2009). Why Pay? An Introduction to Payments Economics. *Journal of Financial Intermediation* 18 (1): 1–23.
- Kahn, R.E. (1972). Communications Principles for Operating Systems. Internal BBN memorandum, January.
- Kaser, G., Hardy, D.R., Moelg, T., Bradley, R.S. & Hyera, T.M. (2004). Modern Glacier Retreat on Kilimanjaro as Evidence of Climate Change: Observations and Facts. *International Journal of Climatology*, 24, 329–339.
- Kasper, J. (2017). Evolution of Bitcoin: Volatility Comparisons with Least Developed Countries' Currencies. *Journal of Internet Banking and Commerce (JIBC)*, vol. 22, no. 3.
- Katsiampa, P. (2017). Volatility estimation for Bitcoin: A comparison of GARCH models. *Economics Letters*, 158, 3–6.
- Kelly, B. (2014). *The Bitcoin Big Bang: How Alternative Currencies Are About to Change the World*. Wiley, 240 pages (ISBN: 978-1-118-96366-1).
- Kindleberger, C.P. (1973). *The World in Depression, 1929-1939*. Univ. of California Press.
- Kindleberger, C.P. (1978). *Manias, Panics, and Crashes: A History of Financial Crises*. New York: Basic Books.
- Kindleberger, C.P. (1986). *The World in Depression, 1929-1939*. 2nd ad. University of California Press.
- King, M.R (1999). Challenges for Monetary Policy: New and Old. Speech delivered at a symposium sponsored by the Federal Reserve Bank of Kansas City, Jackson Hole, WY August 27.
- Kirshner, J. (2008). Dollar primacy and American power: What's at stake? *Review of International Political Economy*, vol. 15(3), pp. 418 – 438.
- Kleinrock, L. (1961). Information Flow in Large Communication Nets. RLE Quarterly Progress Report, July.
- Kleinrock, L. (1964). *Communication Nets: Stochastic Message Flow and Delay*. McGraw-Hill (New York).
- Kleinrock, L. (1976). *Queueing Systems: Vol II, Computer Applications*. John Wiley and Sons (New York).
- Kocherlakota, N.R. (1998). Money Is Memory. *Journal of Economic Theory* 81(2): 232–51, August.
- Kristoufek, L. (2015). What Are the Main Drivers of the Bitcoin Price? Evidence from Wavelet Coherence Analysis. *PLOS ONE*, 10(4).
- Laeven, L. & Valencia, F. (2008). *Systemic Banking Crises: A New Database*. IMF Working Paper WP/08/224.
- Lardner, J. (2009). A Brief History of the Glass Steagall Act. Demos Background Paper, 11/10/09.
- Laurie, B. & Clayton, R. (2004). Proof-of-work proves not to work. WEIS.

- Levine, R. (1997). Financial Development and Economic Growth: Views and Agenda. *Journal of Economic Literature*, Vol. XXXV (June 1997), pp. 688–726.
- Licklider, J.C.R & Clark, W. (1962), On-Line Man Computer Communication. August.
- Mancini-Griffoli, T., Peria, M. S. M., Agur, I., Ari, A., Kiff, J., Popescu, A. & Rochon, C. (2018). Casting Light on Central Bank Digital Currency”, IMF Staff Discussion Note SDN/18/08, November.
- Marples, D.R. (2004). *The Collapse of the Soviet Union 1985—1991*. UK: Pearson Education Limited. p. 2. ISBN 0-582-50599-2.
- Massias, H., Avila, X. S. & Quisquater, J-J. (1999). Design of a Secure Timestamping Service with Minimal Trust Requirements. In 20th Symposium on Information Theory in the Benelux, May.
- McCarthy, J. J., Canziani, O. F., Leary, N. A., Dokken, D.J. & White, K.S. (2001). *Climate Change 2001: Impacts, Adaptation, and Vulnerability*. IPCC, Cambridge University Press, UK.
- McDonough, M.D. (2011). *Manifestly Uncertain Destiny: The Debate over American Expansionism, 1803–1848*. PhD dissertation, Kansas State University.
- Meltzer, A.H. (1976). Monetary and Other Explanations of the Start of the Great Depression. *Journal of Monetary Economics* (November 1976), pp. 455-71.
- Merkle, R.C. (1980). Protocols for Public Key Cryptosystems. In *Proc. 1980 Symposium on Security and Privacy*, IEEE Computer Society, pages 122-133, April.
- Merkle, R.C. (1987). A Digital Signature Based on a Conventional Encryption Function. In *Advances in Cryptology - CRYPTO '87, A Conference on the Theory and Applications of Cryptographic Techniques*, Santa Barbara, California, USA, August 16-20, 1987, Proceedings, pp. 369–378.
- Mersch, Y. (2017). Digital Base Money: An Assessment from the ECB’s Perspective. Available at: <https://www.ecb.europa.eu/press/key/date/2017/html/sp170116.en.html>.
- Mester, L.J. (1996). Repealing Glass Steagall: The Past Points the Way to the Future. *Business Review*, July/August.
- Moore, T. & Christin, N. (2013). Beware the Middleman: Empirical Analysis of Bitcoin-Exchange Risk. In *Financial Cryptography*.
- Nakamoto, S. (2008). *Bitcoin: A Peer-to-Peer Electronic Cash System*. Unpublished paper, available at <https://bitcoin.org/bitcoin.pdf>.
- Nanto, D.K. (1998). The 1997-98 Asian Financial Crisis. CRS Report for Congress, February 6.
- Narayanan, A., Bonneau, J., Felten, E., Miller, A. & Goldfeder, S. (2016). *Bitcoin and Cryptocurrency Technologies*. Princeton University Press: Princeton, NJ, USA.
- NCDC (National Climatic Data Center). (2004). Climate of 2003 in Historical Perspective. <http://www.ncdc.noaa.gov/oa/climate/research/2003/ann/ann03.html#Major%20highlight>
- Oerlemans, J. (2005). Extracting a Climate Signal from 169 Glacier Records. *Science*, 308, 675–677.
- Okamoto, T. & Ohta, K. (1992). Universal Electronic Cash. *CRYPTO*.
- O’Neill, B.C. & Oppenheimer, M. (2002). Dangerous Climate Impacts and the Kyoto Protocol. *Science*, 296, 1971–1972.
- Pike, C. (2018). Blockchain Technology and Competition Policy. OECD Issues Paper on Competition, [https://one.oecd.org/document/DAF/COMP/WD\(2018\)47/en/pdf](https://one.oecd.org/document/DAF/COMP/WD(2018)47/en/pdf).
- Pounds, J.A., Fogden, M.P.L & Campbell, J.H. (1999). Ecology: Clouded futures. *Nature* 398: 611-615.

- Poyser, O. (2017). Exploring the Determinants of Bitcoin's Price: an Application of Bayesian Structural Time Series. Dissertation.
- Reinhart, C.M. & Rogoff, K.S. (2009). This Time is Different: Eight Centuries of Financial Folly. Princeton University Press, p. 496, ISBN 978-0-691-14216-6.
- Reinhart, C.M., Rogoff, K.S. & Savastano, M.A. (2003). Addicted to Dollars. NBER Working Paper No. 10015, October.
- Reitman, R. (2019). Could Regulatory Backlash Entrench Facebook's New Cryptocurrency Libra? Electronic Frontier Foundation (EFF) (July 10), <https://www.eff.org/deeplinks/2019/07/could-regulatory-backlash-entrench-facebooks-new-cryptocurrency-libra>.
- Rivest, R.L. (2004). Peppercoin Micropayments. Financial Cryptography.
- Rivest, R.L. & Shamir, A. (1997). PayWord and MicroMint: Two Simple Micropayment Schemes. Security Protocols Workshop.
- Roberts, L. (1967). Multiple Computer Networks and Intercomputer Communication. ACM Gatlinburg Conf., October.
- Roberts, L. & Merrill, T. (1966). Toward a Cooperative Network of Time-Shared Computers. Fall AFIPS Conf., October.
- Rose, A.K. (1999). One Money, One Market: Estimating the Effect of Common Currencies on Trade. National Bureau of Economic Research Working Paper No. 7432.
- Sander, T. & Ta-Shma, A. (1999). Auditable, Anonymous Electronic Cash. CRYPTO.
- Saurina, J. & Trucharte, C. (2007). An Assessment of Basel II Procyclicality in Mortgage Portfolios. Journal of Financial Services Research, 32 (1), 81–101.
- Schaller, M. (1976). American Air Strategy in China, 1939-1941: The Origins of Clandestine Air Warfare. American Quarterly 28.1 (1976): 3-19.
- Schwartz, A. 2009. Original of the Financial Market Crisis of 2008. Cato Journal, 29 (1), 19-23.
- Schoenmakers, B. (1998). Security Aspects of the Ecash™ Payment System. State of the Art in Applied Cryptography.
- Shannon, T.R. (1996). An Introduction to the World-System Perspective (p.16). Boulder: Westview Press, ISBN 0-8133-2452-1.
- Souza, M.C., Souza, E.T.C., & Pereira, H.C.I. (2017). Cryptocurrencies Bubbles: New Evidences. The Empirical Economics Letters (ISSN 1681-8997), 16(7), July.
- Sovbetov, Y. (2018). Factors Influencing Cryptocurrency Prices: Evidence from Bitcoin, Ethereum, Dash, Litecoin, and Monero. Journal of Economics and Financial Analysis, Vol. 2 (2), pp. 1-27, Tripal Publishing House.
- Steiner, Z. (2011). The Triumph of the Dark: European International History, 1933–1939, pp 690–92, 738–41.
- Stott, P.A., Stone, D.A. & Allen, M.R. (2004). Human Contribution to the European Heatwave of 2003. Nature, 432, 610–614.
- Taskinsoy, J. (2012a). Relevancy of Corporate Financial Policies and the Profit Maximization View of Islamic Banks. Journal of Social and Development Sciences, Vol. 3(6), pp. 184-193, June, ISSN 2221-1152.

- Taskinsoy, J. (2012b). The Investment Rate of Return (IRR) to Tertiary Education in Turkey. *Journal of Education and Vocational Research*, Vol. 3(5), pp. 154-164, May, ISSN 2221-2590.
- Taskinsoy, J. (2012c). The Return of Investment on Tertiary Education in Malaysia. *Journal of Education and Vocational Research*, Vol. 3(6), pp. 183-192, ISSN 2221-2590.
- Taskinsoy, J. (2013a). Rigorous Capital Requirements under Basel III: Possible impact on Turkey's financial sector. *Journal of WEI Business and Economics*, Vol. 2(1), pp. 1-30, April.
- Taskinsoy, J. (2013b). Basel III: Road to Resilient Banking, Impact on Turkey's Financial Sector. LAP LAMBERT Academic Publishing, 237 pages, ISBN 13: 978-3-659-30696-9.
- Taskinsoy, J. (2013c). Economic & Ecological Implications of Hydraulic Fracturing. *West East Journal of Social Sciences*, Vol. 2(1), pp. 11-39, April.
- Taskinsoy, J. (2018a). Bitcoin Mania: An End to the US Dollar's Hegemony or another Cryptocurrency Experiment Destined to Fail? (December 1, 2018). Available at SSRN: <https://ssrn.com/abstract=3311989> or <http://dx.doi.org/10.2139/ssrn.3311989>.
- Taskinsoy, J. (2018b). Effects of Basel III Higher Capital and Liquidity Requirements on Banking Sectors across the Main South East Asian Nations. *International Journal of Scientific & Engineering Research (IJSER)*, Vol. 9(4), pp. 214-37, April, ISSN 2229-5518.
- Taskinsoy, J. (2018c). The Cost Impact of Basel III across ASEAN-5: Macro Stress Testing of Malaysia's Banking Sector. LAP LAMBERT Academic Publishing, 369 pages, ISBN-13 978-613-9-90012-1.
- Taskinsoy, J. (2018d). A Macro Stress Testing Framework for Assessing Financial Stability: Evidence from Malaysia. *Journal of Accounting, Finance and Auditing Studies (JAFAS)*, Vol. 4(3), July, ISSN 2149-0996
- Taskinsoy, J. (2019a). Ever More Financial Instability notwithstanding the Basel Standards and the IMF's Financial Sector Assessment Program (February 4, 2019). Available at SSRN: <https://ssrn.com/abstract=3328473> or <http://dx.doi.org/10.2139/ssrn.3328473>.
- Taskinsoy, J. (2019b). Turkish Lira – A Fiat Currency that Resembles the Volatility of Cryptocurrencies: The Effects of Exchange Rate Volatility on Turkish Economy (February 15, 2019). Available at SSRN: <https://ssrn.com/abstract=3335545> or <http://dx.doi.org/10.2139/ssrn.3335545>.
- Taskinsoy, J. (2019c). The Transition from Barter Trade to Impediments of the Dollar System: One Nation, One Currency, One Monopoly (March 6, 2019). Available at SSRN: <https://ssrn.com/abstract=3348119> or <http://dx.doi.org/10.2139/ssrn.3348119>
- Taskinsoy, J. (2019d). Stress Testing Made Easy: No More US Banks Stumbling and Facing Public Embarrassment Due to the Federal Reserve's Qualitative Objection (March 17, 2019). Available at SSRN: <https://ssrn.com/abstract=3354018> or <http://dx.doi.org/10.2139/ssrn.3354018>.
- Taskinsoy, J. (2019e). Typology of Stress Testing: Microprudential vs. Macroprudential Stress Testing of Risk Exposures (March 28, 2019). Available at SSRN: <https://ssrn.com/abstract=3361528> or <http://dx.doi.org/10.2139/ssrn.3361528>.
- Taskinsoy, J. (2019f). Pure Gold for Economic Freedom: A Supranational Medium of Exchange to End American Monetary Hegemony as the World's Main Reserve Currency (April 25, 2019). Available at SSRN: <https://ssrn.com/abstract=3377904> or <http://dx.doi.org/10.2139/ssrn.3377904>.

- Taskinsoy, J. (2019g). Asian Miracle, Asian Tiger, or Asian Myth? Financial Sector and Risk Assessment through FSAP Experience: Enhancing Bank Supervision in Thailand (May 9, 2019). Available at SSRN: <https://ssrn.com/abstract=3385337> or <http://dx.doi.org/10.2139/ssrn.3385337>.
- Taskinsoy, J. (2019h). Higher Capital and Liquidity Regulations of Basel Standards Have Made Banks and Banking Systems Become More Prone to Financial and Economic Crises (June 9, 2019). Available at SSRN: <https://ssrn.com/abstract=3401378> or <http://dx.doi.org/10.2139/ssrn.3401378>.
- Taskinsoy, J. (2019i). A Delicate Moment in Turkey's Economic Transition: Can Turkey Survive Mounting Economic Problems without the IMF's Bailout Package? (June 22, 2019). Available at SSRN: <https://ssrn.com/abstract=3408520> or <http://dx.doi.org/10.2139/ssrn.3408520>.
- Taskinsoy, J. (2019j). We Need No Dime from the IMF: Is This a Temporary Illusion or Can the Turkish Economy Recover from the Current Recession without the IMF Loans? (July 9, 2019). Available at SSRN: <https://ssrn.com/abstract=3417431> or <http://dx.doi.org/10.2139/ssrn.3417431>.
- Taskinsoy, J. (2019k). Facebook's Project Libra: Will Libra Sputter Out or Spur Central Banks to Introduce Their Own Unique Cryptocurrency Projects? (July 20, 2019). Available at SSRN: <https://ssrn.com/abstract=3423453> or <http://dx.doi.org/10.2139/ssrn.3423453>.
- Taskinsoy, J. (2019l). Is Facebook's Libra Project Already a Miscarriage? (August 15, 2019). Available at SSRN: <https://ssrn.com/abstract=3437857> or <http://dx.doi.org/10.2139/ssrn.3437857>.
- Taskinsoy, J. (2019m). Facebook's Libra: Big Bang or Big Crunch? A Technical Perspective and Challenges for Cryptocurrencies (August 29, 2019). Available at SSRN: <https://ssrn.com/abstract=3445150> or <http://dx.doi.org/10.2139/ssrn.3445150>.
- Taskinsoy, J. (2019n). Turkey's Unravelling Economy Resembles the Ottoman Empire's Financial Collapse (September 29, 2019). Available at SSRN: <https://ssrn.com/abstract=3461216> or <http://dx.doi.org/10.2139/ssrn>.
- Taskinsoy, J. (2019o). A Hiccup in Turkey's Prolonged Credit Fueled Economic Transition: A Comparative Analysis of Before and After the August Rout. *Journal of Accounting, Finance and Auditing Studies (JAFAS)*, Vol. 5(4), October, ISSN 2149-0996. Doi: 10.32602/jafas.2019.37.
- Taskinsoy, J. (2019p). This Time Is Different: Facebook's Libra Can Improve Both Financial Inclusion and Global Financial Stability as a Viable Alternative Currency to the U.S. Dollar. *Journal of Accounting, Finance and Auditing Studies (JAFAS)*, Vol. 5(4), October, ISSN 2149-0996. Doi: 10.32602/jafas.2019.38.
- Taskinsoy, J. (2019q). Blockchain: A Misunderstood Digital Revolution. Things You Need to Know about Blockchain (October 8, 2019). Available at SSRN: <https://ssrn.com/abstract=3466480> or <http://dx.doi.org/10.2139/ssrn.3466480>.
- Taskinsoy, J. (2019r). Blockchain: Moving Beyond Bitcoin into a Digitalized World (October 17, 2019). Available at SSRN: <https://ssrn.com/abstract=3471413> or <http://dx.doi.org/10.2139/ssrn.3471413>.
- Taskinsoy, J. (2019s). Blockchain: An Unorthodox Solution to Reduce Global Warming (October 24, 2019). Available at SSRN: <https://ssrn.com/abstract=3475144> or <http://dx.doi.org/10.2139/ssrn.3475144>.
- Taskinsoy, J. (2019t). Bitcoin and Turkey: A Good Match or a Perfect Storm? (October 30, 2019). Available at SSRN: <https://ssrn.com/abstract=3477849> or <http://dx.doi.org/10.2139/ssrn.3477849>.

- Taskinsoy, J. (2019u). Facebook's Libra: Why Does US Government Fear Price Stable Cryptocurrency? (November 7, 2019). Available at SSRN: <https://ssrn.com/abstract=3482441> or <http://dx.doi.org/10.2139/ssrn.3482441>.
- Taskinsoy, J. (2019v). Global Cooling through Blockchain to Avoid Catastrophic Climate Changes by 2050 (November 29, 2019). Available at SSRN: <https://ssrn.com/abstract=3495674> or <http://dx.doi.org/10.2139/ssrn.3495674>.
- Taskinsoy, J. (2019w). The Global Competitiveness Index: A Comparative Analysis between Turkey and G8 Nations (December 8, 2019). Available at SSRN: <https://ssrn.com/abstract=3500542> or <http://dx.doi.org/10.2139/ssrn.3500542>.
- Taskinsoy, J. (2019x). Bitcoin: The Longest Running Mania – Tulips of the 21st Century? (December 18, 2019). Available at SSRN: <https://ssrn.com/abstract=3505953> or <http://dx.doi.org/10.2139/ssrn.3505953>.
- Taskinsoy, J. (2019y). Widening Development Gaps between ASEAN-5 and Advanced Nations (December 30, 2019). Available at SSRN: <https://ssrn.com/abstract=3511482> or <http://dx.doi.org/10.2139/ssrn.3511482>.
- Taskinsoy, J. (2020a). Malaysia: 'Crouching Tiger, Hidden Dragon' (January 19, 2020). Available at SSRN: <https://ssrn.com/abstract=3522206> or <http://dx.doi.org/10.2139/ssrn.3522206>.
- Taskinsoy, J. (2020b). No Brainer, Tackle Climate Change by 2030 or Await the Domsday by 2100 (February 5, 2020). Available at SSRN: <https://ssrn.com/abstract=3532709> or <http://dx.doi.org/10.2139/ssrn.3532709>.
- Taskinsoy, J. (2020c). From Primitive Barter to Inflationary Dollar: A Warless Economic Weapon of Mass Destruction (February 21, 2020). Available at SSRN: <https://ssrn.com/abstract=3542145> or <http://dx.doi.org/10.2139/ssrn.3542145>.
- Taskinsoy, J. (2020d). Stress Testing: A Measure of Financial Stability across ASEAN-5 (February 25, 2020). Available at SSRN: <https://ssrn.com/abstract=3543955> or <http://dx.doi.org/10.2139/ssrn.3543955>.
- Taskinsoy, J. (2020e). Diminishing Dollar Hegemony: What Wars and Sanctions Failed to Accomplish, COVID-19 Has (April 7, 2020). Available at SSRN: <https://ssrn.com/abstract=3570910> or <http://dx.doi.org/10.2139/ssrn.3570910>.
- Taskinsoy, J. (2020f). Old and New Methods of Risk Measurements for Financial Stability amid the Great Outbreak (April 27, 2020). Available at SSRN: <https://ssrn.com/abstract=3587150> or <http://dx.doi.org/10.2139/ssrn.3587150>.
- Taskinsoy, J. (2020g). COVID-19: Is the Great Outbreak a Sign of What the Future Has Stowed for the Human Race? (May 10, 2020). Available at SSRN: <https://ssrn.com/abstract=3597434> or <http://dx.doi.org/10.2139/ssrn.3597434>.
- Taskinsoy, J. (2020h). Cost Implications of the Great Lockdown due to the Novel Coronavirus Outbreak (May 18, 2020). Available at SSRN: <https://ssrn.com/abstract=3604573> or <http://dx.doi.org/10.2139/ssrn.3604573>.
- Taskinsoy, J. (2020i). Coronavirus Pandemic: The U.S. Induced Great Panic of the 21st Century (May 26, 2020). Available at SSRN: <https://ssrn.com/abstract=3610749> or <http://dx.doi.org/10.2139/ssrn.3610749>.

- Taskinsoy, J. (2020j). COVID-19 Could Cause Bigger Cracks in Turkey's Fragile Crisis Prone Economy (May 28, 2020). Available at SSRN: <https://ssrn.com/abstract=3613367> or <http://dx.doi.org/10.2139/ssrn.3613367>.
- Taskinsoy, J. (2020k). Is Turkey Backsliding on Global Competitiveness and Democracy Amid Its EU Bid in Limbo? (July 10, 2020). Available at SSRN: <https://ssrn.com/abstract=3648210> or <http://dx.doi.org/10.2139/ssrn.3648210>.
- Taskinsoy, J. (2020l). Diminishing American Power: The US - China Trade War, Sanctions, and Coronavirus (August 20, 2020). Available at SSRN: <https://ssrn.com/abstract=3677841> or <http://dx.doi.org/10.2139/ssrn.3677841>.
- Taskinsoy, J. (2020m). The Great Pandemic of the 21st Century: The Stolen Lives (September 9, 2020). Available at SSRN: <https://ssrn.com/abstract=3689993> or <http://dx.doi.org/10.2139/ssrn.3689993>.
- Taskinsoy, J. (2020n). Bitcoin Could Be the First Cryptocurrency to Reach a Market Capitalization of One Trillion Dollars (September 16, 2020). Available at SSRN: <https://ssrn.com/abstract=3693765> or <http://dx.doi.org/10.2139/ssrn.3693765>.
- Taskinsoy, J. (2020o). The World Is at a Dangerous Crossroads on 'China Virus' and US 'Political Virus' (October 17, 2020). Available at SSRN: <https://ssrn.com/abstract=3713745> or <http://dx.doi.org/10.2139/ssrn.3713745>.
- Taskinsoy, J. (2020p). A Move towards a Cashless Society Accelerates With the Novel Coronavirus Induced Global Lockdown (December 12, 2020). Available at SSRN: <https://ssrn.com/abstract=3747750> or <http://dx.doi.org/10.2139/ssrn.3747750>.
- Taskinsoy, J. (2021a). Turkey's Hidden Economic Disaster Fostered by Unorthodox Policies in Isolation (February 8, 2021). Available at SSRN: <https://ssrn.com/abstract=3781491> or <http://dx.doi.org/10.2139/ssrn.3781491>.
- Taskinsoy, J. (2021b). Bitcoin Nation: The World's New 17th Largest Economy (February 28, 2021). Available at SSRN: <https://ssrn.com/abstract=3794634> or <http://dx.doi.org/10.2139/ssrn.3794634>.
- Taskinsoy, J. (2021c). Bitcoinmania: A Ticking Time Bomb Waiting to Explode (April 13, 2021). Available at SSRN: <https://ssrn.com/abstract=3861836> or <http://dx.doi.org/10.2139/ssrn.3861836>.
- Taskinsoy, J. (2021d). The Famous New Bubbles of the 21st Century: Cases of Irrational Exuberance (May 13, 2021). Available at SSRN: <https://ssrn.com/abstract=3845422> or <http://dx.doi.org/10.2139/ssrn.3845422>.
- Taskinsoy, J. (2021e). Turkey's Melancholy Economic Indicators Camouflaged by Militarization as a Foreign Policy (August 18, 2021). Available at SSRN: <https://ssrn.com/abstract=3907590> or <http://dx.doi.org/10.2139/ssrn.3907590>.
- Taskinsoy, J. (2021f). This Time Is Different: Bitcoin Has More Reasons to Reach the Price of \$100,000 (August 30, 2021). Available at SSRN: <https://ssrn.com/abstract=3914299> or <http://dx.doi.org/10.2139/ssrn.3914299>.
- Taskinsoy, J. (2021g). Bitcoin: A New Digital Gold Standard in the 21st Century? (October 13, 2021). Available at SSRN: <https://ssrn.com/abstract=3941857> or <http://dx.doi.org/10.2139/ssrn.3941857>.

- Taskinsoy, J. (2021h). Say Good Bye to Physical Cash and Welcome to Central Bank Digital Currency (November 27, 2021). Available at SSRN: <https://ssrn.com/abstract=3972858> or <http://dx.doi.org/10.2139/ssrn.3972858>
- Taskinsoy, J. (2022a). Turkish Lira's Unprecedented Free Fall: Can a Financial Coup Reverse Lira's Fastest Devaluation in Turkish History Induced by a Decade-Long Unorthodox Economic, Foreign and Defense Policies (February 1, 2022). Available at SSRN: <https://ssrn.com/abstract=4023696> or <http://dx.doi.org/10.2139/ssrn.4023696>
- Taskinsoy, J. (2022b). Stress Testing Financial Systems: Macro and Micro Stress Tests, Basel Standards and Value-at-Risk as Financial Stability Measures (February 11, 2022). Available at SSRN: <https://ssrn.com/abstract=4032869> or <http://dx.doi.org/10.2139/ssrn.4032869>
- Taskinsoy, J. (2022c). Financial Crises Continue to Strike amid Accelerated Evolution of Risk Management (February 19, 2022). Available at SSRN: <https://ssrn.com/abstract=4038732> or <http://dx.doi.org/10.2139/ssrn.4038732>
- Taskinsoy, J. (2022d). Elongated Addiction of Peripheries to Dollar: A Weapon of Mass Economic Destruction (March 10, 2022). Available at SSRN: <https://ssrn.com/abstract=4054317> or <http://dx.doi.org/10.2139/ssrn.4054317>
- Taskinsoy, J. (2022e). Basel I & Basel II Have Aggravated the Man-Made Dollar Virus (April 11, 2022). Available at SSRN: <https://ssrn.com/abstract=4080888> or <http://dx.doi.org/10.2139/ssrn.4080888>
- Taskinsoy, J. (2022f). Economic Meltdown in Turkey: Great Divergence Fueled by Erdogan's Quixotic Policies (August 15, 2022). Available at SSRN: <https://ssrn.com/abstract=4190664> or <http://dx.doi.org/10.2139/ssrn.4190664>
- Temin, P. (1976). Did Monetary Forces Cause the Great Depression? New York: W.W. Norton & Company.
- UNEP (United Nations Environment Programme) (2003). Division of Early Warning and Assessment (DEWA) and Global Resource Information Database (GRID), Impacts of summer 2003 heat wave in Europe, http://www.grid.unep.ch/product/publication/download/ew_heat_wave.pdf.
- Vishnumurthy, V., Chandrakumar, S. & Sirer, E.G. (2003). Karma: A Secure Economic Framework for Peer-to-Peer Resource Sharing. Workshop on Economics of Peer-to-Peer Systems.
- Wallace, B. (2011). The Rise and Fall of Bitcoin. Wired, November 23.
- Wallerstein, I. (1976). The Modern World-System: Capitalist Agriculture and the Origins of the European World-Economy in the Sixteenth Century. New York City: Academic Press. ISBN 0-12-785920-9.
- Wenninger, J. (2000). The Emerging Role of Banks in E-Commerce. Current Issues in Economics and Finance, Vol. 6(3), March. Federal Reserve Bank of New York.
- Wheelock, D.C. (1998). Federal Reserve Bank of St. Louis, Monetary Policy in the Great Depression and Beyond, Source of the Fed's Inflation Bias, published in The Economics of the Great Depression, ed. Mark Wheeler, Kalamazoo, Michigan: W.E. Upjohn Institute for Employment Research, 1998.
- Wigley, T.M.L. (2005). The Climate Change Commitment. Science, 307, 1766-1769.
- Wolla, S.A. (2018). Bitcoin: Money or Financial Investment? Page One Economics. Federal Reserve Bank of St. Louis, March.

- Woodford, M. (2000). Monetary Policy in a World without Money. *International Finance* 3(2), 229–60.
- Yin, M., Malkhi, D., Reiter, M. K., Golan, G. G. & Abraham, I. (2018). Hotstuff: BFT Consensus in the Lens of Blockchain. arXiv preprint arXiv:1803.05069.